

ANIMAL BITE INCIDENT MAPPING AND CLASSIFICATION WITH DECISION
SUPPORT SYSTEM

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By
ANTHONY A. ANGELES
MEG RYAN D. ROJO
JULIE ANN D. PANOLINO

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APPROVAL SHEET

The Capstone Project entitled “**ANIMAL BITE INCIDENT MAPPING AND CLASSIFICATION WITH DECISION SUPPORT**”

Prepared and submitted by **Anthony A. Angeles, Meg Ryan D. Rojo, and Julie Ann D. Panolino** in partial fulfillment of the requirements for the Degree of Bachelor of Science in Information Systems who are hereby recommended for Oral Examination.

JOSE LEO G. REDOBLO, PhD
Adviser

Approved and accepted by the Committee on Oral Examination with a rating of PASSED on January 12, 2025.

CHARWIN PADILLA
Member

JUSTIN T. OLIVAS
Member

ARGIE POLICARPIO
Member

TISHA MAY B. TIZON
Member

DANES P. TUMABING, MIT
Chairman

Accepted in partial fulfillment of the requirements for the Degree of Bachelor of Science in Information Systems.

CRISTINE V. REDOBLO, PhD
Capstone Project Professor

JOE MARIE D. DORMIDO, DIT
Dean, College of Computer Studies



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8. Abstract:



8.1. Abstract

The Animal Bite Incident Mapping and Classification with Decision Support System (ABIMC-DSS) was a web-based platform developed using the V-Model Software Development Life Cycle (SDLC) to strengthen rabies surveillance, monitoring, and decision-making at the Talisay City Health Animal Bite Center. It aimed to address issues such as delayed reporting, inconsistent manual records, and limited visibility of spatial trends by digitizing animal bite documentation, profiling bite victims, and automating case classification by animal type and bite severity. Built with PHP, MySQL, Bootstrap, HTML, CSS, Leaflet.js, and Chart.js, the system provided GIS-based mapping and heat mapping of incidents, analytics dashboards for trend monitoring, and automated report generation to support vaccination planning and preventive interventions. White-box and black-box testing confirmed that the major functions met the defined requirements and operated correctly. Evaluation by IT experts using the ISO/IEC 25010:2011 Software Product Quality Model yielded a Very High overall mean of 4.83, with the highest mean of 4.98 in portability and the lowest mean of 4.72 in compatibility, while all other criteria were likewise interpreted as Very High. The study concluded that ABIMC-DSS improves data accuracy, accelerates reporting, enhances spatial analysis of bite cases, and supports timely, informed public health actions. It recommends full adoption by the



Talisay City Health Office and further enhancements such as expanded analytics and interoperability to enable replication in other local government units.

8.2. Rationale / Background of the Study

Despite ongoing national and local efforts to control rabies, animal bite incidents continue to pose a serious public health concern in many areas of the Philippines. In Talisay City, the Animal Bite Center remains dependent on manual and paper-based methods for recording, monitoring, and reporting animal bite cases. These practices have resulted in delays in data retrieval, difficulty in tracking case trends, and limited capacity to identify high-risk areas for timely intervention. Health workers are required to manually review physical records when preparing reports or planning preventive measures, increasing the risk of errors and inefficiencies. As a result, decision-making related to rabies prevention, vaccination planning, and resource allocation becomes more challenging. To address these concerns, the Animal Bite Incident Mapping and Classification with Decision Support System was developed to digitize case recording, automate classification by animal type and bite severity, and integrate GIS-based mapping and analytical tools. The system aims to support health workers and local government units by improving data accuracy, enhancing monitoring efficiency, and enabling evidence-based public health decision-making in Talisay Cit



8.3. Summary

This study focused on the design and development of an Animal Bite Incident Mapping and Classification with Decision Support System for the Talisay City Health Animal Bite Center. The system was developed to address challenges related to manual record-keeping, fragmented data, delayed reporting, and the lack of spatial analysis in managing animal bite cases.

Using the V-Model Software Development Life Cycle, the system incorporated features such as digital victim profiling, classification of bite incidents by animal type and severity, GIS-based heat mapping of bite locations, automated report generation, and decision-support tools to assist in planning public health interventions. The system was implemented using web-based and open-source technologies including PHP, MySQL, Bootstrap, HTML, CSS, Leaflet.js, and Chart.js.

System functionality was evaluated through black-box and white-box testing, while overall software quality was assessed using the ISO/IEC 25010:2011 standard. Evaluation results indicated that the system effectively supports accurate data management, timely visualization of bite trends, and informed decision-making. The developed system demonstrates its potential to enhance animal bite surveillance, improve



operational efficiency, and strengthen rabies prevention and control efforts in Talisay City.

8.4. Findings

The researchers developed the Animal Bite Incident Mapping and Classification with Decision Support System for the Talisay City Health Animal Bite Center and evaluated its effectiveness through system testing and feedback from City Health Workers, Barangay Health Workers, and IT experts.

The findings presented below are:

1. The developed system successfully integrated digital animal bite incident recording, automated classification by animal type and bite severity, and GIS-based mapping, allowing health workers and IT experts to access and monitor animal bite data through a centralized web-based platform.
2. Functional testing results from black-box and white-box testing indicated that the system features performed as intended, confirming that the system is reliable and operationally stable.
3. System evaluation based on the ISO/IEC 25010:2011 Software Quality Model, as assessed by health workers and IT experts, showed that the system met the required quality standards in terms of functional suitability, performance efficiency, usability, reliability, security, maintainability, and portability.



4. The system effectively addressed existing challenges related to manual record-keeping, fragmented data storage, and difficulty in tracking animal bite trends, thereby improving data accuracy, accessibility, and monitoring efficiency.
5. The integration of GIS-based heat mapping and decision-support features enhanced the ability of health workers, local government units, and IT experts to identify high-risk areas and support evidence-based planning for rabies prevention and animal bite management in Talisay City.

8.5. Conclusions

Based on the findings, the conclusions are the following:

1. The Animal Bite Incident Mapping and Classification with Decision Support System effectively meets the operational needs of the City Health Office of Talisay City by providing a centralized, organized, and data-driven platform for recording and monitoring animal bite cases.
2. The integration of digital patient profiling, bite classification by animal type and severity, heatmap visualization of high-risk areas, automated report generation, and decision support features improves the ability of health workers to monitor cases, identify trends, and plan appropriate public health responses.



3. The successful results of both alpha and beta testing confirm that the system is functional, usable, reliable, and capable of performing its intended tasks without errors during actual use.
4. The evaluation using the ISO/IEC 25010:2011 Software Quality Model validated the overall quality of the system, yielding a “Very High” rating across all major characteristics, which demonstrates that the system meets international software quality standards and fulfills its functional objectives.
5. Overall, the system contributes significantly to improving the documentation, monitoring, and management of animal bite incidents, supporting better decision-making and strengthening rabies prevention and public health efforts in Talisay City.

8.6. Recommendations

Based on the conclusions, the recommendations below are:

1. The City Health Office of Talisay City may adopt and utilize the Animal Bite Incident Mapping and Classification with Decision Support System as an official tool for recording, monitoring, and managing animal bite cases across all barangays.
2. Future enhancements may include improving system compatibility to allow integration with national health monitoring databases for rabies surveillance and public health reporting.



3. Additional features such as automated SMS or email notifications for follow-up schedules and vaccination reminders may be included to improve patient monitoring and treatment compliance.
4. Other Local Government Units (LGUs) may consider replicating or adapting the system to support animal bite surveillance and rabies prevention efforts in their respective areas.
5. Continuous system maintenance, regular data updates, and periodic user training are recommended to ensure accurate data entry, proper system usage, and long-term effectiveness.
6. Future improvements may include advanced data analytics, predictive trend analysis, and enhanced mapping features to support better planning and early response to high-risk animal bite areas.



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Anthony A. Angeles

Julie Ann D. Panolino

Meg Ryan D. Rojo



DEDICATION

This capstone project is dedicated to our families, whose love, sacrifices, and constant support inspired us to pursue and successfully complete this academic endeavor.

We also dedicate this work to the faculty and mentors of the College of Computer Studies, Carlos Hilado Memorial State University, whose guidance and commitment to excellence helped shape our knowledge, skills, and professional values throughout this journey.

Most importantly, this project is dedicated to the City Health Office and Animal Bite Treatment Center of Talisay City, as well as the community it serves, with the hope that this system may contribute to improved animal bite monitoring, better public health response, and stronger rabies prevention efforts within the city.

Anthony A. Angeles

Julie Ann D. Panolino

Meg Ryan D. Rojo



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CHAPTER 1

INTRODUCTION

The rapid advancement of digital technology had revolutionized data management and decision-making across various sectors, including public health. However, rabies remains a serious public health issue. According to the Department of Health (DOH), a total of 426 Filipinos died from rabies in 2024, the highest number of deaths recorded in over a decade (Department of Health, 2024). Awareness of rabies disease was crucial because it engages communities and empowers people to save themselves by seeking the care they need (World Health Organization [WHO], 2025.).

Despite ongoing rabies awareness campaigns, many cities in the Philippines still face challenges in tracking animal bite incidents efficiently. The absence of a centralized digital system limits efficient data collection, analysis, and visualization, making it difficult for health authorities to respond effectively. Studies indicate that geospatial analysis plays a vital role in tracking rabies outbreaks and identifying high-risk areas (Cruz et al., 2024).

Talisay City faced rising animal bite cases but lacked a system for efficient monitoring. Health workers do not have an existing system with data recording, hindering trend analysis and prevention. The proposed Animal Bite Incident Mapping and Classification with Decision Support System integrates geospatial



mapping, classification, and decision support to help LGUs and health officials mitigate rabies risks.

In Talisay City, the Animal Bite Center still relied on manual reporting, hampering efficient data collection and timely responses. Geospatial analysis can effectively map disease distribution and highlight high-risk zones, underscoring the need for a digital surveillance system (Hu et al., 2023).

The purpose of this study was to develop an Animal Bite Incident Mapping and Classification with Decision Support System for tracking cases in Talisay City. The study aimed to improve monitoring efficiency by automating mapping and classification, which cuts down the time needed to collect, analyze, and display data. This helps health workers quickly identify high-risk areas and respond faster, enhancing public health and reducing rabies outbreaks.

1.1 Background of the Study

The Talisay City Health Animal Bite Center serves as the primary facility responsible for managing and responding to animal bite incidents within Talisay City. With a growing population and increasing interaction between humans and animals, the center plays a vital role in monitoring bite cases, administering post-exposure treatments, and preventing the spread of rabies and other zoonotic diseases. The center was managed by trained healthcare workers who are tasked with recording bite incidents, profiling patients, and reporting relevant data to local health authorities.



Currently, the center relied on manual documentation of animal bite cases. When individuals report bite incidents, healthcare personnel record the victim's details, type of animal involved, and nature of the incident using paper forms. These records are stored physically in folders or filing cabinets. When data was needed for reporting, trend analysis, or referrals health workers must manually search through files, which was time consuming and prone to error. There was no centralized database or digital classification method in place to efficiently track incidents by animal type, frequency, or geographic location.

In response to this, the researchers proposed the Animal Bite Incident Mapping and Classification with Decision Support System to address the center's challenges in managing animal bite cases. The system was designed to digitize the recording process, classify incidents based on the type and severity of the animal bite, and generate geospatial maps to identify high-risk areas. Through this solution, health workers will be able to access and analyze data more efficiently, enabling faster response, accurate reporting, and more strategic public health interventions.

1.2 Objectives of the Study

The purpose of this study was to develop an Animal Bite Incident Mapping and Classification with Decision Support System

Specifically, this study aimed to:



1. Design and develop a Animal Bite Incident Mapping and Classification with Decision Support System

with the following technical features:

- 1.1. encoding bite victim details into system;
- 1.2. classification by animal type and bite severity;
- 1.3. heat mapping to identify high-risk bite areas;
- 1.4. automated generation of incident reports; and
- 1.5. decision support for health planning and response.

2. Test the functionality of the developed system.

3. Evaluate the quality of the system using the ISO 25010:2011 Criteria:

- 3.1. functional suitability;
- 3.2. performance efficiency;
- 3.3. compatibility;
- 3.4. usability;
- 3.5. reliability;
- 3.6. security;
- 3.7. maintainability; and
- 3.8. portability.

4. Develop a User's Guide of the developed system.



1.3 Significance of the Study

This study was significant as it developed a localized, data-driven Animal Bite Incident Mapping and Classification with Decision Support System that enhances the ability of Talisay City to monitor, analyze, and respond to animal bite incidents associated with rabies. By integrating digital case recording, automated bite classification, GIS-based mapping, and decision-support features into a centralized platform, the system improves the accuracy, accessibility, and timeliness of animal bite data. Its development contributes to more efficient surveillance, faster identification of high-risk areas, improved planning of vaccination and prevention strategies, and informed public health decision-making at the local government level.

The results of this study was significant to the following;

Animal Bite Treatment Center. The result of this study will make a significant contribution to the Animal Bite Center by improving the accuracy and speed of recording animal bite cases, allowing health workers to respond more effectively and prioritize high-risk incidents. This addresses the current challenge of manual data recording, which is time-consuming and error-prone. With a digital system in place, the center could generate real-time reports, spot trends, and allocate medical resources more efficiently. It will also be supported long-term improvements in rabies prevention strategies and evidence-based planning.

Local Government Units (LGUs). The result of this study will be supported LGUs by equipping them with data-driven insights for planning and implementing



rabies control programs. Through geospatial mapping and classification tools, LGUs can easily identify areas with frequent bite incidents and focus their interventions accordingly. This addresses the existing lack of timely and accurate data, which hinders effective resource allocation. In the long term, the system can guide policy development, budget planning, and public health campaigns aimed at reducing animal bite cases in Talisay City.

Public Health Researchers and Policy Makers. The result of this study provided researchers and policymakers with a reliable source of categorized and location-specific data on animal bite cases. This improves the quality of research and supports the development of effective public health policies. It addresses the challenge of limited and disorganized data that makes it difficult to evaluate the impact of rabies programs. Over time, the system can serve as a reference model for other cities and support national efforts in controlling and eliminating rabies.

Future Researchers. The result of this study was beneficial for other proponents interested in developing systems related to public health, animal bite monitoring, and decision support. This can serve as a foundation of knowledge for them, providing essential background information and insights on geospatial mapping, incident classification, and automated health recommendations that can guide their own studies.



1.4 Scope and Limitation

This study focuses on the development of an Animal Bite Incident Mapping and Classification with Decision Support System for the Talisay City Health Animal Bite Center. The system was designed to improve how animal bite cases are recorded, categorized, and analyzed, helping health workers respond more efficiently and plan preventive measures. It had two interfaces one for Barangay Health Workers and another for City Health Workers at the Animal Bite Center. These interfaces enabled them to efficiently manage, classify, and monitor reported animal bite cases as part of their daily operations. Key features of the system include digital victim profiling, classification by animal type and severity, geospatial mapping of high-risk zones, automated report generation, and decision support tools to help plan vaccination campaigns and other responses. The system was developed using the V-Model Software Development Life Cycle (SDLC), which ensures a structured and step-by-step process with corresponding testing at each development phase. For the technology stack, the proponents used Bootstrap, HTML, and CSS for the frontend design; PHP for backend development; MySQL for data storage; Leaflet.js for geospatial mapping; and Chart.js or D3.js for data visualization. The system was responsive and accessible via desktop. The study spanned eight months and will involve testing by City Health Workers and Barangay Health Workers.

While the system offers several helpful features, there were some limitations to consider. The accuracy of the decision support and mapping tools depends



heavily on the quality and completeness of the data entered by health workers. Mistakes in data entry or missing information could affect the reliability of the reports and recommendations. In addition, the system is designed solely as a web-based platform for desktop use, as all data encoding will be handled by Barangay Health Workers and City Health Workers on behalf of the residents, and it still requires a stable internet connection for proper use.

1.5 Definition of Terms

The following key terms have been operationally defined based on their specific usage within the context of this study. These definitions aim to provide clarity and ensure a consistent understanding of concepts central to the research, particularly those related to animal bite reporting, geospatial mapping, incident classification, and public health decision support within Talisay City.

Animal Bite Incident Mapping and Classification System. In this study, this term refers to a system designed to a digital platform designed to record, organize, and visualize data related to animal bite cases, enabling efficient tracking, classification, and analysis of incidents.

Automated Generation of Incident Reports. In this study, this term refers to the system's capability to automatically compile and generate summaries, and reports of the collected data for easier monitoring and decision-making.



Classification by Animal Type and Severity. In this study, this term refers to the categorization of bite incidents based on the kind of animal involved (dog, cat, or other) and the seriousness of the injury (minor, moderate, or severe).

Digital Profiling of Bite Victims. In this study, this term refers to the process of collecting and storing detailed personal and incident-related information about individuals who have been bitten, including demographics, location, and medical details.

Decision Support for Health Planning and Response. In this study, this term refers to a tool and insights provided by the system to assist public health officials in making data-driven decisions regarding prevention strategies, resource allocation, and emergency responses related to animal bites.

Mapping of High-Risk Areas. In this study, this term refers to the use of geographic information systems (GIS) to visually display locations with frequent or severe animal bite cases, helping to identify patterns and clusters of incidents.

V-Model System Development Life Cycle. In this study, this term refers to a software development approach that emphasizes a step-by-step process where each development phase directly linked with a corresponding testing phase. It follows a “V-shaped” structure, ensuring that verification and validation occur simultaneously, leading to early detection of errors, improved quality, and a more systematic approach to building reliable systems



Barangay Health Worker (BHW). In this study, this term refers to a community health representative responsible for encoding patient and bite incident data at the barangay level into the system.

City Health Worker (CHW). In this study, this term refers to the personnel assigned at the Talisay City Animal Bite Center who act as system administrators. They manage encoded data, user accounts, reports, and system settings within the Animal Bite Incident Mapping and Classification with DSS.



CHAPTER II

REVIEW OF RELATED LITERATURE/STUDIES

This chapter presents a brief review of related literature, studies, and the interrelated system as the basis of making and developing the Animal Bite Incident Mapping and Classification with Decision Support System.

2.1 Foreign Literature

Exploring the Spatial Pattern of Animal Bites in Iran

This study investigated the geographic distribution and epidemiological characteristics of animal bite incidents in Iran, emphasizing their public health significance due to the risk of rabies particularly in high-burden regions such as South-East Asia, Africa, and the Middle East. The findings showed that bites were largely attributed to dogs and cats, with domestic dogs identified as the primary source. Most cases involved young males living in rural areas, and the study noted an increasing trend in reported bite incidents over time, alongside clear spatial disparities across provinces.

To better understand where risk was concentrated, the researchers applied spatial epidemiological methods, including the Besag, York, and Mollié (BYM) model and spatial scan statistics, to detect high-risk clusters that may extend beyond administrative boundaries. The study also highlighted key methodological challenges such as ecological fallacy, the modifiable areal unit problem (MAUP),



and spatial autocorrelation that can influence results if not properly addressed. Overall, the research stressed that mapping and analyzing spatial patterns of animal bites is essential for guiding targeted prevention efforts and designing more effective public health interventions (Gholampoor-Sigaroodi et al., 2023).

Mapping rabies distribution in China: a geospatial analysis of national surveillance data

This study analyzed the spatial distribution of rabies in China using provincial-level national surveillance data, with the goal of identifying geographic patterns of risk and examining how nonmedical ecological factors influence rabies occurrence. The research emphasized that although rabies cases have declined nationally, the disease remains a public health concern in specific provinces where risk persists or is emerging.

Findings showed that rabies risk was significantly associated with environmental and socioeconomic conditions, particularly temperature, economic status, and population density. Several provinces such as Guizhou, Guangxi, Hunan, and Hainan consistently recorded high standardized morbidity ratios, while Yunnan and Anhui were identified as newly emerging high-risk areas in recent years. The study quantified these relationships, reporting that a 1°C increase in temperature was linked to a 28% rise in rabies risk, and a 10,000-yuan increase in GDP per capita was associated with a 17% increase in standardized morbidity ratios.



Overall, the study highlighted the usefulness of geospatial modeling for improving disease monitoring and supporting targeted interventions, including strengthening dog vaccination programs and enhancing surveillance in vulnerable areas. It also recommended adopting a One Health approach that integrates human, animal, and environmental health to support long-term rabies elimination efforts in China (Hu et al., 2023).

Spatial Distribution and Population Estimation of Dogs in Thailand: Implications for Rabies Prevention and Control

This study examined the spatial distribution and population characteristics of stray dogs in Thailand and discussed their implications for rabies prevention and control. The researchers described stray dogs as those not under direct supervision or those allowed to roam freely, often originating from owned dogs permitted to wander, abandoned pets, or unowned dogs that reproduce without regulation. The study emphasized that growing stray dog populations create major public health concerns, including the spread of zoonotic diseases, increased dog bite cases, and even traffic-related accidents.

Findings highlighted that dog, mostly strays, account for 97% of animal bite injuries in Thailand, with challenges observed in both urban and rural settings. The problem is intensified by the presence of free-roaming dogs that are frequently unvaccinated and unsterilized, yet are still supported through community feeding practices, which can sustain transmission cycles and hinder rabies control efforts.



Rabies was identified as still endemic in the country, with 46 reported human deaths between 2010 and 2015 and over 600,000 individuals requiring post-exposure prophylaxis each year.

Overall, the study stressed the importance of localized and ecologically informed data for designing sustainable dog population management strategies and strengthening rabies prevention programs, particularly in high-risk communities (Aiyarak et al., 2021).

The epidemiology of dog bite injuries in Greece: preliminary results of a surveillance scheme

The study by Chatzis et al. (2023) presented preliminary findings from a surveillance system established to record and analyze dog bite incidents in Greece. Emphasizing the close relationship between humans and dogs, the research discussed how human to dog interactions can also create health risks, especially in communities with large numbers of unsupervised or ownerless dogs. Dogs were identified as the most common species involved in animal bite injuries, which may lead to pain, emotional distress, functional impairment, and potential exposure to zoonotic infections such as rabies.

The surveillance initiative was developed in response to the limited availability of systematic national data on dog bite injuries. By gathering information through healthcare facility reporting, the study aimed to improve epidemiological understanding and strengthen evidence-based planning. Overall,



the findings underscored the value of structured monitoring systems in supporting public health decision-making and policy development related to dog bite prevention and response (Chatzis et al., 2023).

Overview of snakebite in Brazil: Possible drivers and a tool for risk mapping

The study by Silva et al. (2021) examined snakebite as a continuing public health concern in Brazil, where medically important snake species are widely distributed across the country. To understand patterns of snakebite morbidity and mortality, the researchers integrated snake distribution data with environmental indicators and health system factors, allowing them to assess community vulnerability and identify conditions that may increase risk.

Results showed that forest loss, low gross domestic product per capita, and higher rural population density were significant predictors of elevated snakebite risk. Spatial cluster analysis further identified municipalities with consistently high risk, particularly those located in remote and forested areas where access to timely medical care may be limited. Building on these findings, the study developed a risk categorization tool intended to guide practical interventions such as prioritizing antivenom distribution, improving medical training, and strengthening public awareness programs.

Overall, the study highlighted that effective snakebite prevention and control requires geographically targeted strategies and recommended a One Health



approach that connects environmental change, animal ecology, and human health outcomes to reduce snakebite burden (Silva et al., 2021).

2.2 Local Literature

Spatio-temporal analysis of animal bite cases

This study from Nueva Vizcaya, Philippines, highlighted the growing concern of animal bite cases and emphasized the importance of identifying high-risk areas through spatio-temporal analysis. By applying Global Moran's I and Getis-Ord G_i^* statistical tools, the researchers were able to map animal bite hotspots, revealing that a majority of municipalities, including Bambang, Bayombong, Solano, and Bagabag, exhibited significantly high incidence rates both during the day and at night in 2019. The study underscored the need for targeted public health interventions such as localized awareness campaigns, stricter implementation of ordinances, and increased community education. These findings support the relevance of developing systems such as the Animal Bite Incident Mapping and Classification with Decision Support System, which similarly aims to identify high-risk zones and assist health authorities in responding more efficiently to animal bite incidents (Taclay et al., 2023).

Clinical, epidemiological, and spatial features of human rabies cases in Metro Manila, the Philippines from 2006 to 2015

This study provided valuable insights into the persistent burden of rabies in the Philippines despite the availability of post-exposure prophylaxis (PEP). Through a retrospective review of 575 clinical rabies cases recorded from 2006 to



2015 in a Manila referral hospital, the researchers found that the majority of patients were adult males, with most exposures occurring in the National Capital Region and surrounding regions. Spatial analysis using case mapping and heatmaps revealed that human rabies cases repeatedly occurred in the same geographic areas throughout the ten-year period, underscoring the need for localized surveillance and targeted intervention. Alarming, very few patients had received even a single dose of the rabies vaccine, and two suspected PEP failures were identified. These findings support the importance of implementing systems such as the Animal Bite Incident Mapping and Classification with Decision Support System, which can help visualize high-risk areas, improve vaccination tracking, and support more proactive rabies control strategies at the community level (Guzman et al., 2023).

Risk Factors Affecting Severity of Rabies Exposure in Dog Bite Victims

This study analyzed 1,658 dog bite cases collected from the Davao City Health Office to determine the risk factors affecting the severity of rabies exposure in humans. Using probit and logistic regression models, the researchers examined associations between rabies exposure severity and variables such as the age and gender of the victim, the location of the biting incident, ownership status of the biting dog, and the location of the bite on the victim's body. The results showed that younger victims, particularly toddlers aged 0–4 years, had the highest probability of severe rabies exposure, while older individuals were at lower risk. Additionally, victims bitten on the arm, hand, leg, thigh, or foot were less likely to



be severely exposed compared to bites in other body parts. The study emphasized the importance of targeted information, education, and communication (IEC) campaigns for parents and schools, as well as prompt first aid and medical treatment following dog bites. These findings support the relevance of systems such as the Animal Bite Incident Mapping and Classification with Decision Support System, which can help authorities identify high-risk groups, target educational campaigns, and improve timely medical interventions (Mata et al., 2020).

Animal Bite Patterns and Implementation of Rabies Prevention and Control Program

This study employed a retrospective descriptive comparative quantitative design to analyze animal bite cases in Quezon Province and assess the implementation of the Rabies Prevention and Control Program (RPCP) in both government and private Animal Bite Treatment Centers (ABTCs). Using 200 patient records and survey responses from 10 healthcare workers, the study found that animal bite incidents peaked in May, July, and August, with an ascending trend observed from 2016 to 2017. Females and individuals aged above 15 years were identified as the most common victims, and dogs were the primary biting animals, with most cases classified as Category II bites. While the RPCP was frequently implemented in government and private centers, significant differences in implementation levels were observed. Despite proper program implementation, animal bite cases continued to increase, highlighting areas that require improvement. These findings support the need for systems such as the Animal



Bite Incident Mapping and Classification with Decision Support System, which can monitor trends, identify high-risk areas, and assist authorities in improving program implementation (Amparo & Jayme, 2023).

Spatial Distribution of Canine Rabies

This study analyzed the spatial distribution of canine rabies infections in Pampanga from 2014 to 2018 using QGIS-generated heatmaps. Data on rabies cases were obtained from the Regional Animal Disease Diagnostic Laboratory, while temperature, rainfall, and humidity data were sourced from the Philippine Atmospheric, Geophysical, and Astronomical Services Administration. The results showed that rabies cases increased over the five-year period, with higher incidence observed between February and June. The spatial distribution of cases appeared scattered across the province, showing no clear clustering pattern. Among the climatic factors examined, only rainfall showed a significant negative association with rabies cases. These findings support the need for spatially informed surveillance and targeted interventions, aligning with the objectives of the Animal Bite Incident Mapping and Classification with Decision Support System to improve rabies control and resource allocation (Santos, 2021).

2.3 Foreign Systems

Bite Case Tracker (BCT)

The Bite Case Tracker (BCT) is a web and mobile surveillance tool from the Global Alliance for Rabies Control that tracks and maps animal and snake bite



incidents at the community level in real time. It captures key case details such as the bite location, victim information, and bite circumstances, then generates mapped outputs to show where bite cases are occurring and which areas may need targeted action. Because it visualizes bite-prone locations, BCT supports decision-making for focused interventions such as bite-risk education, vaccination prioritization, and improved monitoring in high-risk communities (Global Alliance for Rabies Control, 2022).

DHIS2 Human Rabies Surveillance toolkit

The DHIS2 Human Rabies Surveillance toolkit is a web-based surveillance package designed to standardize reporting and support decision-making for rabies control programs. It structures data capture for case classification, exposure details, post-exposure prophylaxis administration, and outcomes, and it can be paired with DHIS2 analytics and mapping features to help identify trends and geographic areas needing stronger response. By organizing rabies surveillance indicators into dashboards and visual outputs, the toolkit supports evidence-based planning and targeted interventions (DHIS2, 2024).

Rabies in San Diego County: Rabies Local Data Dashboard

The County of San Diego maintains a web-based rabies data dashboard that presents animals reported to the Epidemiology Unit and submitted for rabies testing to the county public health laboratory. The platform provides time-based



filtering (for example, by year) and interactive visual summaries to help users interpret trends and support timely public health actions. By organizing surveillance outputs into an accessible dashboard, the system helps guide monitoring and response planning for rabies risk within the jurisdiction (County of San Diego, 2025).

Rabies Dashboard (Franklin County, North Carolina)

Franklin County operates an online rabies dashboard that consolidates rabies surveillance visuals, including mapped case locations and summary graphics for animal exposures. The dashboard presents confirmed rabies case locations and also summarizes animal bites and severe scratches across multiple years, helping users identify where exposure reports and confirmed cases are concentrated. By combining bite and exposure indicators with geographic displays, the platform supports local decision-making for prevention messaging, investigation prioritization, and vaccination-related actions (Franklin County, 2025).

WHO Integrated Data Platform for Rabies (Global Health Observatory data improvement initiative)

The World Health Organization supports rabies data reporting through its Integrated Data Platform, which was designed to strengthen global rabies surveillance by standardizing and improving how countries submit rabies-related indicators. The initiative emphasizes improving the quality and completeness of



rabies data and describes dashboard-style outputs that can support interpretation and planning, including the use of mapped and visual analytics for monitoring patterns and targeting interventions. This supports a One Health, data-driven approach to rabies control and elimination planning (World Health Organization, 2023).

2.4 Local Systems

Mapping of Animal Bites in Cagayan Province

This study developed a decision-support system that visualizes animal bite cases in Cagayan Province using an online mapping platform. Implemented with Google Maps and a database system, the platform enables health workers to identify clusters of animal bite incidents and determine areas requiring immediate intervention. Interviews with Animal Bite Treatment Center personnel revealed that many victims seek medical attention only when bites have already reached high-risk levels and that vaccination efforts are often delayed due to budget constraints and supply shortages. By consolidating bite data, geographic patterns, and resource gaps into a single dashboard, the system supports improved planning of dog vaccination campaigns, pre-positioning of vaccine supplies, and faster response to emerging cases (Guiquing, 2023).

Spatial Analysis of African Swine Fever Outbreak: A Basis for Developing a Web-Based Decision Support System using Business Analytics and Geospatial Technology



This study focused on the African Swine Fever outbreak in Region VI and proposed a web-based decision support system that integrates Geographic Information Systems, business analytics, and the DBSCAN clustering method to monitor disease spread among hog populations. The system was designed to map outbreak hotspots in real time, analyze spatial patterns, and provide timely information to veterinarians, farmers, and government agencies. By combining spatial and non-spatial data, the system supports informed decision-making and efficient resource allocation. Although the study centers on African Swine Fever, it demonstrates how GIS-based decision support systems can enhance outbreak response and public sector planning, supporting the relevance of similar approaches in animal bite surveillance and public health monitoring (Mission & Rubio, 2024).

One Health Surveillance for Rabies: A Case Study of Integrated Bite Case Management

The implementation of Integrated Bite Case Management (IBCM) in Albay Province demonstrated that systematic classification of bite cases significantly improved rabies detection and response. The approach increased rabies case detection by approximately fourfold through the integration of patient interviews, follow-ups, and field investigations. Despite high bite incidence, the study found that many patients received unnecessary post-exposure prophylaxis, as only a small proportion were confirmed rabies exposures. The findings revealed that rabies transmission was localized, reinforcing the importance of targeted



surveillance systems. The study supports the development of systems such as the Animal Bite Incident Mapping and Classification with Decision Support System to improve triage, identify high-risk areas, and strengthen coordination under the One Health framework (Rysava et al., 2022).

Enhanced Rabies Surveillance Using Integrated Bite Case Management

This study applied Integrated Bite Case Management in Oriental Mindoro over a three-year period from 2020 to 2022 to enhance rabies surveillance. A probabilistic decision tree model was used to estimate rabies burden, assess surveillance performance, and analyze the cost-effectiveness of prevention strategies. IBCM increased rabies case detection nearly fourfold compared to routine surveillance and revealed that a significant proportion of exposed individuals did not seek medical care (Tan et al., 2023).

Genomics-Informed Surveillance for Rabies Elimination

This study developed and validated a cost-effective whole-genome sequencing workflow using nanopore technology to enhance rabies surveillance in the Philippines. By integrating genomic data with epidemiological information, the research identified four major rabies virus clades corresponding to the country's main island groups. The findings emphasized the importance of combining genomic surveillance with traditional epidemiological approaches and Integrated Bite Case Management to strengthen rabies detection and support region-specific control strategies (Bautista, 2025).



2.5 SYNTHESIS

Table 1

Synthesis of Related Literatures

Related Studies	Features and Functionalities	Developed System
Exploring the Spatial Pattern of Animal Bites in Iran	• Spatial cluster detection • Bayesian spatial modeling	Added bite cluster detection and spatial risk classification for unequal-risk areas
Mapping Rabies: A Geospatial Analysis of National Surveillance Data	• Spatiotemporal rabies modeling • Links cases with environment • Identifies high-risk zones over time	Inspired trend analysis + forecasting using time-based maps and environment overlays
Spatial Distribution and Population Estimation of Dogs in Thailand	• Dog population estimation • Separates stray vs. owned • Supports control planning	Led to dog profiling/tagging (household vs. stray) for vaccination planning
The Epidemiology of Dog Bite Injuries in Greece	• Routine bite surveillance reporting • Basic epidemiologic profiling • Supports trend tracking	Strengthened case profiling (demographic/clinical) to improve classification and triage
Overview of Snakebite in Brazil: Drivers and a Tool for Risk Mapping	• Risk mapping approach • Uses environmental drivers • Guides targeted interventions	Encouraged risk maps using location + environmental layers for targeted response
Spatio-Temporal Analysis of Animal Bite Cases in Nueva Vizcaya	• Time-based filtering • Spatial clustering • Supports campaign scheduling	Guided spatiotemporal filters to show incidence changes across months/seasons
Clinical, Epidemiological, and Spatial Features of Human Rabies in Metro Manila	• Heatmap visualization • Spatial overlays • Highlights vaccine gaps.	Added heatmaps + overlays (severity, vaccine status, repeat cases) for stronger monitoring
Risk Factors Affecting Severity of Rabies Exposure in Davao City)	• Identifies severity predictors • Uses regression analysis • Highlights high-risk groups	Supported high-risk group detection to guide education and faster medical response
Animal Bite Patterns and Implementation of RPCP in Quezon Province	• Bite trend patterns • Peak months + demographics • Program implementation differences	Reinforced trend monitoring + hotspot identification to support program improvement
Spatial Distribution of Canine Rabies in Pampanga, Philippines	• GIS heatmaps • Environmental link (rainfall) • Spatial pattern assessment	Supports GIS heatmap surveillance for targeted, location-based interventions



The Animal Bite Incident Mapping and Classification System represents an important development in improving how animal bite cases were monitored, analyzed, and addressed. Based on the reviewed studies, the use of spatial mapping, statistical modeling, and trend analysis has proven effective in identifying bite hotspots, determining risk levels, and guiding intervention efforts. Literature also highlights the importance of dog population estimation, timely reporting, and visual data presentation to support planning, surveillance, and public health response. These studies show that systems equipped with geospatial tools, heatmaps, and classification features can help authorities understand patterns more clearly and make informed decisions.

In summary, the proposed system provides a complete solution for mapping bite cases, separating households from stray dog incidents, visualizing hotspots, and showing trends over time. It aligns with the direction of the reviewed literature by offering a centralized and easy-to-use platform that supports risk detection, data monitoring, and strategic planning. With continued development and use, the system can enhance surveillance efforts, improve decision-making, and serve as a reliable tool for public health management and response.



Table 2

Synthesis of Related Systems

Related Literature	Features and Functionalities	Developed System
Bite Case Tracker (BCT) Bite Case Tracker (BCT)	• Records bite details + GPS • Victim + animal info • Generates maps for high-risk areas	Guided geotagging, incident mapping, and hotspot detection for priority intervention zones
DHIS2 Human Rabies Surveillance toolkit	• Standardized rabies reporting • Case classification + exposure details • PEP tracking + outcomes	Inspired structured bite classification fields and dashboard analytics for trend monitoring and planning
Rabies in San Diego County: Rabies Local Data Dashboard	• Public dashboard • Time filters • Interactive summaries (reported/tested animals)	Supported an interactive dashboard with filters and trend summaries for situational awareness
Rabies Dashboard (Franklin County, North Carolina)	• Map-based exposure display • Multi-year summaries • Tracks bites/severe scratches indicators	Reinforced map viewing + historical summaries to spot recurring risk locations and exposure patterns
WHO Integrated Data Platform for Rabies	• Standardized rabies indicators • Data completeness/quality focus • Dashboard-style outputs	Encouraged standard reporting fields, data quality checks, and One Health-ready outputs for decision support
Mapping of Animal Bites in Cagayan Province	• GIS visualization of bite cases • Highlights high-incidence areas	Supported interactive GIS mapping of bite locations + frequency to identify LGU-level hotspots
Spatial Analysis of African Swine Fever Outbreak	• Real-time monitoring dashboard • Geospatial visualization	Inspired real-time plotting and dynamic monitoring dashboards for immediate action and trend checks
One Health Surveillance for Rabies: IBCM in Albay	• One Health integration • Bite classification workflow • Socio-environmental analysis	Led to comprehensive classification, socioeconomic tagging
Enhanced Rabies Surveillance Using IBCM in Oriental Mindoro	• Multi-year IBCM implementation • Improves detection • Reveals PEP-seeking gaps	Supports improved monitoring, resource allocation, and intervention planning using better case tracking
Genomics-Informed Surveillance for Rabies Elimination in the	• Genome sequencing workflow • Linked with epidemiologic data	Reinforces stronger surveillance + case tracking



The Animal Bite Incident Mapping and Classification System was designed to support health authorities in monitoring animal bite cases through mapping, classification, and trend visualization. From the reviewed systems, features such as hotspot mapping, automated reporting, and interactive dashboards were shown to improve incident monitoring and response. Systems like Bite Case Tracker, Animal Control Manager Software, and other GIS-based surveillance platforms highlight the usefulness of geomapping, real-time visualization, and centralized databases in identifying high-risk locations and guiding intervention strategies. These systems demonstrate how combining spatial mapping with reporting tools can help health agencies respond faster and more effectively.

In summary, the developed system provides a complete and accessible platform for plotting bite cases, identifying hotspots, classifying incidents, and generating analytical insights. It aligns with the capabilities observed in related systems by offering mapping, data integration, and real-time monitoring in one environment. This system supports better public health decision-making, improves surveillance efforts, and helps authorities plan targeted responses to reduce animal bite incident.



CHAPTER III

Methodology

This chapter introduces the systematic approach and tools used to develop and evaluate the Animal Bite Incident Mapping and Classification with Decision Support System. The study adopts the Systems Development Life Cycle (SDLC) framework, which provides a structured process for planning, analyzing, designing, building, testing, and implementing the system. The chapter outlines the operational framework that guided the project and presents key modeling tools including the Context Diagram and Data Flow Diagrams (DFD) to illustrate system processes, the Entity-Relationship Diagram (ERD) to define data structures, and the Unified Modeling Language (UML) Use Case Diagram and scenarios to describe user interactions. A complete Data Dictionary was provided to define all data elements used in the system. The chapter also specified the necessary hardware and software requirements, presented the project timeline, and discussed the evaluation process, which includes system testing and assessment using the ISO/IEC 25010:2011 Software Quality Model to ensure functionality, reliability, and overall user experience. The chapter concluded with the identification of respondents involved in the system assessment and the ethical considerations observed throughout the study.



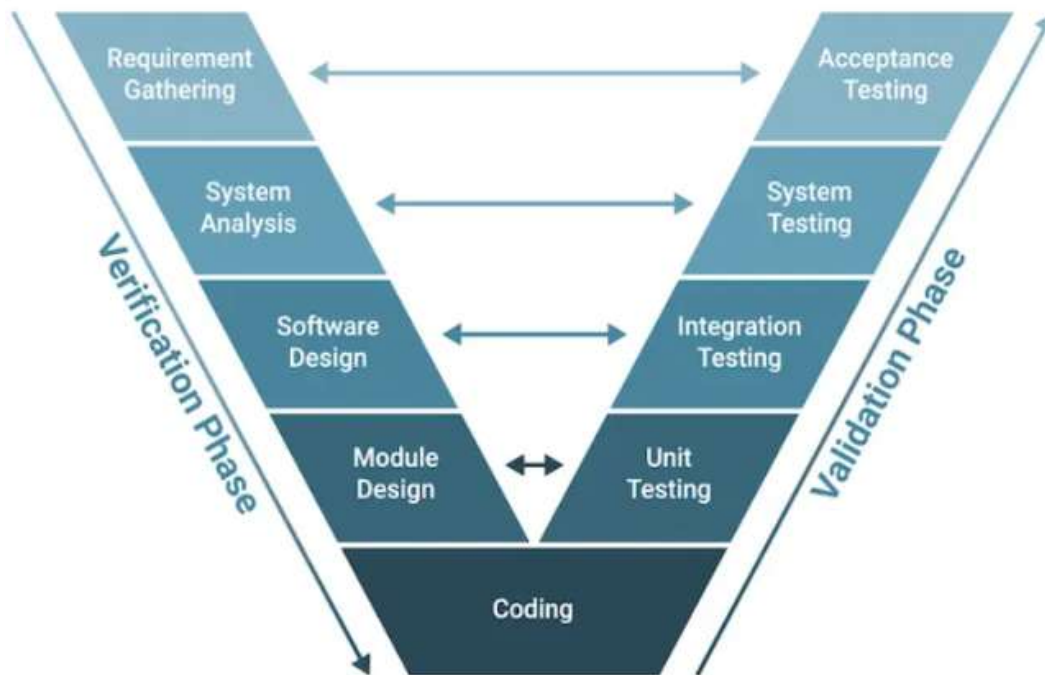
3.1 Systems Development Method

Systems Development Method refers to the framework used to structure, plan, and control the information system development process. This study employed both descriptive and developmental research methods, where descriptive research was used to evaluate the system's functionality, usability, and performance through testing procedures and feedback from City Health Workers and IT experts, while developmental research focused on the design, development, testing, and refinement of the Animal Bite Incident Mapping and Classification with Decision Support System. To guide the system development process, the proponents adopted the V-Model Software Development Life Cycle, a structured SDLC approach that pairs each development phase with a corresponding testing phase and emphasizes verification and validation throughout the system development process (Simran, 2025).



Figure 1

V-Model Methodology Model



The figure above shows the V-Model Software Development Life Cycle adapted from Simran (2025). The system development life cycle guided the proponents in developing the Animal Bite Incident Mapping and Classification with Decision Support System. This software development life cycle model included the phases of requirements analysis, system design, architectural design, module design, and coding, along with their corresponding testing phases, namely unit testing, integration testing, system testing, and acceptance testing (Simran, 2025). The study employed a combination of developmental research, which focuses on the iterative creation and refinement of practical systems (Richey & Klein, 2007), and descriptive research, which aims to systematically observe and evaluate



existing conditions and user feedback without intervention (Creswell & Creswell, 2018). This approach was appropriate for the development of the system due to its emphasis on verification and validation at every stage of development, ensuring that each phase produces deliverables that were thoroughly tested before proceeding to the next stage. As a result, this structured methodology helped maintain accuracy and reliability throughout the development process.

Requirements Gathering Phase

In this phase, the proponents conducted interviews with City Health Workers at the Animal Bite Center to identify existing processes, challenges, and system needs in handling animal bite incidents. Information was gathered on how cases are recorded, categorized, monitored, and reported. These findings were used to define the system's core functions, including bite incident recording, automatic classification, GIS-based mapping, report generation, and decision support features.

System Analysis

Based on the gathered requirements, the proponents analyzed the current workflow and identified system limitations related to manual record-keeping and data fragmentation. Functional and non-functional requirements were established, including data accuracy, usability, security, and performance needs. This phase clarified how the system should support Barangay Health Workers and City Health Workers in monitoring and managing animal bite cases.



Software Design

In this phase, the overall software design of the system was planned. This included designing the user interface for health workers, defining how data would be entered, processed, and stored, and planning system functionalities. Technologies such as PHP for backend development, MySQL for database management, Bootstrap, HTML, and CSS for frontend design, Leaflet.js for geospatial mapping, and Chart.js for data visualization were selected to support the system requirements.

Module Design Phase

Each feature of the system was designed in detail. For example, the classification module was created to categorize bite cases automatically based on severity and exposure. The mapping module was designed to show bite cases using location pins and heatmaps. Diagrams such as data flow diagrams and entity relationship diagrams were created to guide development.

Coding Phase

During this phase, the system was built based on the approved designs. Each module, such as the victim form, classification engine, and mapping feature, was developed using the chosen programming tools. Coding was done in stages, with each component being built and tested to ensure it matched the designs.



Unit Testing Phase

Each part of the system was tested separately to confirm that it worked as intended. For example, the classification feature was tested with different bite case scenarios to see if it returned the correct category. Bugs or errors were fixed before combining the modules.

Integration Testing Phase

After testing individual modules, the system was tested as a whole to make sure the parts work properly together. For instance, it was verified that data entered by Barangay Health Workers could be classified and automatically displayed on the City Health Workers' map and report dashboard.

System Testing Phase

In this phase, the complete system was tested in an environment similar to real use. All features were checked from encoding patient data to generating reports and visualizing mapped cases to ensure they functioned as a whole and responded well to user actions.

Acceptance Testing Phase

Finally, the finished system was presented to City Health Workers for actual testing. Their feedback confirmed that the system met their needs and worked well in their daily operations. Based on their comments, the proponents made final adjustments before considering the system ready for use.



3.2 Operational Framework

Figure 2

Operational Framework of Animal Bite Incident Mapping and Classification with Decision Support System

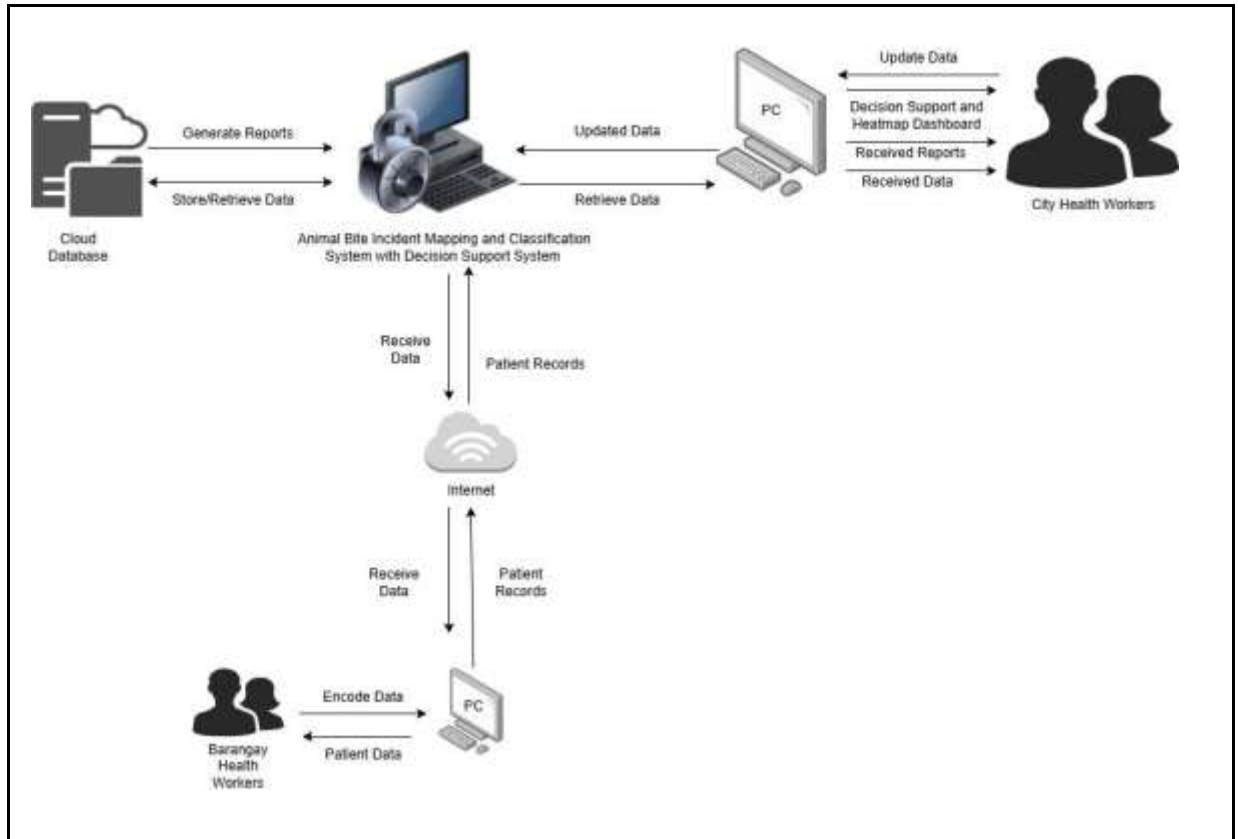


Figure 2 shows the Operational Framework of the Animal Bite Incident Mapping and Classification with Decision Support System, which illustrates how bite incidents were reported and managed for timely and effective response. When a bite incident occurs, the affected resident proceeds to the barangay to report the case. Barangay Health Workers would encode the patient data into the system using a computer. The data was transmitted through the internet to the central system, where it was stored in a cloud database, classified by type and severity,



and visualized for analysis. City Health Workers at the Animal Bite Center retrieve, review, and update this data as needed to plan appropriate interventions, such as vaccine distribution or health campaigns. This process ensures organized tracking and faster decision-making in managing animal bite cases in Talisay City.

3.3 Data Flow Diagram

Figure 3

Context Diagram of Animal Bite Incident Mapping and Classification with Decision Support System

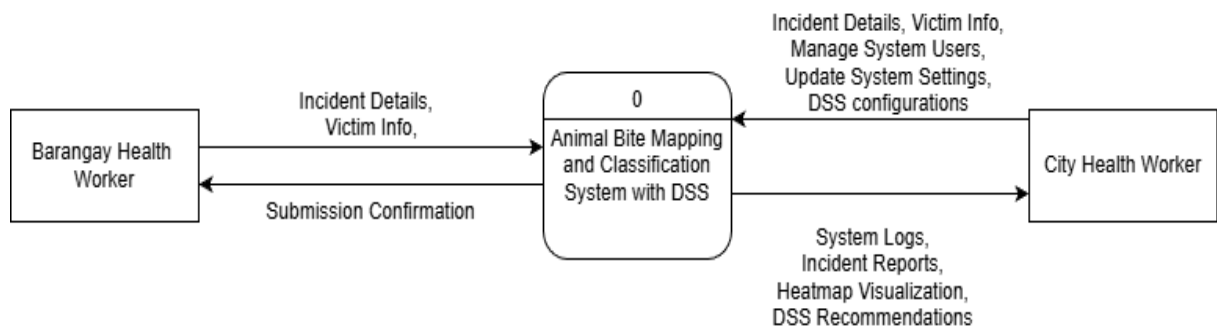


Figure 3 shows the context diagram of the system, highlighting how Barangay Health Workers and City Health Workers interact with it. Barangay Health Workers provide victim information, and incident details, while City Health Workers manage users, check data accuracy, and approve reports. The system sends back reports, updates, and system data to help both types of health workers perform their duties properly and keep bite case records organized.



Figure 4
Data Flow Diagram (Level 1)

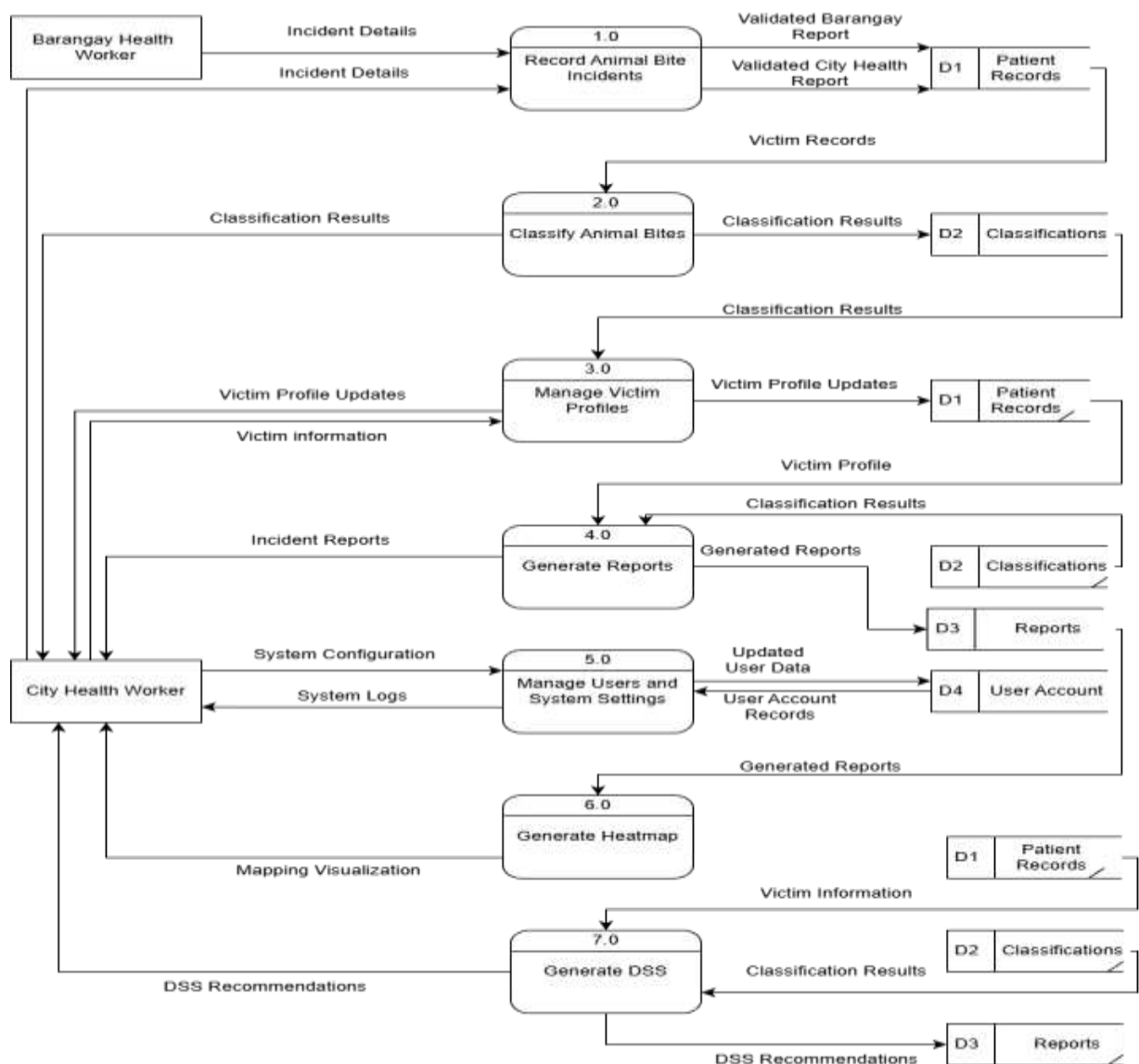
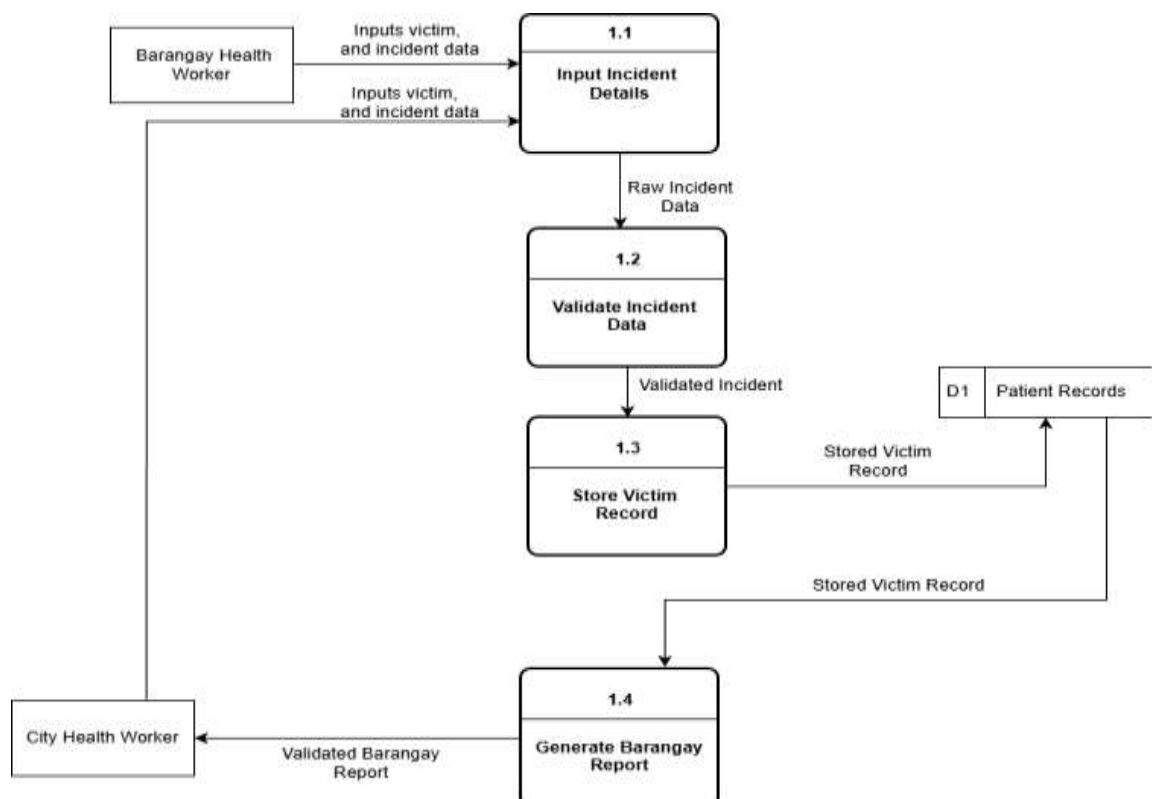


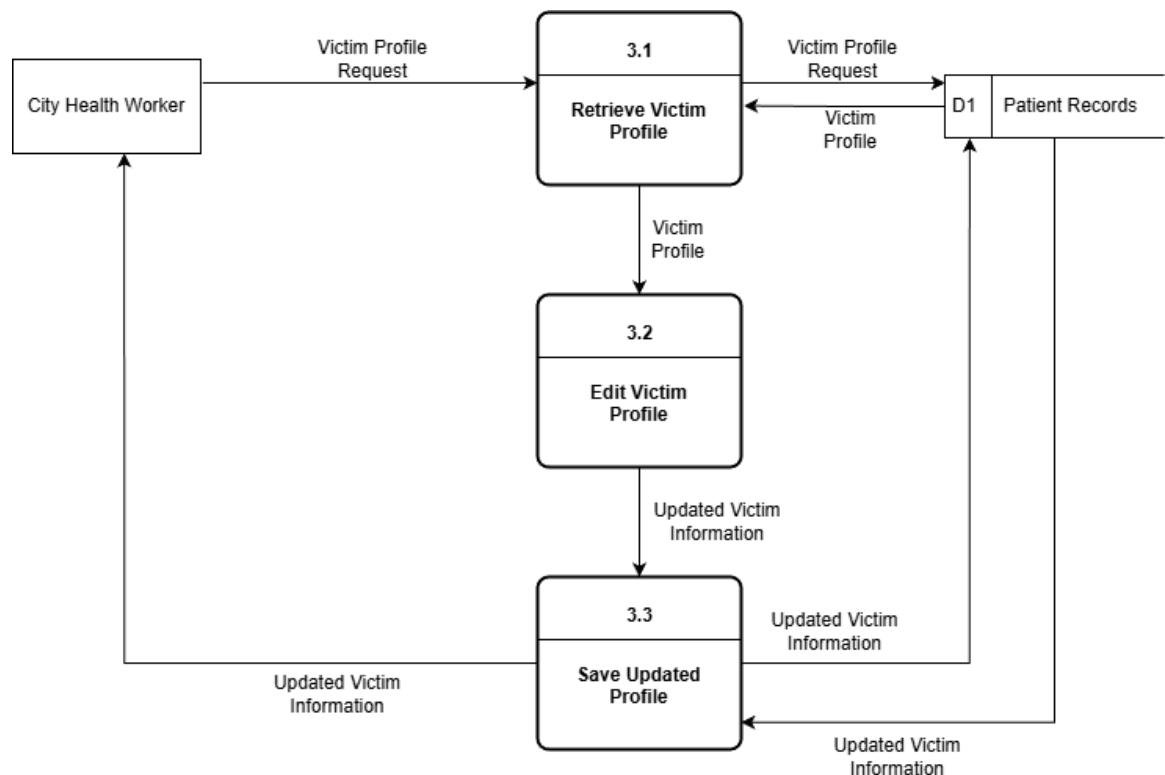
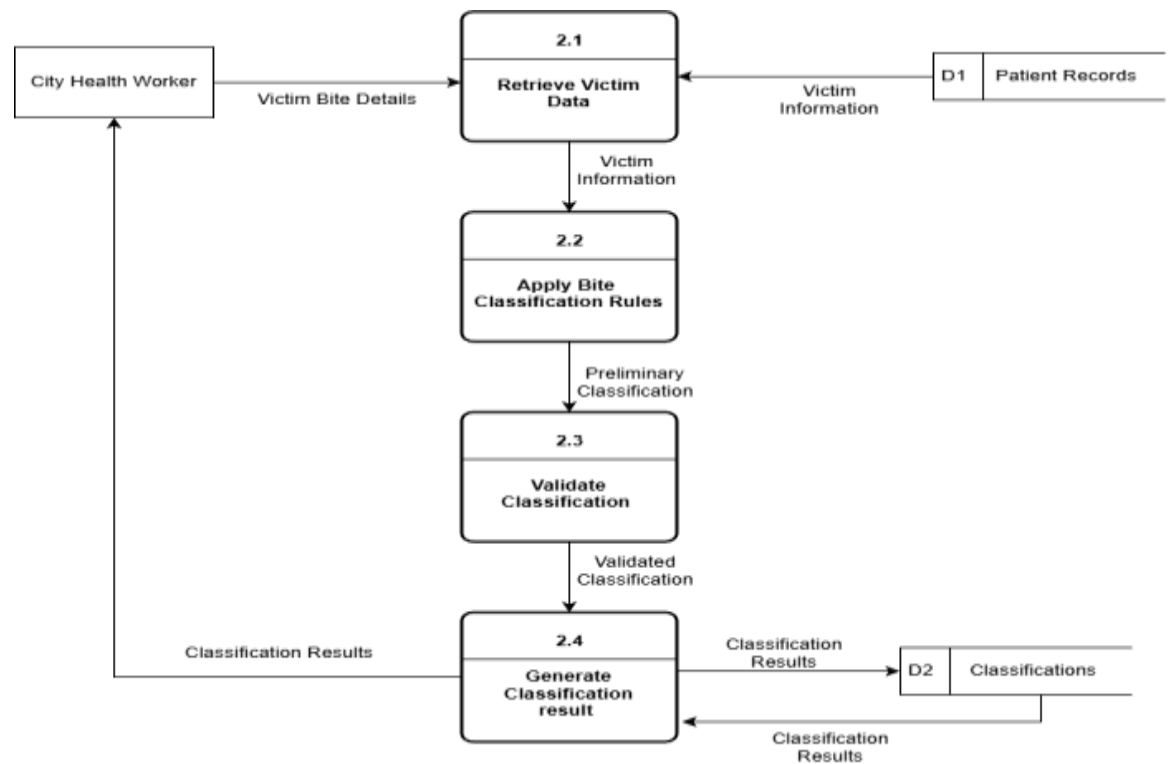
Figure 4 illustrates the Level 1 Data Flow Diagram of the Animal Bite Mapping and Classification System with Decision Support. It expands the main process into seven interconnected functions that show how data moves throughout the system. Barangay Health Workers begin the workflow by recording animal bite

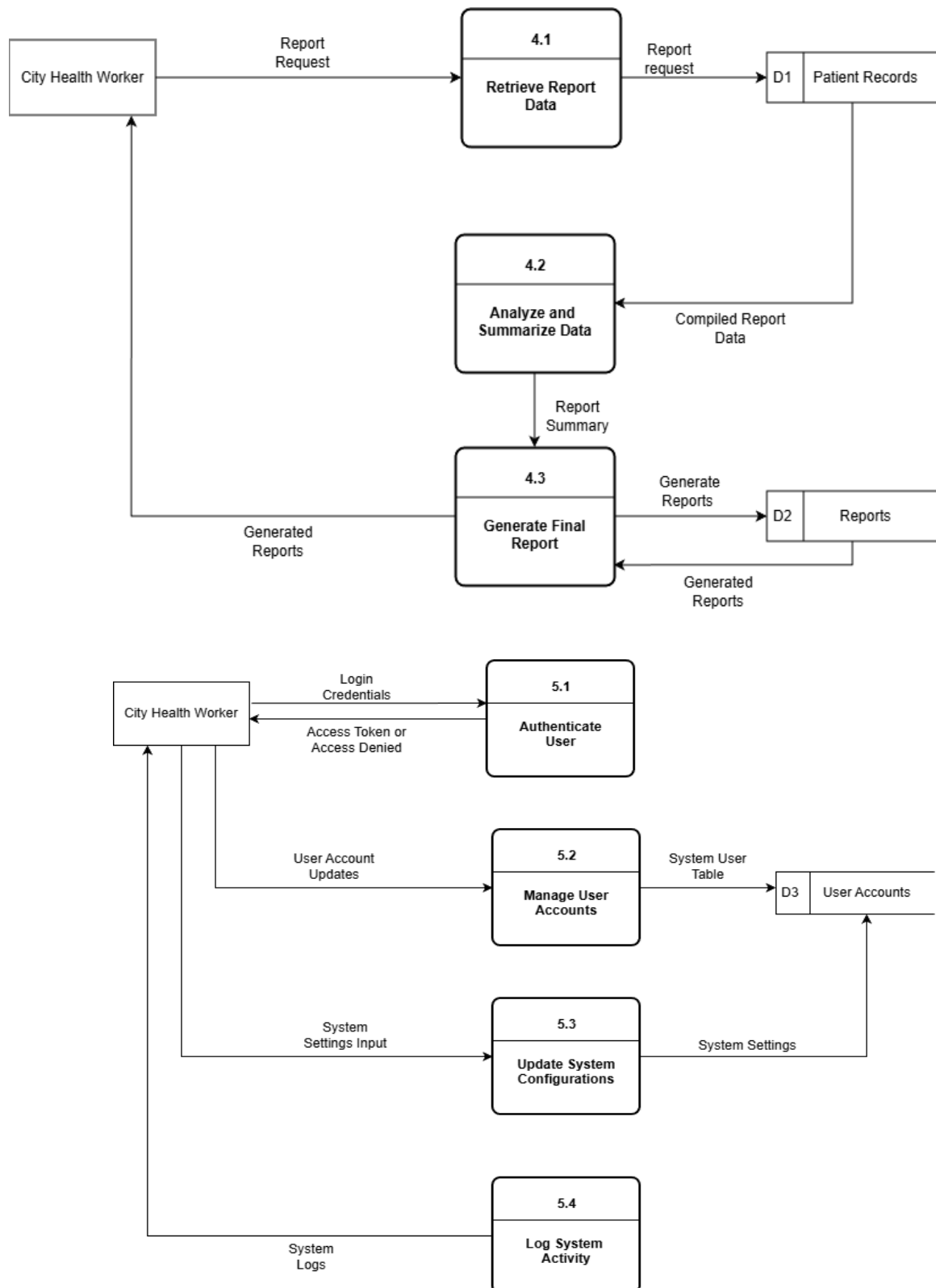


incidents. These details were stored as victim records and passed to the classification process, where each case was categorized. The system then updates victim profiles to ensure that all records remain accurate and complete. City Health Workers interact with the system by generating reports, managing user accounts, and adjusting system settings. The system provides them with updated user data, system logs, and incident reports to support monitoring and decision-making. Additional processes generate heatmaps for geolocation-based visualization and produce decision-support outputs based on classifications and location data.

Figure 5
Data Flow Diagram (Level 2)







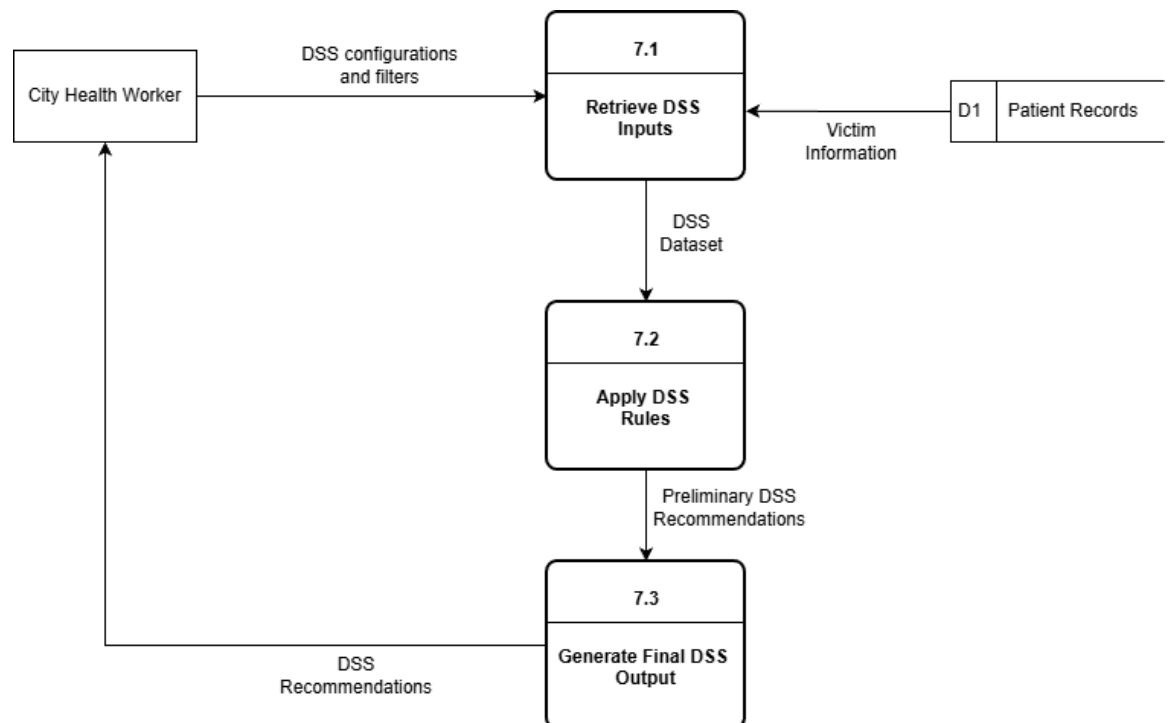
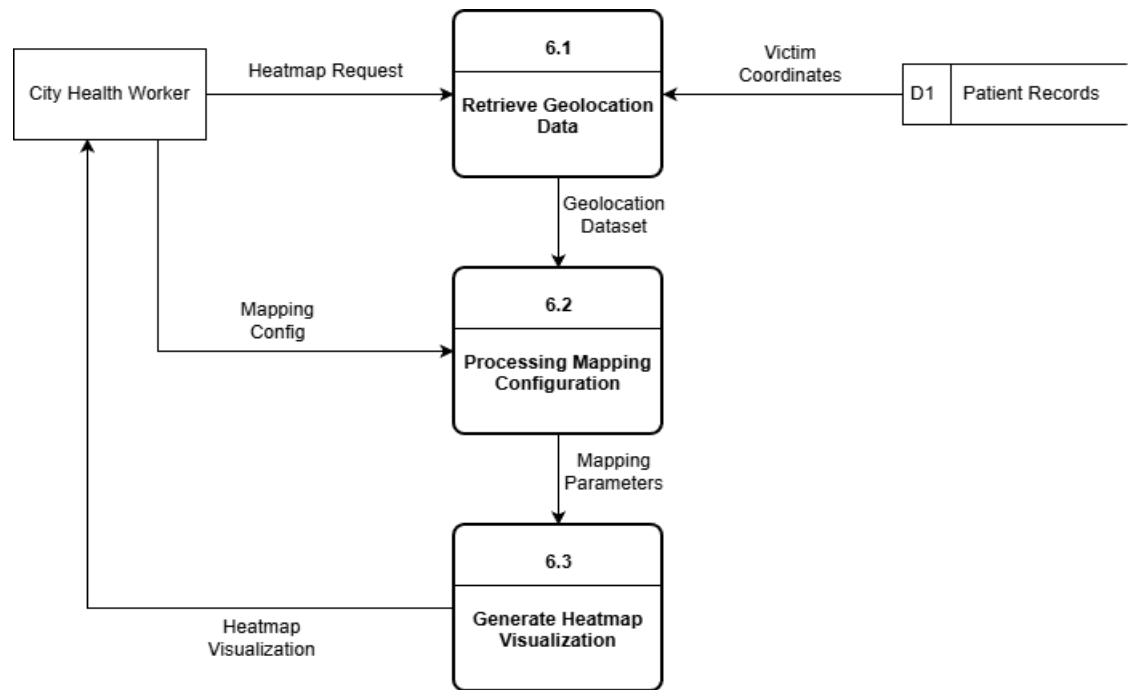




Figure 5 from the previous page presents the Level 2 Data Flow Diagram of the Animal Bite Incident Mapping and Classification with Decision Support System, showing how barangay and city health workers record incidents, classify cases, manage victim profiles, generate reports, create heatmaps, and produce decision support outputs. The process begins with entering and validating incident details before storing them in the patient records database. The system then applies and validates bite classification rules and saves the final results. It also allows profile updates, report generation, heatmap creation using geolocation data, and the production of recommendations based on recorded cases. The diagram shows a clear flow of data to support accurate recording, proper classification, and informed public health decisions.



Figure 6
Data Flow Diagram (Level 3)

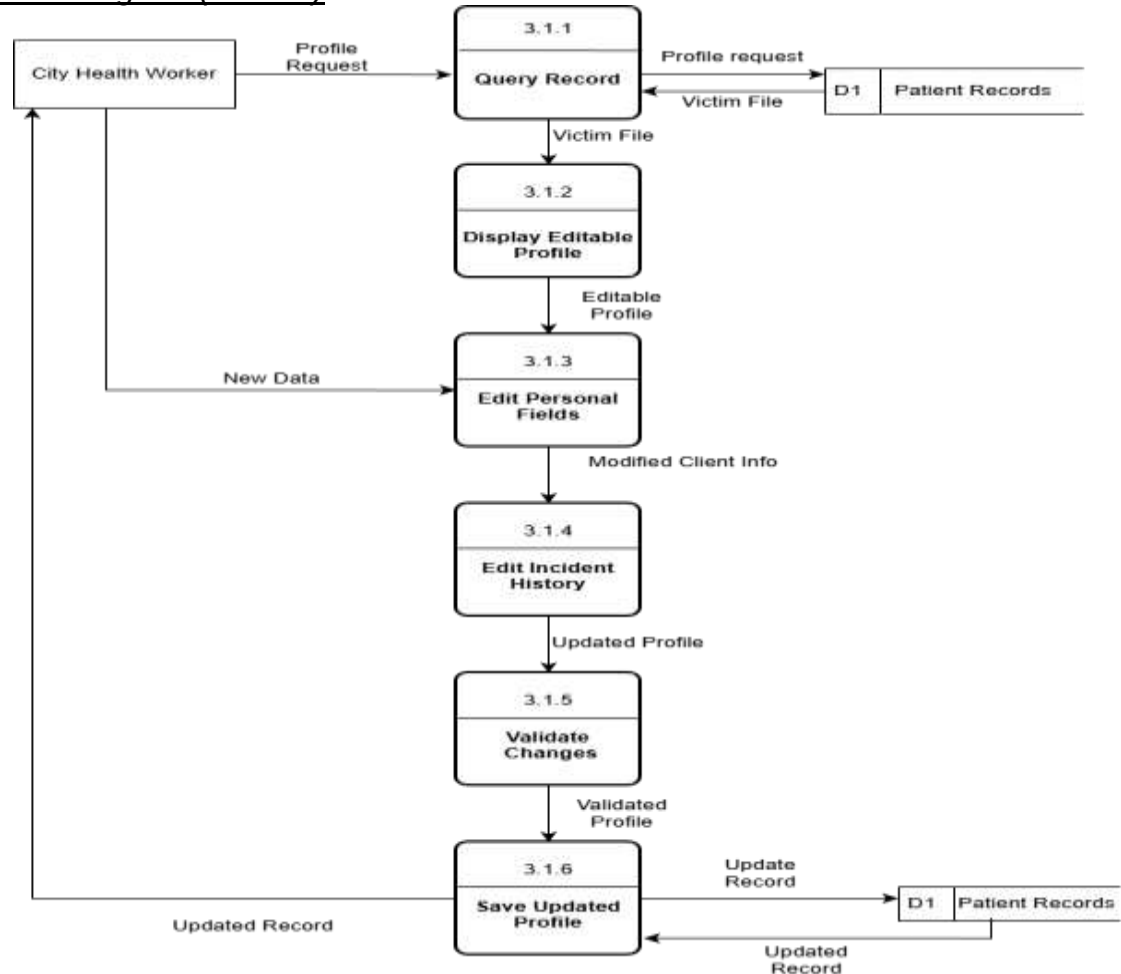


Figure 6 illustrates the Level 3 decomposition of the Victim Profile Management process, detailing how the system handles profile retrieval and updates. The workflow begins with validating a profile request to ensure the input was correct before searching the victim index for a matching record. If a match was found, the system retrieves the corresponding victim profile and generates a response file that becomes the basis for displaying an editable version of the



profile. The city health worker can then modify personal details and update the incident history as needed. After all changes were applied, the system validates the updated information to confirm accuracy and consistency. Once validated, the final step saves the updated profile into the Victim Records data store. This detailed breakdown ensures reliable searching, accurate editing, and secure updating of victim information within the system.

Figure 7
Data Flow Diagram (Level 4)

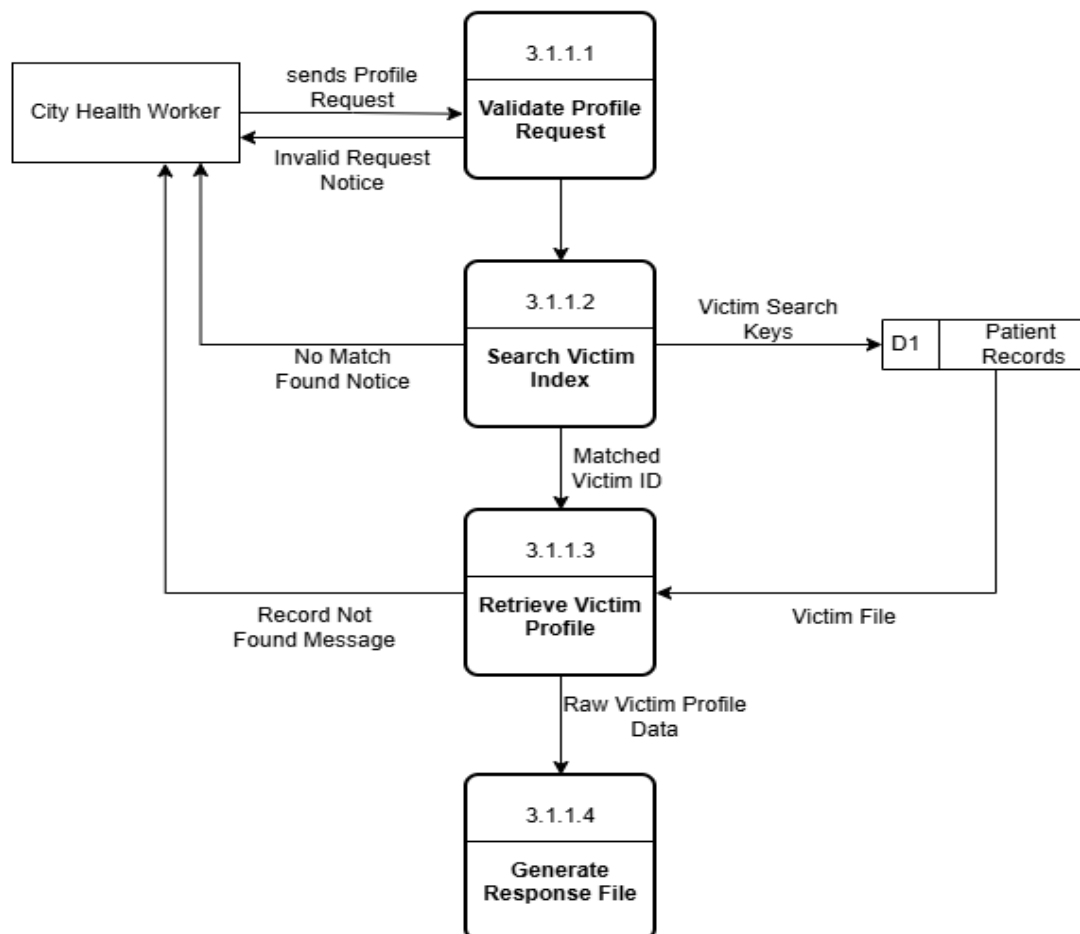




Figure 7 from the previous page shows the Level 4 expansion of the initial stages of the Victim Profile Management process, focusing on validating and locating a victim record. It begins with the system checking the profile request to ensure that the search inputs were complete and acceptable. Once validated, the system proceeds to search the victim index using the provided search keys, comparing them against the stored records in the Victim Records data store. If a matching entry was found, the process continues to the next steps of retrieving and preparing the profile; however, if no match was located or the request was invalid, the system immediately notifies the city health worker. This level emphasizes accurate request validation and efficient record searching as the foundation for reliable victim profile retrieval.



3.4 Entity Relationship Diagram

Figure 8

Entity Relationship Diagram of Animal Bite Incident Mapping and Classification with Decision Support System

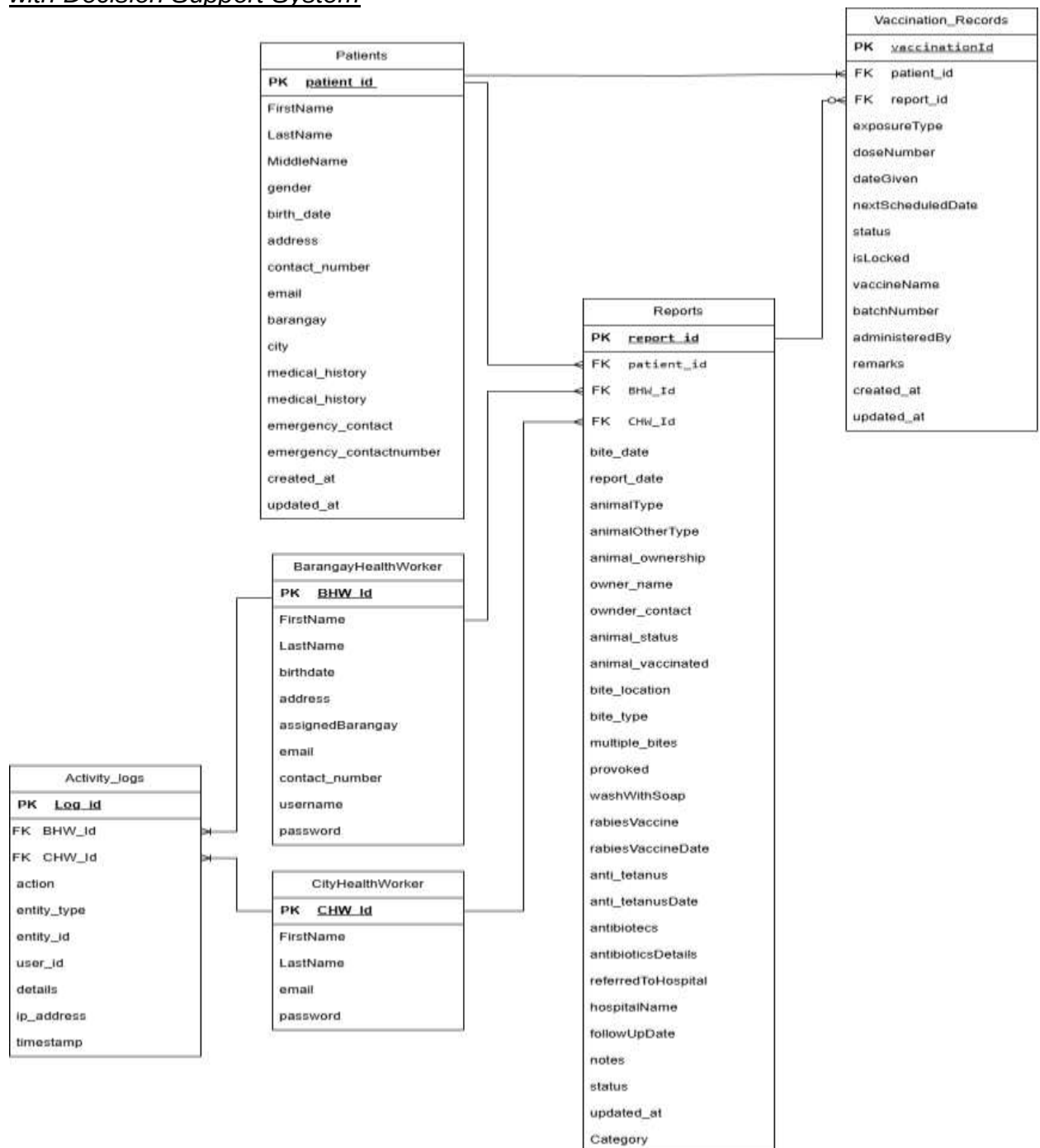




Figure 8 from the previous page shows the Entity Relationship Diagram (ERD) of the Animal Bite Incident Mapping and Classification with Decision Support System, included the main entities such as Patients, Barangay Health Worker, City Health Worker, Reports, Vaccination_Records, and Activity_Logs. The Reports table connects to patients to record the details of each bite victim and to Barangay Health Worker and City Health Worker to identify the health personnel responsible for encoding, updating, or managing the report. The Vaccination_Records table was linked to both Patients and Reports to document vaccination information, dates, and follow-up schedules for bite cases. The Activity_Logs table stores user actions performed within the system, allowing tracking of activities for security and accountability purposes. The ERD illustrates how each table was related and how data moves between them to support accurate case documentation, monitoring, and decision-making for animal bite incidents.



3.5 UML Use Case Diagram

Figure 9

UML Use Case Diagram of Animal Bite Incident Mapping and Classification with Decision Support System

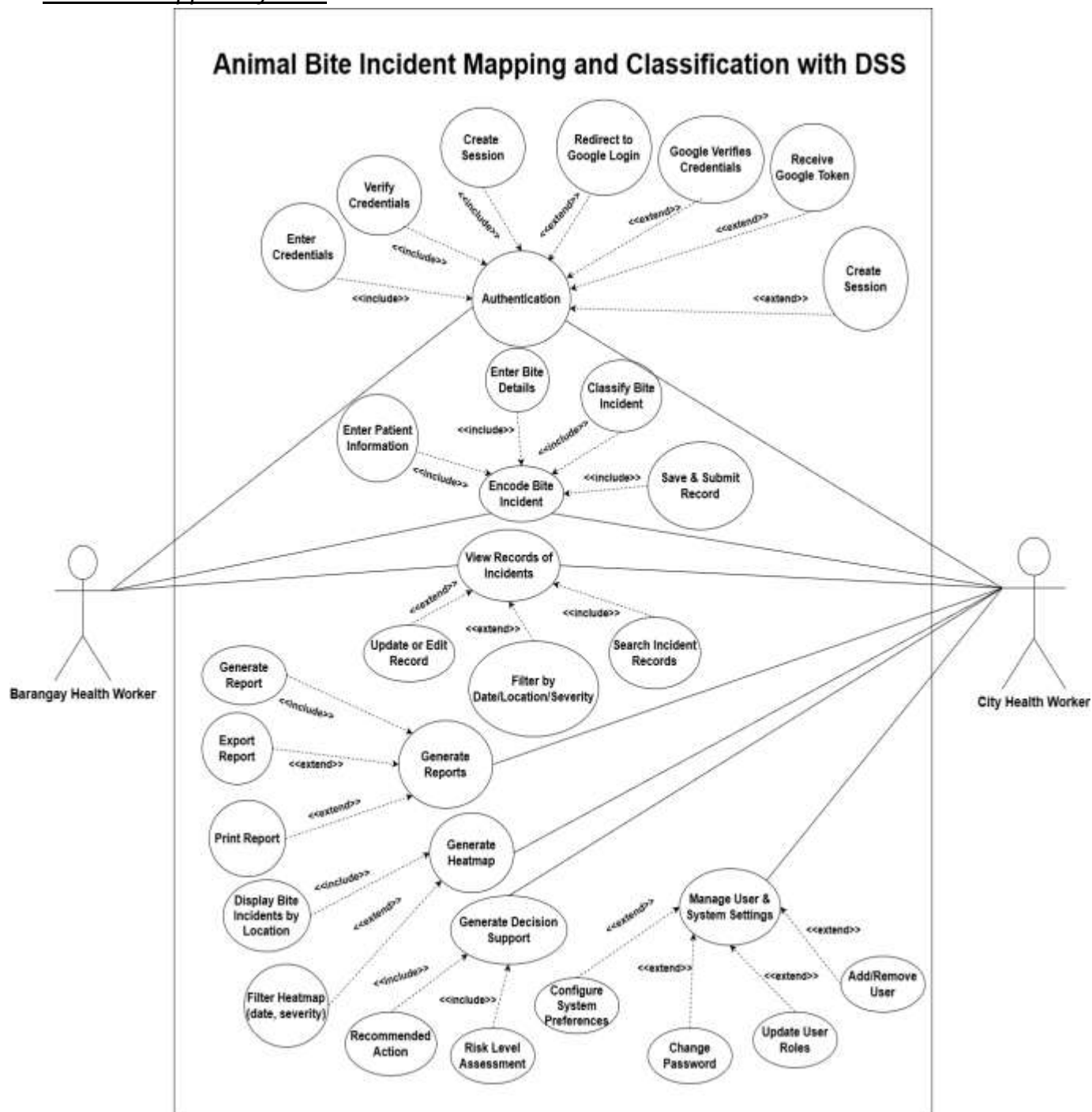




Figure 9 from the previous page shows the UML Use Case Diagram of the Animal Bite Incident Mapping and Classification with Decision Support System. It illustrates how the Barangay Health Worker and the City Health Worker interact with all major features of the system. The system supports two ways of logging in, which were the traditional username and password method and the Google Login option, allowing users to access the system in a secure and convenient way. Both actors can encode bite incidents and view existing records, while the City Health Worker has additional abilities such as generating reports, creating heatmaps, and using the decision support feature. The City Health Worker can also manage user accounts and system settings. Overall, the diagram presents the different roles, permissions, and system processes that help ensure efficient monitoring, analysis, and management of animal bite incidents in the community.



3.6 Use Case Scenario

Table 3

Authentication

Use Case 1: Authentication

Use Case Goal: To allow authorized users to securely access the Animal Bite Incident Mapping and Classification with DSS.

Scenario 1: Successful login and access to dashboard.

Actor: Barangay Health Worker / City Health Worker

Pre-condition: The system is running and accessible. The user has an active registered account.

Main Flow:

Step 1: The user enters a registered username/email and password.

Step 2: The system validates the entered credentials against the user records in the database.

Step 3: The user clicks the Login button.

Step 4: The system authenticates the user and determines the assigned role.

Step 5: The system creates a user session and redirects the user to the appropriate dashboard based on role.

User Action: Enters username/email and password.

System Response: Validates the provided credentials.

User Action: Clicks "Login."

System Response: Confirms that the credentials are correct, creates a user session, and redirects the user to their assigned dashboard.

Table 4

Encode Bite Incidents

Use Case 2: Encode Bite Incident

Use Case Goal: To allow authorized users to record a new animal bite incident into the system.

Scenario 1: Successful encoding and saving of bite incident details.

Actor: Barangay Health Worker / City Health Worker

Pre-condition: The system is running and accessible, the user is authenticated and authorized to encode incidents, and all required reference data such as barangay lists, animal types, and severity categories is available.

Main Flow:

Step 1: The user navigates to the Encode Bite Incident module.

Step 2: The system displays the incident entry form containing fields for victim information, incident details, animal details, and severity level.

Step 3: The user enters victim details, incident date and time, location, animal type, ownership status, and severity level.

Step 4: The system validates all required fields and data formats.

Step 5: The user uploads supporting files if necessary and clicks Submit.

Step 6: The system saves the incident record, assigns a unique incident ID, and confirms successful encoding.

User Action: Navigates to the Encode Bite Incident module and enters incident details.

System Response: Displays the incident entry form with fields for victim information, incident details, animal details, and severity level.

User Action: Clicks Submit.

System Response: Saves the record, generates an incident ID, and confirms successful encoding.



Table 5

View Records of Incidents

Use Case 3: View Records of Incidents

Use Case Goal: To allow users to view, filter, and access detailed information about recorded animal bite incidents.

Scenario 1: Successful viewing and filtering of incident records.

Actor: Barangay Health Worker / City Health Worker

Pre-condition: The user is logged in, authorized to view records, and incident data exists in the system.

Main Flow:

Step 1: The user navigates to the View Records of Incidents module.

Step 2: The system displays a list of all recorded animal bite incidents.

Step 3: The user applies filters such as date range, barangay, animal type, or severity.

Step 4: The system updates the incident list based on the selected filters.

Step 5: The user selects a specific incident record.

Step 6: The system displays complete incident details, including victim information, incident location, animal data, severity, and map placement.

User Action: Applies filters such as date range, barangay, animal type, or severity.

System Response: Updates the list and displays detailed incident information.

Table 6

Generate Reports

Use Case 4: Generate Reports

Use Case Goal: To allow the City Health Worker to generate summarized reports of animal bite incidents for monitoring and analysis.

Scenario 1: Successful generation and display of bite incident reports.

Actor: City Health Worker

Pre-condition: The user is logged in with reporting permissions, bite incident records exist in the database, and filters or parameters for generating reports are available.

Main Flow:

Step 1: The user navigates to the Reports module.

Step 2: The system displays report parameters such as date range, barangay, severity level, and animal type.

Step 3: The user selects report parameters and clicks Generate Report.

Step 4: The system retrieves the relevant incident data and generates the report.

Step 5: The user views the generated report.

Step 6: The system provides options to download or print the report in PDF or Excel format.

User Action: Selects report parameters and clicks "Generate Report."

System Response: Generates the report and provides export options.



Table 7

Generate Heatmap

Use Case 5: Generate Heatmap

Use Case Goal: To allow the City Health Worker to visualize the geographic distribution of animal bite incidents using a heatmap.

Scenario 1: Successful generation and display of the heatmap.

Actor: City Health Worker

Pre-condition: The user is logged in with permission to access mapping tools, the system has recorded incidents with valid location data, and map services are available.

Main Flow:

Step 1: The user navigates to the Heatmap module.

Step 2: The system displays filter options such as date range, barangay, animal type, and severity.

Step 3: The user selects filter criteria and clicks Generate Heatmap.

Step 4: The system retrieves geolocation-enabled incident records and processes them for mapping.

Step 5: The system displays the heatmap with color intensity representing case density.

Step 6: The user interacts with the map by zooming, panning, or viewing incident counts.

User Action: Selects filters and clicks Generate Heatmap.

System Response: Displays an interactive heatmap with density visualization.

Table 8

Generate Decision Support

Use Case 6: Generate Decision Support

Use Case Goal: To allow the City Health Worker to generate analytical insights and recommendations based on recorded animal bite incidents.

Scenario 1: Successful generation of decision support output.

Actor: City Health Worker

Pre-condition: The user is logged in with access to decision-support tools, sufficient incident data exists in the system, and classification and location details are available for analysis.

Main Flow:

Step 1: The user navigates to the Decision Support module.

Step 2: The system displays analysis parameters such as date range, barangay, animal type, and severity.

Step 3: The user selects analysis parameters and clicks Generate Decision Support.

Step 4: The system processes the data using decision-support logic.

Step 5: The system displays analytical results including trends, risk zones, severity patterns, and recommendations.

Step 6: The user views the results and may export or print them.

User Action: Selects analysis parameters and clicks Generate Decision Support.

System Response: Displays analytical insights and recommendations.



Table 9

Manage User and System Settings

Use Case 7: Manage User and System Settings

Use Case Goal: To allow the City Health Worker to manage user accounts and configure system settings to ensure proper system operation and data accuracy.

Scenario 1: Successful update of user accounts and system configurations.

Actor: City Health Worker

Pre-condition: The user is logged in with administrator privileges, the system is accessible, and existing reference data is available in the database.

Main Flow:

Step 1: The user navigates to the Settings module.

Step 2: The system displays options such as User Management, Reference Data, and System Preferences.

Step 3: The user selects User Management and chooses to add, edit, or deactivate a user account.

Step 4: The system validates the input and updates the account details.

Step 5: The user modifies system reference data such as barangay names, animal types, or severity categories.

Step 6: The system saves the changes, updates the database, and logs the action.

User Action: Updates user accounts or system reference data.

System Response: Validates input, saves changes, and records the action for audit purposes.



3.7 Data Dictionary

Data Dictionary furnishes detailed information about the business data, like standard definitions of data elements, their meaning, and allowable values.

Table 10
Patients Data Table

Field Name	Data Type	Data Format	Field Size	Description	Example
patient_id	int		6	Unique identifier for the patient	100001
first_name	varchar	Text	100	Patient's first name	Juan
last_name	varchar	Text	100	Patient's last name	Dela Cruz
middle_name	varchar	Text	100	Patient's middle name	Santos
gender	varchar	M/F	10	Patient's gender	Male
birth_date	date	DD/MM/YYYY	10	Patient's date of birth	05/06/1998
address	text	text		Patient's complete address	Brgy. Zone 2, Talisay
contact_number	int	Numeric	11	Patient's contact number	09171234567
email	varchar	Email	100	Patient's email address	juan@gmail.com
barangay	varchar	Text	100	References barangay.barangay_id	Zone 2
city	varchar	Text	100	Patient's city	Talisay
province	varchar	Text	50	Patient's province	Negros Occidental
medical_history	text	text		Patient's medical background	Allergic to antibiotics
emergency_contact	varchar	Numeric	11	Emergency contact person's number	09181234567
emergencyPhone	varchar	Numeric	11	Secondary emergency number	09074561234
allergies	TEXT	Text	255+	Patient known allergies	Penicillin
previousRabiesVaccine	boolean	Yes/No	1	Indicates if patient had rabies vaccination before	Yes
created_at	timestamp	YYYY-MM-DD HH:MM:SS	19	Record creation timestamp	2025-09-11 08:00:00
updated_at	timestamp	YYYY-MM-DD HH:MM:SS	19	Last record update timestamp	2025-09-11 09:15:00



Table 11
Barangay Health Worker Data Table

Field Name	Data Type	Data Format	Field Size	Description	Example
BHW_Id	int		3	Unique identifier	001
first_name	varchar	Text	100	First name	Maria
last_name	varchar	Text	100	Last name	Lopez
birth_date	date	DD/MM/YYYY	10	Date of birth	12/03/1985
address	text	Text	255	Staff address	Zone 3, Talisay
assignedBarangay	varchar	Text	255	Barangay assigned to the staff for case management	Concepcion
email	varchar	Email	100	Email address	maria@gmail.com
contact_number	int	Numeric	11	Contact number	09193456789
username	varchar	Alphanumeric	100	Username for login	mlopez
password	varchar	Encrypted	100	Encrypted password	*****

Table 12
City Health Worker Data Table

Field Name	Data Type	Data Format	Field Size	Description	Example
CHW_Id	int		2	Unique identifier	01
first_name	varchar	Text	100	First name	John
last_name	varchar	Text	100	Last name	Reyes
email	varchar	Email	100	Email address	john@gmail.com
password	varchar	Encrypted	100	Encrypted password	*****

Table 13
Reports

Field Name	Data Type	Data Format	Field Size	Description	Example
report_id	int		11	Unique identifier for the report	10001
patient_id	int		6	References Patients.patient_id	100001
staff_id	int		3	References BarangayHealthWorker	002
CHW_Id	int		2	References CityHealthWorker	01
bite_date	date	DD/MM/YYYY	10	Date of bite incident	01/08/2025
report_date	date	DD/MM/YYYY	10	Date report was created	02/08/2025



animalType	Enum	Predefined values	N/A	Type of biting animal (Dog, Cat, Rat, Monkey, Bat, Other)	Dog
animalOtherType	varchar	Text	50	Specific animal type if "Other" is selected	Pig
animal_ownership	varchar	Text	100	Ownership status of animal	Owned
owner_name	varchar	Text	100	Animal owner's name	Pedro Santos
owner_contact	varchar	Numeric	11	Animal owner's contact	09182345678
animal_status	varchar	Text	100	Status of animal	Alive
animal_vaccinated	boolean	Yes/No	1	Whether animal was vaccinated	Yes
bite_location	varchar	Text	100	Body part bitten	Left Arm
bite_type	varchar	Text	100	Type of bite	Scratch
multiple_bites	boolean	Yes/No	1	If multiple bites occurred	No
provoked	boolean	Yes/No	1	If bite was provoked	Yes
washWithSoap	boolean	Yes/No	1	Wound washed with soap?	Yes
rabies_vaccine	boolean	Yes/No	1	If rabies vaccine was given	Yes
vaccine_date	date	DD/MM/YYYY	10	Date vaccine was given	02/08/2025
anti_tetanus	boolean	Yes/No	1	If anti-tetanus was given	Yes
antiTetanusDate	DATE	YYYY-MM-DD	10	Date of anti-tetanus administration	2025-11-21
antibiotics	boolean	Yes/No	1	If antibiotics prescribed	No
antibiotics_Details	varchar	Text	200	Antibiotics details	Amoxicillin
referred_To_Hospital	boolean	Yes/No	1	If referred to hospital	Yes
hospital_Name	varchar	Text	100	Hospital name	Talisay Hospital
follow_Up_Date	date	DD/MM/YYYY	10	Follow-up date	10/08/2025
notes	text			Additional notes	Patient stable
status	varchar	Text	100	Report status (open/closed)	Open
updated_at	timestamp	YYYY-MM-DD HH:MM:SS	19	Last updated timestamp	2025-09-11 09:15:00
category	varchar	Text	20	Derived bite category (Category I, II, or III)	Category II



Table 14
Vaccination Records Data Table

Field Name	Data Type	Data Format	Field Size	Description	Example
vaccinationId	int	Numeric	11	Unique identifier for the vaccination record	3001
report_id	int	Numeric	11	References reports.report_id	10001
patientId	int	Numeric	11	References patient who received the vaccine	100001
exposureType	enum	Predefined values	N/A	Type of exposure: PrEP (pre-exposure) or PEP (post-exposure)	PEP
doseNumber	int	Numeric	11	Dose number in the series (e.g., 1, 2, 3, 4)	1
dateGiven	date	YYYY-MM-DD	10	Date when the vaccine dose was administered	2025-11-21
nextScheduled Date	date	YYYY-MM-DD	10	Next scheduled vaccination date	2025-11-24
status	enum	Predefined values	N/A	Status of dose (Completed, Upcoming, Missed, Overdue, NotScheduled)	Upcoming
isLocked	tinyint	Numeric (0 or 1)	1	Indicates if record is locked from editing (0 = No, 1 = Yes)	0
vaccineName	varchar	Text	100	Name or brand of the vaccine	Verorab
batchNumber	varchar	Text	100	Batch or lot number of the vaccine	BATCH-2025-001
administeredBy	varchar	Text	100	Name of health worker who administered the vaccine	Nurse Ana Santos
remarks	Text	Text	1000	Additional remarks about vaccination	Patient tolerated dose well
created_at	timestamp	YYYY-MM-DD HH:MM:SS	19	Date and time when record was created	2025-11-21 09:05:00
updated_at	timestamp	YYYY-MM-DD HH:MM:SS	19	Date and time when record was last updated	2025-11-21 09:10:30



Table 15
Activity Logs Data Table

Field Name	Data Type	Data Format	Field Size	Description	Example
log_Id	int	Numeric	11	Unique identifier for the activity log entry	7001
CHW_Id	int	Numeric	2	Unique identifier for the CityHealth Worker	01
BHW_Id	int	Numeric	3	Unique identifier for the Barangay Health Worker	001
action	varchar	Text	50	Type of action performed (e.g., create, update, delete, login)	update_report
entity_type	int	Numeric	11	Name of the entity/table affected (patients, reports, staff, etc.)	reports
entity_id	int	Numeric	11	ID of the record in the referenced entity	50001
details	LongText	Text / JSON	N/A	Detailed description of the action or changes made	Changed status from pending to completed
ip_address	varchar	Text (IPv4/IPv6)	45	IP address where the action originated	192.168.1.10
timestamp	DATETIME	YYYY-MM-DD HH:MM:SS	19	Date and time when the action was logged	2025-11-30 08:45:00

3.8 Hardware and Software Requirements

Hardware and Software Requirements provide detailed minimum requirements needed to support the developed system's functions and executions

Recommended Minimum Hardware Requirements

- Computer: A desktop or laptop computer to host the system
- Processor Speed: 3.00GHz
- Processor: Intel I3 Processor and above



- RAM: At least 8 GB of RAM for smooth system operation
- Storage: HDD/SSD - 256Gb Storage

Recommended Software Requirements

- Operating System: Windows 11
- Web Server: Google Chrome
- Text Editor/IDE - Visual Studio Code
- Database: MySQL for data management as primary as the database for data management
- Mapping Tool - Open Street Map API, Leaflet
- Programming Language - PHP, JavaScript

3.9 Project Timeline

The development of the Animal Bite Incident Mapping and Classification with Decision Support System (ABIMC-DSS) officially began on March 24, 2025, following the approval of the project and the start of formal requirements activities shown in the Gantt Chart. The project was successfully completed by November 28, 2025, after the final documentation was finished. Overall, the system development covered 249 total project days, spanning the full lifecycle from requirements gathering up to user acceptance and closeout. The project followed the V-Model Software Development Life Cycle (SDLC), which emphasized



structured progression from requirements to design and implementation, with corresponding validation and verification activities during testing.

The Planning and Requirements Phase ran from March 24 to April 19, 2025. During this period, the team conducted Requirements Analysis and Interviews (March 24 - April 12) to identify the operational gaps in animal bite reporting, surveillance, and record handling. This was reinforced by Process Observation and Documentation (March 27- April 19), which ensured that the existing workflows and reporting practices of the health office were fully understood and properly documented. The completion of this phase established a clear baseline for all succeeding activities.

From April 13 to June 3, 2025, the project entered the System Design Phase, where the overall structure of ABIMC-DSS was planned and validated. The team worked on the UI, data flow, and database structure design (April 13 - May 4), conducted technology selection and system architecture review (April 15 - May 7), and finalized core technical blueprints through the architectural design phase (May 1 - May 20). This phase also included data flow finalization for the BHW–CHO–Dashboard workflow (May 8 - May 29) and the decision support workflow design (May 10–June 3), ensuring that the reporting process, visualization needs, and decision-support logic were aligned before development began.

The Module Design Phase was carried out from June 4 to July 5, 2025, focusing on detailed specifications of the system's major components. This included classification module design (June 4 - June 24), mapping module design



(June 8–June 28), and form and reporting module design (June 10 - July 1). Supporting documentation and technical artifacts such as ERD, DFDs, and system diagrams were consolidated from June 15 to July 5, serving as the blueprint for structured coding and integration.

The Coding Phase (the most intensive stage) ran from July 8 to September 15, 2025, with overlapping activities to maximize development efficiency. Implementation included front-end coding (July 8 - August 5), back-end coding (July 10 - August 28), and database setup and integration (July 12 - August 20). Core intelligent features were developed through the classification engine (July 15 - August 30), mapping feature development (July 20 - September 10), and the reports and dashboard module (July 25 - September 15). This phase produced the working ABIMC-DSS platform using the selected web technologies and prepared it for formal validation.

Testing and refinement followed the V-Model validation flow. The Unit Testing Phase ran from September 16 to October 12, 2025, consisting of module-based unit testing (September 16 - October 5) alongside bug fixes per module (September 18 - October 12). The Integration Testing Phase then proceeded from October 6 to October 29, 2025, including module integration (October 6 - October 22) and system flow verification (October 10 - October 29) to confirm that all modules worked seamlessly as one system. The System Testing Phase was completed through full system testing (November 10 - November 13) and



performance and stress testing (November 14 - November 18), ensuring stability and readiness for end-user validation.

The final stage focused on acceptance and closeout. User Acceptance Testing (UAT) was conducted from November 19 to November 20, 2025, followed by feedback integration (November 21 - November 23) to apply necessary revisions based on user input. The project concluded with final documentation from November 24 to November 28, 2025, marking the completion of deliverables and the readiness of the system for turnover. Dependency management was carefully observed throughout: design activities depended on verified requirements, implementation relied on approved module designs, and acceptance testing required a stable integrated build. The combination of overlapping development tasks and structured testing checkpoints ensured on-time completion and quality assurance across all phases. (See Appendix J “Gantt Chart”)

3.10 System Evaluation Tools

White-Box Testing

We conducted White-Box Testing using Pest PHP to evaluate the internal logic and backend functionality of the Animal Bite Incident Mapping and Classification System. This testing approach focused on examining authentication processes, database operations, input validation, data handling, and session management by executing unit and feature tests that directly inspected the system’s code behavior. Pest PHP enabled us to validate admin and staff



authentication workflows, detect incorrect logic in error handling, and ensure secure processing of login attempts. We also conducted thorough database connection and query execution tests, including edge cases and error scenarios, to confirm that the system handled valid queries correctly and gracefully responded to malformed or unexpected inputs. Additional tests assessed input validation rules, password security, date and time processing, and transaction handling to ensure consistency and robustness across system modules. The report submission and staff management components were also validated to ensure correct logic, proper data storage, and reliable session behavior. Overall, White-Box Testing allowed us to identify hidden logical errors, verify the accuracy of backend operations, and ensure that core system functions performed reliably according to the intended program structure.

Black-Box Testing

We conducted Black-Box Testing using Cypress to evaluate the Animal Bite Incident Mapping and Classification System from a user's perspective, focusing on both admin and staff functionalities. The tests simulated real interactions such as logging in, navigating dashboards, viewing reports, managing patients and staff, applying filters, performing searches, and using the geomapping features. Through this process, the system successfully displayed most UI components, navigation menus, tables, statistics cards, and filtering tools across admin and staff modules. However, several issues were identified, including missing action buttons (View,



Edit, Delete), search buttons that failed to trigger filtering, missing or incomplete advanced filter elements, and login forms that did not redirect correctly. These failures revealed inconsistencies between expected and actual behavior in key modules such as Patients, Staff, and Reports. After documenting these defects, the development team used the results to trace and correct the underlying problems. Retesting confirmed that corrected components performed properly, ensuring the system handled navigation, data display, and feature interactions more reliably. Overall, the test cycle verified that a majority of functions passed successfully while also identifying critical areas for improvement to enhance system stability and user experience.

System Usability Instrument

This section discusses the instrument used to evaluate the usability of the developed system. The System Usability Instrument was utilized to measure users' perceptions regarding the system's ease of use, learnability, efficiency, and overall satisfaction. The findings from this evaluation were used to determine whether the system met user expectations and effectively supported the completion of tasks in an efficient manner. (See Appendix A for reference)



Table 16

Likert Scale, Range, and Verbal Interpretation for ISO 25010:2011 System Software Usability Evaluations

Responses	Scale	Range	Verbal Interpretation
Strongly Agree	5	4.20 - 5.00	Very High
Agree	4	3.40 - 4.19	High
Undecided	3	2.60 - 3.39	Average
Disagree	2	1.80 - 2.59	Low
Strongly Disagree	1	1.00 - 1.79	Very Low

The ISO/IEC 25010:2011 Software Quality Model was used as the evaluation tool for assessing the quality of the developed Animal Bite Incident Mapping and Classification with Decision Support System because it provides a standardized and comprehensive way of measuring software performance. This model evaluates systems based on eight characteristics such as Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. These characteristics help determine how well a system meets user needs and performs under real operational conditions.

Each characteristic focuses on a specific aspect of software quality. Functional Suitability checks if the system performs its intended tasks correctly. Performance Efficiency examines the speed of the system and how it uses resources. Compatibility measures how well the system operates alongside other platforms or software. Usability focuses on how easy the system was to learn and use. Reliability ensures that the system works consistently without errors. Security protects information and limits access to authorized users. Maintainability refers to



how easily the system can be improved or updated. Portability evaluates the system's ability to function on different devices or environments.

In this study, respondents rated the system using a five points Likert scale ranging from Very Low to Very High based on their experience with functions such as digital profiling, incident encoding, heatmap visualization, and automated reporting. The ratings were analyzed using the ISO/IEC 25010:2011 criteria. The results showed that the system achieved very high performance across most areas, with compatibility identified as the area that could be further improved to enhance future interoperability with other platforms and systems.

3.11 Respondents of the Study

Table 17
Respondents of the Study

Respondent	Frequency	Percent
Barangay Health Workers	16	64%
City Health Workers	4	16%
IT Experts	5	20%
Total	25	100%

The research involved a carefully selected group of 25 respondents who interacted with or evaluated the Animal Bite Incident Mapping and Classification



with Decision Support System. These respondents include Barangay Health Workers who encode patient and incident data, City Health Workers who also handle encoding and report management at the Animal Bite Treatment Center, and IT experts who assess the system's technical performance. They were chosen based on their direct involvement in animal bite reporting, public health monitoring, or software evaluation, ensuring meaningful and informed feedback on the system's usability and effectiveness.

3.12 Ethical Consideration

The researchers prepared a formal request letter to conduct interviews and system evaluations with the intended beneficiaries of the study. Before the research activities began, the proponents sought and obtained the participation of the Talisay City Animal Bite Treatment Center and selected barangays within the city. The purpose, scope, and objectives of the study were clearly explained to the health offices involved, and they willingly agreed to participate.

The researchers strictly upheld the confidentiality of all information provided by respondents. Any personal or sensitive data, including patient details, health worker information, and system evaluation responses, was securely handled, protected from unauthorized access, and used solely for academic purposes within this study. Appropriate data protection measures were implemented to maintain respondent trust and uphold ethical research standards.



In compliance with the Data Privacy Act of 2012, informed consent was obtained from all participants prior to data collection. The researches would explain how participant data would be used, stored, processed, and protected. Regular checks would be conducted to ensure that all procedures follow ethical guidelines and legal requirements, thereby maintaining data integrity, confidentiality, and responsible research conduct.



CHAPTER IV

RESULTS AND DISCUSSION

This chapter presented the results and discussions based on the development of the Animal Bite Incident Mapping and Classification with Decision Support System, aligned with the research objectives. It includes screenshots of the system's interfaces and modules, as well as the evaluation results using the ISO/IEC 25010:2011 Software Quality Model, which outlines key software quality characteristics and their sub-characteristics. Data were gathered from the identified respondents using a 5-point Likert scale, with responses ranging from 1 (Strongly Disagree) to 5 (Strongly Agree). The evaluation covers the eight quality characteristics defined by the standard: Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability.

This chapter also discussed the findings of the system evaluation conducted for the Animal Bite Incident Mapping and Classification with Decision Support System. The assessment utilized a standardized questionnaire based on the ISO/IEC 25010:2011 model and was administered to Barangay Health Workers, City Health Workers, and IT experts who interacted with the developed system. The results provide an overview of the system's effectiveness, usability, and overall quality. Furthermore, this chapter explained the implications of the findings, highlighting areas of strength and identifying opportunities for future improvements and system enhancements.



4.1 Results of the Study

The first objective of this study aimed to design and develop an Animal Bite Incident Mapping and Classification with Decision Support System with the following technical features: digitized documentation of bite victim details, classification of incidents based on animal type and bite severity, heat mapping to identify high risk areas, automated generation of incident reports, and decision support functions to assist health workers in planning and response.

The result of this objective was the successful development of the Animal Bite Incident Mapping and Classification with Decision Support System, which fully integrated all the intended technical features. The system was capable of encoding and storing victim and incident information, classifying bite cases accurately, generating heatmaps to visualize risk locations, producing automated reports, and providing decision support insights for health interventions. This demonstrates that the system has achieved the functional requirements set by the researchers and was ready for use in real world health monitoring and decision making.

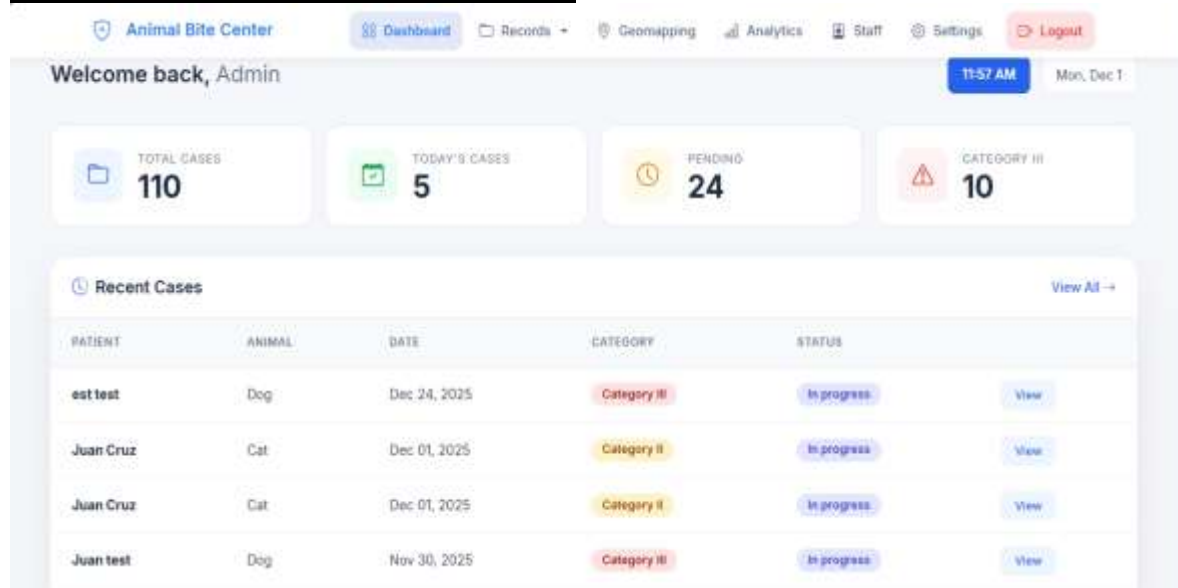


Figure 10
City Health Worker Login Interface

This figure shows the Admin Login interface of the Animal Bite Incident Mapping and Classification with Decision Support System. This page serves as the entry point for City Health Workers who have authorized access to the system. The interface provides fields for entering an email address and password, ensuring that only registered users can access administrative features. It also offers an alternative sign in option using a Google account, providing flexibility and convenience during login.



Figure 11
Animal Bite Center Dashboard Interface



This figure 11 shows the main dashboard of the Animal Bite Incident Mapping and Classification with Decision Support System, which provides City Health Workers with a real time summary of reported bite cases. The dashboard displays essential information such as the total number of cases, pending cases, Category III incidents, and cases recorded for the day, enabling quick monitoring of bite trends within the city. It also presents a list of recent bite incidents with details including patient name, animal type, incident date, category, and status for easy reference. Through its simple and well-arranged layout, the dashboard becomes the main area where users can easily reach all system functions. From here, they can manage records, view maps of animal bite cases, check analytics, and perform other necessary tasks without opening many different pages.



Figure 12
Patient Records Interface

Animal Bite Center | Dashboard | Records | Geomapping | Analytics | Staff | Settings | Logout

Patients | 103 total records

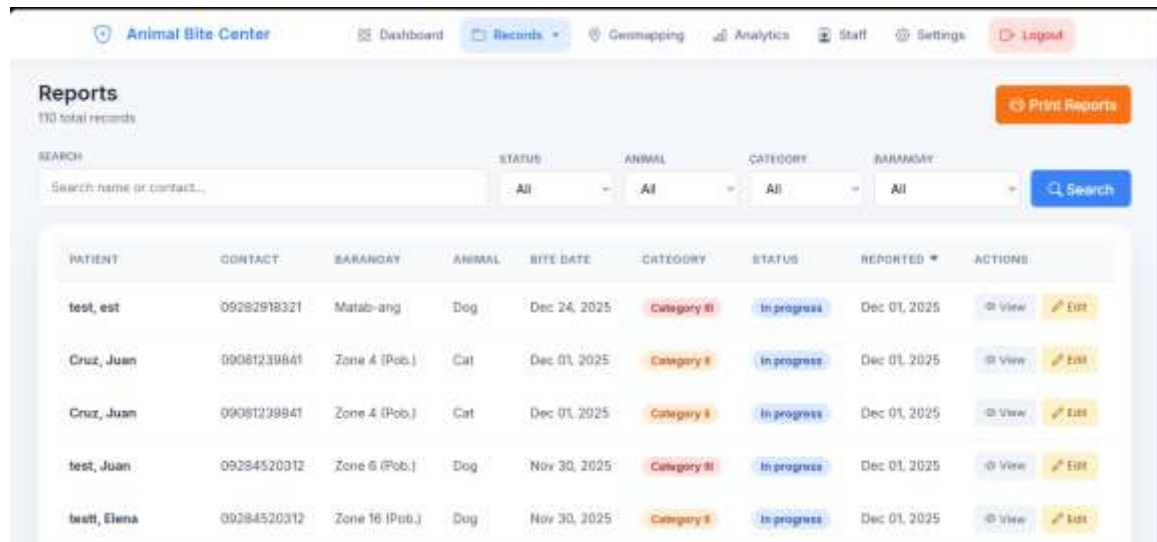
Search name, contact, or ID... | All Genders | All Barangays | Search

NAME	GENDER	AGE	CONTACT	BARANGAY	REPORTS	LAST VISIT	ACTIONS
Panolino, Anthony	Male	8	09284530312	Zone 8 (Pob.)	1	Nov 19, 2025	View
cahilig, cristel ann	Female	19	09365067993	Concepcion	1	Nov 11, 2025	View
rojo, meg	Other	19	12345678910	Katilingan	1	Nov 11, 2025	View
Angeles, Anthony	Other	22	09171234567	Zone 5 (Pob.)	1	May 21, 2025	View
Ramos, Andres	Male	31	09566631782	Zone 16 (Pob.)	1	May 20, 2025	View
Santino, Miguel	Male	65	09211745793	Zone 16 (Pob.)	1	May 20, 2025	View

This figure displays the Patient Records interface of the Animal Bite Incident Mapping and Classification with Decision Support System. This module allows City Health Workers to view and manage the list of individuals who have reported animal bite incidents. The interface provides essential patient information including name, gender, age, contact number, barangay, number of reports, and the date of the last visit. A search bar and filter options for gender and barangay were included to help users quickly locate specific records or narrow down data based on selected criteria.



Figure 13
Animal Bite Reports Interface

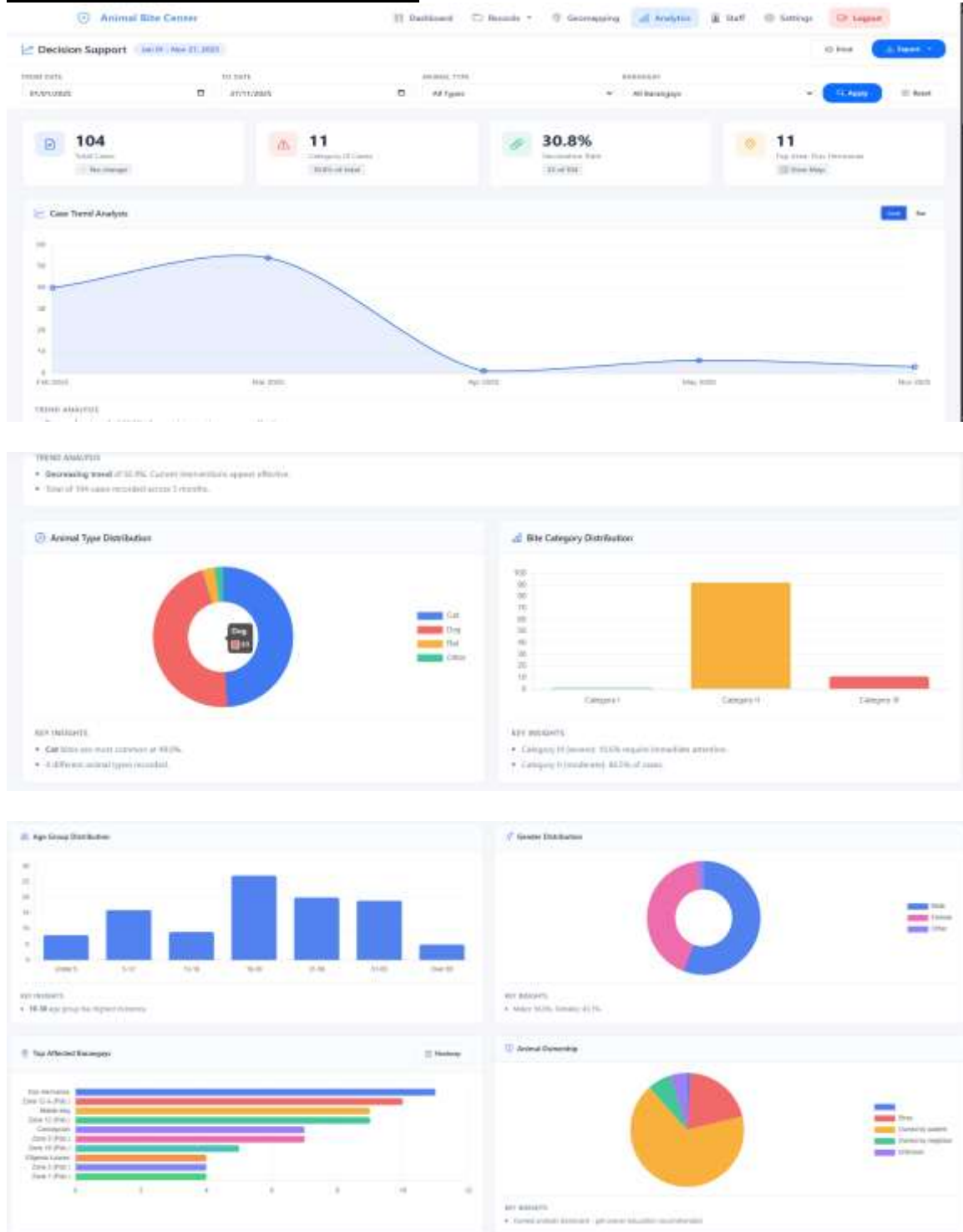


PATIENT	CONTACT	BARANGAY	ANIMAL	BITE DATE	CATEGORY	STATUS	REPORTED	ACTIONS
test, est	09282918321	Matab-ang	Dog	Dec 24, 2025	Category III	In progress	Dec 01, 2025	View Edit
Cruz, Juan	09081239841	Zone 4 (Pob.)	Cat	Dec 01, 2025	Category I	In progress	Dec 01, 2025	View Edit
Cruz, Juan	09081239841	Zone 4 (Pob.)	Cat	Dec 01, 2025	Category I	In progress	Dec 01, 2025	View Edit
test, Juan	09284520312	Zone 6 (Pob.)	Dog	Nov 30, 2025	Category III	In progress	Dec 01, 2025	View Edit
test, Elena	09284520312	Zone 16 (Pob.)	Dog	Nov 30, 2025	Category I	In progress	Dec 01, 2025	View Edit

This figure shows the Reports interface of the Animal Bite Incident Mapping and Classification with Decision Support System, where City Health Workers can view and monitor all recorded bite incidents. The interface presents important information such as patient name, contact number, barangay, animal involved, bite date, bite category, case status, and the date it was reported. It includes a search bar and multiple filter options for status, animal type, category, and barangay, allowing users to quickly locate specific records without browsing through all entries. A “Print Reports” button was also provided for generating hard copies of report summaries when needed.



Figure 14
Decision Support System Interface





This figure presents the Decision Support page of the Animal Bite Incident Mapping and Classification with Decision Support System, which assists city health workers in analyzing recorded bite incidents and generating data-driven insights. The interface includes interactive filters for date range, animal type, and barangay, allowing users to refine data according to specific needs. It provides key summary indicators such as total reported cases, Category III incidents, vaccination rate, and most affected barangay, enabling quick situational assessment. The system displays various visual analytics including trend graphs and distribution charts for animal type, bite category, age group, gender, vaccination status, ownership type, and treatment measures. It also generates automated insights and recommendations based on observed data patterns to support strategic health interventions. Additionally, the interface features Print and Export options, enabling users to save analytical reports in external formats or produce hard copies for administrative reporting and documentation.



Figure 15
Staff Management Interface

The screenshot displays the Staff Management interface. At the top, there's a header with the title 'Staff Management' and a subtitle 'View and manage barangay health workers'. Below this, there are three summary cards: 'Total Staff' (2), 'Active Staff' (2), and 'Inactive Staff' (1). A search bar is present with a 'Search' button and a 'Clear' button. Below the search bar is a table listing staff members. The table has columns for 'Staff Information', 'Contact Number', 'Email', 'Birth Date', and 'Actions'. Two staff members are listed: Anthony Angeles (ID: 1) and Julie Ann Paraiso (ID: 2). Each row has 'Edit' and 'Delete' buttons. Below the table, it says 'Showing 1 to 2 of 2 staff members'. To the right of the table are two side panels. The 'Edit Staff' panel is for updating staff member information and includes sections for 'Personal Information' (First Name, Last Name, Birth Date, Address), 'Contact Information' (Contact Number, Email Address), and 'Account Information' (New Password, Confirm New Password). The 'Add New Staff' panel is for registering a new barangay health worker and includes similar sections for 'Personal Information', 'Contact Information', and 'Account Information'. Both panels have 'Cancel' and 'Save Changes'/'Add Staff' buttons at the bottom.

Staff Information	Contact Number	Email	Birth Date	Actions
Anthony Angeles (ID: 1)	1234567890	aaangeles.chmsu@gmail.com	Mar 31, 2020	Edit Delete
Julie Ann Paraiso (ID: 2)	1234567890	jparaiso.chmsu@gmail.com	Jan 19, 2019	Edit Delete

Figure 15 interface displays the Staff Management page of the Animal Bite Incident Mapping and Classification with Decision Support System. It enables the City Health Worker to view, search, add, edit, and remove Barangay Health Workers registered in the system. The page provides a quick summary of the total



number of staff and active members, followed by a searchable table that presents their basic profile information, including name, contact number, email address, and birth date. Functional action buttons such as Add New Staff, Edit, and Delete allow authorized users to manage staff records efficiently. When adding new personnel, the system presents an organized form that collects personal details, contact information, and login credentials. Editing a staff profile opens a similar structured interface, enabling updates to personal data or account credentials without affecting other system records.



Figure 16
New Animal Bite Report Interface

New Animal Bite Report

Patient Information

Patient Selection
☒ Existing Patient ☐ New Patient

Search Patient *
Type to search for a patient

Animal Bite Details

Date of Bite * 27/11/2025
Animal Type * -- Select Animal --
Animal Ownership * -- Select Ownership --
Animal Status * Alive
Animal Vaccinated * Unknown
Was the bite provoked? * Unknown
☐ Multiple Bites

Bite Location & Classification

Bite Location (Body Part) *
e.g., right hand, left leg, face

Bite Category *
-- Select Category --

Bite Categories:
Category I: Touching or feeding of animals, licks on intact skin
Category II: Hissing of uncovered skin, minor scratches or abrasions without bleeding
Category III: Single or multiple transdermal bites or scratches, contamination of mucous membrane or broken skin with saliva from animal licks, abrasions due to direct contact with bats

Treatment Information

☐ Additional Notes

The New Animal Bite Report page serves as the main encoding interface where health workers document all essential details of a bite incident in a structured manner. It begins with selecting or registering a patient and continues with recording the circumstances of the bite, including the date, animal type,



ownership, vaccination status, animal condition, and whether the bite was provoked. The interface further captures the bite location and category to support proper case classification based on severity. It also includes a Treatment Information section where interventions such as wound washing, rabies vaccination, anti-tetanus administration, antibiotic use, and hospital referral are indicated, along with a follow-up date and the current case status. An Additional Notes field was provided for any special remarks or observations regarding the incident. Once submitted, all collected data becomes part of the system's central database, ensuring accurate monitoring, reporting, and support for decision-making.

The second objective of this study aims to Test the Functionality of the aforementioned features.

The result of this objective was that the system passed both alpha and beta testing, confirming that all modules and functions operated as intended. During alpha testing, IT experts verified that core features such as patient profiling, bite incident encoding, automated classification, geomapping, analytics, and the decision support component performed correctly without system errors. Beta testing was conducted with actual end-users, including barangay and city health workers, who were able to navigate the system, input data, generate reports, and view mapped bite cases successfully. All test cases were marked Pass,



demonstrating that the developed system meets its functional requirements and was ready for real-world use.

Table 18
Black-Box Testing of the Alpha Testing of City Health Worker Authentication Case

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-AdminAuth001	Login form visibility test	All CHW login form elements should be visible on page load	All expected form elements were visible	Pass	No issues encountered
TC-AdminAuth002	Branding and text display check	All branding and descriptive text should display correctly	Branding text and page descriptions appeared correctly	Pass	Branding rendered correctly
TC-AdminAuth003	Login submission and redirection validation	User should be redirected to the CHW dashboard after successful login	The system successfully authenticated valid user credentials and redirected the user to the dashboard	Pass	Validation works as intended; incorrect inputs properly rejected

The results in Table VIII demonstrate that the authentication module for the City Health Worker was functioning correctly. All test cases passed, indicating that the login interface elements were visible, the branding and page information were displayed properly, and the system successfully authenticated valid credentials. The redirection to the City Health Worker dashboard after login verifies that the system grants access only to authorized users. These results confirm that the authentication mechanism meets usability and security expectations, providing a stable entry point to the system for CHW users.



Table 19
Black-Box Testing of the Alpha Testing of City Health Worker Dashboard Case

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-AdminD ash004	CHW dashboard layout and navigation visibility test	Navigation bar should display the system title and all main links (Dashboard, Geomapping, Reports, Patients) with "Dashboard" highlighted as active	All navigation bar items were visible and correctly displayed; "Dashboard" was highlighted	Pass	Navigation displayed correctly
TC-AdminD ash005	Welcome section visibility check	Welcome header, description text, and "View All Reports" button should be visible	Welcome section displayed correctly with all expected elements	Pass	No issues encountered
TC-AdminD ash006	Statistics cards display test	All statistics cards should be visible and show corresponding case counts	All statistics cards appeared correctly and showed visible case numbers	Pass	Statistics displayed as expected
TC-AdminD ash007	Patient and staff statistics cards check	Cards showing total patients and total staff should be visible	"Total Patients" and "Total Staff" cards were displayed properly	Pass	Both cards rendered correctly
TC-AdminD ash008	Quick action cards visibility test	Quick action cards should be visible and properly labeled	All quick action cards appeared correctly	Pass	Labels and cards displayed fine
TC-AdminD ash009	Recent bite cases table visibility test	The recent bite cases table and "View All" button should be visible with correct column headers	The recent bite cases table and "View All" button displayed correctly with visible rows	Pass	Table loaded with correct columns

The dashboard testing results show that all interface components required by the City Health Worker role were displayed as intended. Navigation elements,



headers, quick action cards, and statistical summaries appeared correctly with real-time values. The presence of recent bite case records assures that the dashboard provides an accurate operational overview. Since all test cases passed, the dashboard successfully fulfills its function as the main control panel for City Health Workers, enabling them to efficiently monitor system activity and access essential features.

Table 20

Black-Box Testing of the Alpha Testing of City Health Worker Geo Mapping Case

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-AdminG eo010	Geomapping page layout and navigation check	The “Geomapping” link should appear active; the page should display the title “Geomapping Analysis” and a brief description about visualizing cases	The Geomapping link was active, and the title and description appeared correctly	Pass	Page layout displayed properly
TC-AdminG eo011	Map container and control buttons visibility test	The map container and map control buttons should be visible	The map container, Heatmap, and Markers buttons were visible and functional	Pass	Map and controls visible
TC-AdminG eo012	Map filter and statistics controls visibility test	All filter fields, dropdowns, and action buttons should be visible and ready for input	The filter form, dropdowns, and buttons were all displayed properly	Pass	Filters rendered correctly
TC-AdminG eo013	Map interaction behavior validation	The map should remain visible and update when switching between Heatmap and Markers views	The map responded correctly to Heatmap and Markers controls	Pass	Map interaction worked smoothly



TC-AdminG eo014	Map statistics display test	Statistic cards should be visible and show key mapping information	All statistic cards were displayed correctly with proper labels	Pass	Statistic cards displayed correctly
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All geomapping test scenarios passed, confirming that the City Health Worker can access a functional geospatial interface. The map rendered properly, filters and interaction controls worked as expected, and statistical cards provided clear information. Heatmap and marker visualizations responded dynamically to user input, allowing the CHW to identify bite incident clusters and high-risk locations. These results verify that the geomapping module performs reliably and supports evidence-based public health decision-making.

Table 21
Black-Box Testing of the Alpha Testing of City Health Worker Patients Case

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-AdminPatients015	Patients page layout and navigation check	The "Patients" link should be active and the page title should be visible	"Patients" link was active and the page heading displayed properly	Pass	Page layout displayed correctly
TC-AdminPatients016	Patients table and filter visibility test	The table and all filter options should be visible for user interaction	Table and all filter options were visible as expected	Pass	Filters and table rendered correctly
TC-AdminPatients017	Search functionality validation	Page URL should update with the search query and display filtered results	URL updated with search=test and filtered results appeared correctly	Pass	Search executed successfully



TC-AdminPatients018	Patient data table content check	Table should display column headers and at least one patient record	Table displayed correctly with proper headers and visible rows	Pass	Patient records loaded properly
TC-AdminPatients019	Patient row "View" button visibility test	Each patient row should contain a working "View" button	"View" button was visible and functional in the first table row	Pass	Button displayed with no rendering delay
TC-AdminPatients020	Pagination visibility test	Pagination controls should be visible for navigating multiple pages	Pagination controls and page links were visible as expected	Pass	Pagination displayed correctly

Test results confirm that patient management features were fully operational for the City Health Worker. The system correctly displayed patient records, filters, action buttons, and pagination controls. Search functions and detailed record access were also successful, demonstrating smooth data retrieval and interaction. Since all test cases passed, the patient module effectively enables the CHW to view, manage, and verify patient information essential for monitoring treatment and follow-up of bite incidents.

Table 22
Black-Box Testing of the Alpha Testing of City Health Worker Reports Case

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-AdminReports021	Reports page layout and navigation check	The "Reports" link should be active, and the page title "Animal Bite Reports" with a short description should be visible	Navigation link was active; title and description displayed correctly	Pass	Page displayed as expected



TC-AdminReports022	Reports table and filter visibility test	All filter fields and options should be visible to the CHW user	All filters, search fields, and date inputs were visible as expected	Pass	Filters and fields rendered correctly
TC-AdminReports023	Search functionality validation	Page URL should include the search parameter and filter results accordingly	URL updated to include search=test, confirming successful search execution	Pass	Search function worked correctly
TC-AdminReports024	Reports table structure and data check	Table should display correct column headers and visible report records	Table headers and data rows were visible and formatted correctly	Pass	Records displayed properly
TC-AdminPatients025	Patient row "View" button visibility test	Each patient row should contain a working "View" button	"View" button was visible and functional in the first table row	Pass	Button displayed with no rendering delay
TC-AdminReports026	Report action buttons visibility test	Each report entry should have working View and Edit buttons	"View" and "Edit" buttons were visible and loaded correctly	Pass	Buttons functional and responsive
TC-AdminReports027	Pagination controls visibility test	Pagination controls should be visible for navigating multiple report pages	Pagination and page links were displayed properly	Pass	Pagination rendered with no issues

The reporting module performed successfully during alpha testing. Reports were generated correctly, filters responded as expected, and action buttons were functional. Visibility of headers, records, and pagination confirms that the system provides an organized reporting interface. Since the reports module was exclusively available to the City Health Worker, the successful test results validate



that authorized personnel can generate and review analytical outputs required for documentation, monitoring, and public health reporting.

Table 23

Black-Box Testing of the Alpha Testing of Staff Authentication Case

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-StaffAuth028	Staff login form visibility test	Email and password inputs, "Login" button, and "Staff Login" heading should be visible	All form elements and title displayed correctly	Pass	Form components loaded properly
TC-StaffAuth029	Staff login branding display check	The page should display "BHW Portal" logo, "Welcome Back!" heading, and login description	All branding text and the logo were visible as expected	Pass	Branding displayed correctly
TC-StaffAuth030	Staff login form submission validation	After clicking "Login," the system should redirect to staff_login.php for authentication	The login form redirected correctly to staff_login.php and URL updated as expected	Pass	Login processing worked properly

Authentication tests for the Staff (BHW) role passed successfully. Login form elements appeared correctly, branding was consistent, and the authentication process redirected users to the appropriate page. This confirms that the system's role-based access control was functioning properly, ensuring that Barangay Health Workers can enter the system but were limited to permitted features only. The integrity of user access was therefore maintained.



Table 24

Black-Box Testing of the Alpha Testing of Staff Dashboard Case

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-StaffDas h031	Welcome section display test	“Welcome” heading, overview message, and “New Bite Report” button should be visible	Welcome text and “New Bite Report” button displayed correctly	Pass	Welcome elements visible
TC-StaffDas h032	Encoding a new Animal Bite Report	The user fills out all required patient information fields and clicks Save.	The system accepted all input, saved the new bite report, and redirected to the confirmation/records page	Pass	All input fields accepted valid entries, record saved successfully, and no validation or system errors occurred

The results of the Staff Dashboard testing confirm that Barangay Health Workers can access only the functionalities assigned to their role. The dashboard elements, including the welcome message, quick action buttons, and status indicators, were displayed correctly and operated as intended. Administrative, geomapping, and reporting components were not accessible, demonstrating that the system correctly enforces role-based access restrictions. Since all test cases passed, the Staff Dashboard module effectively limits BHW users to patient and bite incident encoding tasks, fully aligning with the system’s updated access control design.



Table 25

White-Box Testing of the Alpha Testng of Admin Authentication Case

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result
TC-AdminA uth001	Function: adminLogin()	Test login with correct username and password	Username:"admin", password="correct123"	Should authenticate user and redirect to dashboard	Matches expected output	Pass
TC-AdminA uth002	Function: adminLogin()	Test login with incorrect password	username="admin", password="wrongpass"	Should return "Invalid credentials"	Matches expected output	Pass
TC-AdminA uth003	Function: adminLogin()	Test empty username handling	username="", password="test123"	Should output "Username required"	Matches expected output	Pass
TC-AdminA uth004	Function: adminLogin()	Test empty password handling	username="admin", password=""	Should output "Password required"	Matches expected output	Pass
TC-AdminA uth005	Function: adminLogin()	Test SQL injection prevention	username="" OR 1=1 --", password="none"	Inputs should be sanitized; login should fail	Matches expected output	Pass
TC-AdminA uth006	Function: adminLogin()	Test login attempt for deactivated admin	username="admin2", password="valid123"	System shows "Account disabled"	Matches expected output	Pass
TC-AdminA uth007	Function: adminLogin()	Test failed login rate-limiting (5 failed attempts)	5 incorrect login attempts	Account should be temporarily locked	Matches expected output	Pass



TC-AdminA uth008	Function: adminLogin()	Test case sensitivity of username	username="ADMIN", password="correct123"	Login should fail if case- sensitive	Matches expected output	Pass
TC-AdminA uth009	Function: adminLogin()	Test trimming of leading/trailing spaces	username="admin", password="correct123"	Should trim spaces and authenticate	Matches expected output	Pass
TC-AdminA uth010	Function: adminLogin()	Handling of special characters	username="admin!@#", password="test!@#"	Should accept special characters and validate normally	Matches expected output	Pass

The results of the Admin Authentication testing confirm that the system reliably validates administrator login attempts and properly enforces security controls. All core authentication behaviors including credential verification, empty field handling, input sanitization, and account status checking functioned exactly as intended. The module successfully prevented unauthorized access by rejecting incorrect passwords, blocking SQL injection inputs, and denying login for disabled accounts. Additional tests, such as whitespace trimming, rate-limiting after repeated failed attempts, and the handling of special characters, further demonstrated the robustness of the login logic. Since all test cases passed, the Admin Authentication component effectively upholds secure access control and ensures that only authorized administrators can enter the system dashboard.



Table 26
White-Box Testing of the Alpha Testing of Staff Authentication Case

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result
TC-StaffAuth001	Function: staffLogin()	Test valid staff login	username="staff1", password="correct123"	Authenticate user and redirect to staff dashboard	Matches expected output	Pass
TC-StaffAuth002	Function: staffLogin()	Handle incorrect password	username="staff1", password="wrongpass"	Display "Invalid credentials"	Matches expected output	Pass
TC-StaffAuth003	Function: staffLogin()	Test login for deactivated staff account	username="staffDeactivated", password="valid123"	System should display "Account disabled"	Matches expected output	Pass
TC-StaffAuth004	Function: staffLogin()	Test SQL injection handling	username="" OR 1=1 --", password="none"	Input sanitized; login must be rejected	Matches expected output	Pass

The results of the Staff Authentication testing confirm that staff login logic was functioning correctly and securely. The system authenticated valid staff credentials while properly rejecting incorrect passwords, SQL injection attempts, and login attempts involving deactivated accounts. Input validation and error-handling components responded accurately across all tested scenarios. All test cases passed, demonstrating that the Staff Authentication module successfully maintains secure access boundaries for staff users and aligns with the system's overall access control policies.



Table 27

White-Box Testing of the Alpha Testing of Authentication Error Handling Case

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result
TC-AuthError001	Function: adminLogin() / staffLogin()	Test handling of missing input fields	username="", password=""	System must return "All fields required"	Matches expected output	Pass
TCAuthError002	Function: adminLogin() / staffLogin()	Test invalid characters in username	username="adm!n<>", password="test123"	Reject and show "Invalid input format"	Matches expected output	Pass
TC-AuthError003	Function: adminLogin() / staffLogin()	Test large input size	username="a"*300, password="b"*300	System must reject overly long inputs and prevent overflow	Matches expected output	Pass

Testing for Authentication Error Handling shows that the system properly manages invalid, missing, or oversized login inputs across both admin and staff login functions. The module displayed the correct error prompts for empty fields, invalid characters, and overly long input values, ensuring consistent communication and protection against malformed data. All test cases passed, confirming that the authentication error-handling logic was robust, user-friendly, and resistant to potential misuse or input-based attacks.



Table 28

White-Box Testing of the Alpha Testing of Database Connection Test

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result
TC-Data Conn 001	Function: Database::connect()	Test successful database connection	Valid DB credentials	Connection should be established without errors	Matches expected output	Pass
TCD ataConn 02	Function: Database::connect()	Test network timeout handling	Incorrect DB password	System should throw connection error	Matches expected output	Pass
TCD ataConn 03	Function: Database::connect()	Test persistent connection reuse	host=""	Should fail and return "Unable to connect to host"	Matches expected output	Pass
TC-Data Conn 004	Function: Database::connect()	Test network timeout handling	Simulated network delay	Should trigger connection timeout error	Matches expected output	Pass

The Database Connection testing confirms that the system can reliably establish and manage communication with the database under various operational conditions. Valid credentials produced successful connections, while incorrect credentials and missing configuration values triggered appropriate error responses. The system also handled network timeouts and demonstrated proper reuse of persistent connections, ensuring efficiency and stability. Since all test cases passed, the Database Connection module was proven to be stable, fault-tolerant, and aligned with best practices for backend data access.



The third objective of this study was to evaluate the overall quality of the Animal Bite Incident Mapping and Classification with Decision Support System using the eight characteristics of the ISO/IEC 25010:2011 Software Quality Model. These characteristics include Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability, which collectively determine whether the developed system meets software quality expectations.

The result of this objective was the successful validation of the system's overall software quality, obtaining an Average Mean Score of 4.83 interpreted as Very High. This indicates that the developed system meets and, in some aspects, exceeds software quality standards. Among the evaluated criteria, Portability received the highest rating (4.98), confirming that the system performs efficiently across different devices and environments, while Compatibility obtained the lowest rating (4.72), suggesting that future enhancements may focus on improved interoperability and integration with other systems. Overall, the results demonstrate that the system was reliable, user friendly, and technically sound based on the ISO/IEC 25010:2011 framework.



Table 29
ISO/IEC 25010:2011 SOFTWARE QUALITY MODEL CHARACTERISTICS

Criteria	Mean Score	Verbal Interpretation
Functionality Suitability Characteristics	4.97	Very High
Performance Efficiency Characteristics	4.92	Very High
Compatibility Characteristics	4.72	Very High
Usability Characteristics	4.75	Very High
Reliability Characteristics	4.76	Very High
Security Characteristics	4.76	Very High
Maintainability Characteristics	4.83	Very High
Portability Characteristics	4.98	Very High
Average Mean	4.83	Very High

The quality evaluation of the Animal Bite Incident Mapping and Classification with Decision Support System shows very strong performance based on the ISO/IEC 25010:2011 Software Quality Model. Among the eight criteria, Portability received the highest mean score of 4.98, indicating that the system works well across different devices and environments. Compatibility obtained the lowest mean score of 4.72, suggesting that while the system functions effectively, improvements can still be made to support smoother interactions with other systems or tools.

These results were similar to a study by Pisueno and Lamis [26], where Portability also received the highest rating and Compatibility was among the lowest. This supports the findings of the present study and shows that web-based systems commonly excel in portability while compatibility remains a frequent



challenge. Overall, the evaluation confirms that the developed system meets high software quality standards and performs consistently with existing research, with compatibility identified as a meaningful area for future enhancement.

Figure 17
User's Guide Cover Page

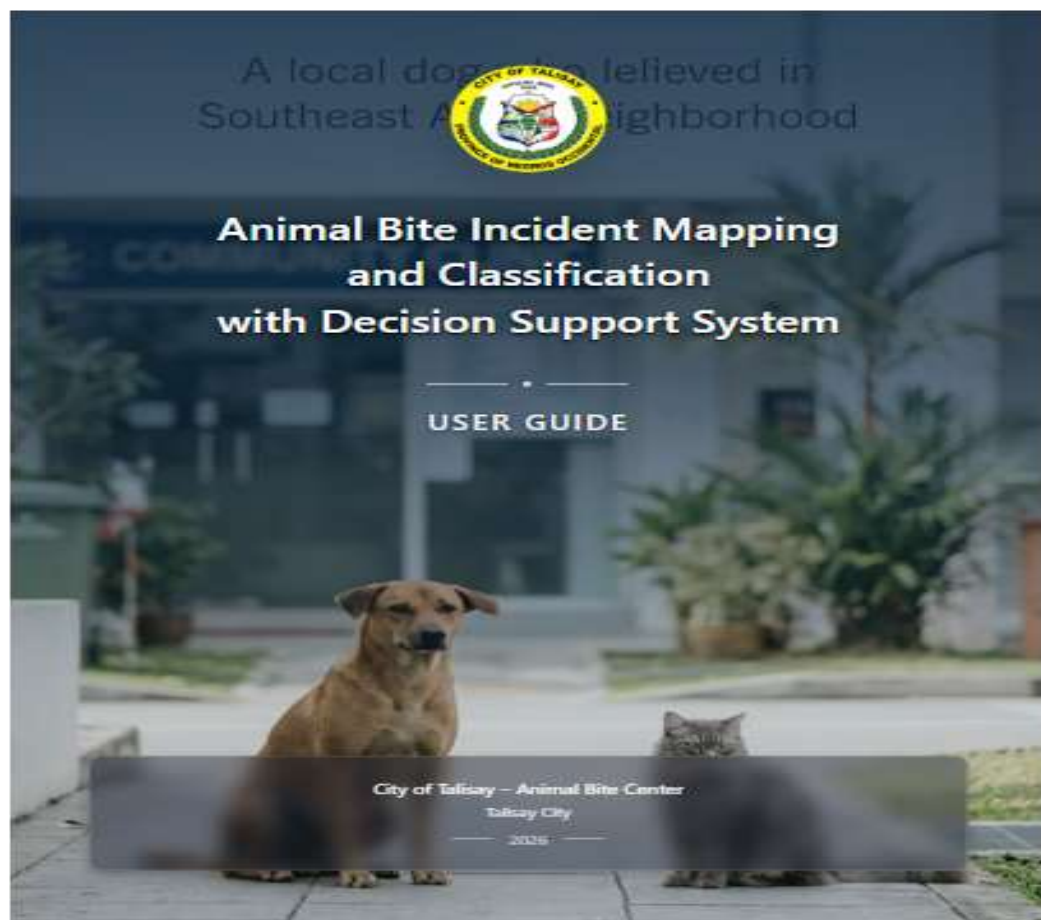


Figure 17 shows the cover page of the User Guide for the Animal Bite Incident Mapping and Classification with Decision Support System, which provides an overview of how the system operates for its different user roles. Barangay



Health Workers have limited access and were primarily responsible for recording animal bite incidents and submitting victim information for monitoring. City Health Workers have broader access, allowing them to review reported cases, classify bite incidents based on risk level, view geospatial maps, and generate reports for analysis and decision-making. System Administrators have full system access and were responsible for managing user accounts, maintaining system data, and ensuring the proper operation of all system features. (See Appendix K, User Manual)



CHAPTER V

CONCLUSION AND RECOMMENDATION

This chapter presents the conclusions drawn from the development, testing, and evaluation of the Animal Bite Incident Mapping and Classification with Decision Support System (ABIMC-DSS) for the Talisay City Health Animal Bite Center. It consolidates the key results of the study and confirms that the developed system effectively addressed the major challenges in animal bite surveillance and management, particularly issues related to delayed reporting, inconsistent manual records, and limited access to timely spatial and statistical insights. The findings highlight the system's contribution to improving rabies monitoring, supporting accurate case classification, and strengthening evidence-based decision-making through GIS visualization and automated reporting. Furthermore, this chapter includes practical recommendations based on the study's outcomes to sustain and enhance the system's implementation.

5.1 Summary of Findings

The findings of this study confirmed that the developed Animal Bite Incident Mapping and Classification with Decision Support System was both technically capable and highly acceptable to its users based on the ISO/IEC 25010:2011 Software Quality Model. The system obtained an Overall Average Mean Score of 4.83, interpreted as Very High, indicating that the respondents strongly agreed that the system performs effectively and meets its intended purpose. Portability



received the highest rating of 4.98, showing that the system can be accessed and used on different devices without technical issues, while Compatibility attained the lowest rating of 4.72, still interpreted as Very High, suggesting that only minimal improvements may be needed for smoother integration with other platforms in the future. In summary, the results demonstrate that the system successfully achieved its objectives and complies with international software quality standards, proving it to be a reliable, functional, and user-friendly tool that supports efficient documentation, monitoring, and analysis of animal bite incidents.

5.2 Conclusions

This study successfully developed, tested, and evaluated the Animal Bite Incident Mapping and Classification with Decision Support System for Talisay City. The system addressed the research problem of inefficient manual documentation and limited data-driven planning related to animal bite cases by providing digital profiling of bite victims, automated incident classification, heatmap visualization of high-risk areas, and a decision support component for public health response. The evaluation using the ISO/IEC 25010:2011 Software Quality Model resulted in an Overall Average Mean Score of 4.83, interpreted as Very High, indicating that users strongly agree with the system's performance, usability, and reliability. Portability received the highest rating, indicating that the system can be used smoothly on different devices and in various work settings. Compatibility, while still



rated Very High, received the lowest score, showing that the system could be improved further by allowing better connection with external platforms, such as national health monitoring databases used for rabies and public health reporting.

The findings of the study confirm that the system was not only technically functional but also highly beneficial for health workers. It helps in organizing bite incident records, reduces mistakes during data entry, makes information easier to find, and provides clear visuals and summaries that support rabies prevention and public health actions. The system allows users to access updated case information quickly and use it as a guide in planning responses and monitoring trends. In summary, the system has achieved its objectives and has shown its potential as a valuable tool for strengthening local health surveillance, improving decision-making, and supporting the long-term control and management of animal bite cases within the city.

5.3 Recommendations

Based on the findings of this study, it was recommended that the City Health Office continue using and further strengthen the Animal Bite Incident Mapping and Classification with Decision Support System to ensure consistent and accurate documentation of bite cases. Training sessions should be regularly conducted for Barangay Health Workers and new personnel so they can fully understand and



use all system features, especially in encoding victim details and interpreting heatmap results. It was also advised that the local government explore ways to integrate the system with national health monitoring databases to support a more unified reporting process and reduce duplicate records.

Future improvements may include adding automated SMS or email notifications for follow-up treatment schedules, enabling offline data entry for barangays with limited internet access, and enhancing compatibility so the system can smoothly connect with other public health platforms. Researchers were encouraged to conduct further studies that evaluate the long-term impact of the system on rabies cases within Talisay City or test the system in other municipalities to measure its wider usability. Continuous review and updates are recommended to ensure the system remains useful, secure, and adaptable to future health information needs.

In closing, the Animal Bite Incident Mapping and Classification with Decision Support System has demonstrated its potential to improve local health response and rabies prevention efforts. With continued support and development, it can become an essential tool for strengthening public health response in Talisay City and may also serve as a model for other communities facing similar challenges.



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APPENDICES

“Appendix A”

ISO/IEC 25010:2011 Software Quality Requirements and Evaluation Questionnaire

Characteristics and Sub-Characteristics	Rating				
Functional Suitability Characteristics	1	2	3	4	5
Functional completeness. The degree to which the set of functions covers all the specified tasks and intended users' objectives.					
Functional correctness. The degree to which a product or system provides accurate results when used by intended users.					
Functional appropriateness. The degree to which the functions facilitate the accomplishment of specified tasks and objectives.					
Performance Efficiency Characteristics	1	2	3	4	5
Time behavior. The degree to which the response time and throughput rates of a product or system, when performing its functions, meet requirements.					
Resource utilization. The degree to which the amounts and types of resources used by a product or system, when performing its functions, meet requirements.					
Capacity. The degree to which the maximum limits of a product or system parameter meet requirements.					



Compatibility Characteristics	1	2	3	4	5
Co-existence. The degree to which a product can perform its required functions efficiently while sharing a common environment and resources with other products, without detrimental impact on any other product.					
Interoperability. The degree to which a system, product or component can exchange information with other products and mutually use the information that has been exchanged					
Usability Characteristics	1	2	3	4	5
Appropriateness recognizability. The degree to which users can recognize whether a product or system is appropriate for their needs.					
Learnability. The degree to which the functions of a product or system can be learnt to be used by specified users within a specified amount of time.					
Operability. The degree to which a product or system has attributes that make it easy to operate and control.					
User error protection. The degree to which a system prevents users against operation errors.					
User Interface Aesthetics. The degree to which a user interface enables pleasing and satisfying interaction for the user.					
Accessibility. The Degree to which a product can be used by people with the widest range of characteristics and capabilities to achieve specified goals in a specified context of use.					
Reliability Characteristics	1	2	3	4	5
Faultlessness. The degree to which a system, product or component performs specified functions without fault under normal operation					



Availability. The degree to which a system, product or component is operational and accessible when required for use.					
Fault tolerance. The degree to which a system, product or component operates as intended despite the presence of hardware or software faults.					
Recoverability. The degree to which, in the event of an interruption or a failure, a product or system can recover the data directly affected and re-establish the desired state of the system.					
Security Characteristics	1	2	3	4	5
Confidentiality. The degree to which a product or system ensures that data are accessible only to those authorized to have access.					
Integrity. The degree to which a system, product or component ensures that the state of its system and data are protected from unauthorized modification or deletion either by malicious action or computer error.					

Non-repudiation. The degree to which actions or events can be proven to have taken place so that the events or actions cannot be repudiated later.					
Accountability. The degree to which the actions of an entity can be traced uniquely to the entity.					
Authenticity. The degree to which the identity of a subject or resource can be proved to be the one claimed.					
Maintainability Characteristics	1	2	3	4	5



Modularity. The degree to which a system or computer program is composed of discrete components such that a change to one component has minimal impact on other components.					
Reusability. The degree to which a product can be used as an asset in more than one system, or in building other assets.					
Analyzability. The degree of effectiveness and efficiency with which it is possible to assess the impact on a product or system of an intended change to one or more of its parts, to diagnose a product for deficiencies or causes of failures, or to identify parts to be modified.					
Modifiability. The degree to which a product or system can be effectively and efficiently modified without introducing defects or degrading existing product quality.					
Testability. The degree of effectiveness and efficiency with which test criteria can be established for a system, product or component and tests can be performed to determine whether those criteria have been met.					
Portability Characteristics	1	2	3	4	5
Adaptability. The degree to which a product or system can effectively and efficiently be adapted for or transferred to different hardware, software or other operational or usage environments.					
Installability. The degree of effectiveness and efficiency with which a product or system can be successfully installed and/or uninstalled in a specified environment.					
Replaceability. The degree to which a product can replace another specified software product for the same purpose in the same environment.					



“Appendix B”

LETTER TO BENEFICIARY

 **Carlos Hilado Memorial State University**
Alijis Campus • Binalbagan Campus • Fortune Towne Campus • Talisay (Main) Campus
A leading GREEN institution of higher learning in the global community by 2030
(Good governance, Research-oriented, Extension-driven, Education for Sustainable Development, and Nation-building)

College of Computer Studies
Carlos Hilado Memorial State University
Talisay City, Negros Occidental
November 2025

Talisay City Health – Animal Bite Center
Bonifacio St, Talisay City, Negros Occidental

Subject: Request for Beta Testing of the Animal Bite Incident Mapping and Classification System with Decision Support

Dear Animal Bite Center Officers,

Good day!

We, the students from the College of Computer Studies of Carlos Hilado Memorial State University, are pleased to inform you that we have developed a system entitled **“Animal Bite Incident Mapping and Classification System with Decision Support.”** This system is designed to assist in tracking, classifying, and managing animal bite incidents, thereby improving response efficiency and supporting timely medical and administrative decisions at your center.

In line with the development process, we would like to formally request your permission to conduct **Beta Testing** of our system within your organization. The purpose of this testing is to evaluate the functionality, usability, and reliability of the system in a real-world environment. Your feedback and observations will be invaluable in helping us improve and finalize the system for full deployment.

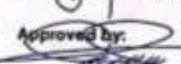
Rest assured that all data collected during the testing period will be treated with the utmost confidentiality and used solely for academic and system improvement purposes.

We sincerely hope for your favorable response and cooperation in allowing us to perform this Beta Testing at your convenience. Should you have any questions or need further clarification, you may contact us at 0907-806-0026 or email us at paarageles.chmsu@gmail.com.

We would be glad to provide any additional details or demonstrations.

Thank you very much for your time and support.

Respectfully yours,
Meg Ryan D. Pajo - Project Manager
Anthony A. Morales - Programmer
Jule Ann D. Panolina - System Analyst

Approved by: 
Jose Leo G. Redobio, PhD
Capstone Adviser


CRISTINE V. REDOBIO, PhD
Capstone Project 2 Professor

 (034) 712 0005

GREEN CHMSU ExCELSIOR!



“Appendix C”

LETTER TO CAPSTONE PROJECT ADVISER



Carlos Hilado Memorial State University
Alijis Campus • Binalbagan Campus • Fortune Towne Campus • Talisay (Main) Campus
A leading GREEN institution of higher learning in the global community by 2030
(Good governance, Research-oriented, Extension-driven, Education for Sustainable Development, and Nation-building)

College of Computer Studies

March 25, 2025

DANES P. TUMABING, MIT
Program Chairperson, BSIS Department
Talisay Main Campus

Thru: **RUBY MAE M. BERMEJO, PhD**
Instructor, Capstone Project 1/ BSIS Faculty

Dear Sir:

Greetings!

We are reaching out regarding the selection of an adviser for our capstone project, titled **Animal Bite Incident Mapping and Classification System with DSS** approved last **March 18, 2025**. Considering our project's needs, we would like to request **Dr. Jose Leo G. Redoblo** to serve as our adviser.

His significant experience and expertise align with our project's needs, and we are confident that his guidance will be invaluable in helping us achieve our capstone project's goals.

We kindly request your approval for this adviser selection. Thank you for considering our request, and we look forward to your positive response.

Very respectfully yours,


MEG-RYAN D. ROJO


JULIE ANN D. PANOLINO


ANTHONY A. ANGELES

BSIS 3D



college.computerstudies@chmsu.edu.ph
(034) 434 8148
chmsu.edu.ph

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“Appendix D”

BLACK BOX TESTING - ALPHA TESTING

BLACK-BOX TESTING OF THE ALPHA TESTING OF CITY HEALTH WORKER AUTHENTICATION CASE

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-AdminAuth001	Login form visibility test	All CHW login form elements should be visible on page load	All expected form elements were visible	Pass	No issues encountered
TC-AdminAuth002	Branding and text display check	All branding and descriptive text should display correctly	Branding text and page descriptions appeared correctly	Pass	Branding rendered correctly
TC-AdminAuth003	Login submission and redirection validation	User should be redirected to the CHW dashboard after successful login	The system successfully authenticated valid user credentials and redirected the user to the dashboard	Pass	Validation works as intended; incorrect inputs properly rejected



*BLACK-BOX TESTING OF THE ALPHA TESTING OF CITY HEALTH WORKER
DASHBOARD CASE*

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-AdminDash004	CHW dashboard layout and navigation visibility test	Top nav shows the system title and main links (Dashboard, Geomapping, Reports, Patients), with Dashboard highlighted as active.	All navigation bar items were visible and correctly displayed; "Dashboard" was highlighted	Pass	Navigation displayed correctly
TC-AdminDash005	Welcome section visibility check	Welcome header, description text, and "View All Reports" button should be visible	Welcome section displayed correctly with all expected elements	Pass	No issues encountered
TC-AdminDash006	Statistics cards display test	All statistics cards should be visible and show corresponding case counts	All statistics cards appeared correctly and showed visible case numbers	Pass	Statistics displayed as expected
TC-AdminDash007	Patient and staff statistics cards check	Cards showing total patients and total staff should be visible	"Total Patients" and "Total Staff" cards were displayed properly	Pass	Both cards rendered correctly
TC-AdminDash008	Quick action cards visibility test	Quick action cards should be visible and properly labeled	All quick action cards appeared correctly	Pass	Labels and cards displayed fine
TC-AdminDash009	Recent bite cases table visibility test	The recent bite cases table and "View All" button should be visible with correct column headers	The recent bite cases table and "View All" button displayed correctly with visible rows	Pass	Table loaded with correct columns



**BLACK-BOX TESTING OF THE ALPHA TESTING OF CITY HEALTH WORKER
GEOMAPPING CASE**

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-AdminGeo010	Geomapping page layout and navigation check	The “Geomapping” link should appear active; the page should display the title “Geomapping Analysis” and a brief description about visualizing cases	The Geomapping link was active, and the title and description appeared correctly	Pass	Page layout displayed properly
TC-AdminGeo011	Map container and control buttons visibility test	The map container and map control buttons should be visible	The map container, Heatmap, and Markers buttons were visible and functional	Pass	Map and controls visible
TC-AdminGeo012	Map filter and statistics controls visibility test	All filter fields, dropdowns, and action buttons should be visible and ready for input	The filter form, dropdowns, and buttons were all displayed properly	Pass	Filters rendered correctly
TC-AdminGeo013	Map interaction behavior validation	The map should remain visible and update when switching between Heatmap and Markers views	The map responded correctly to Heatmap and Markers controls	Pass	Map interaction worked smoothly
TC-AdminGeo014	Map statistics display test	Statistic cards should be visible and show key mapping information	All statistic cards were displayed correctly with proper labels	Pass	Statistic cards displayed correctly



**BLACK-BOX TESTING OF THE ALPHA TESTING OF CITY HEALTH WORKER
GEOMAPPING CASE**

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-AdminGeo010	Geomapping page layout and navigation check	The “Geomapping” link should appear active; the page should display the title “Geomapping Analysis” and a brief description about visualizing cases	The Geomapping link was active, and the title and description appeared correctly	Pass	Page layout displayed properly
TC-AdminGeo011	Map container and control buttons visibility test	The map container and map control buttons should be visible	The map container, Heatmap, and Markers buttons were visible and functional	Pass	Map and controls visible
TC-AdminGeo012	Map filter and statistics controls visibility test	All filter fields, dropdowns, and action buttons should be visible and ready for input	The filter form, dropdowns, and buttons were all displayed properly	Pass	Filters rendered correctly
TC-AdminGeo013	Map interaction behavior validation	The map should remain visible and update when switching between Heatmap and Markers views	The map responded correctly to Heatmap and Markers controls	Pass	Map interaction worked smoothly
TC-AdminGeo014	Map statistics display test	Statistic cards should be visible and show key mapping information	All statistic cards were displayed correctly with proper labels	Pass	Statistic cards displayed correctly



BLACK-BOX TESTING OF THE ALPHA TESTING OF CITY HEALTH WORKER PATIENTS CASE

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-AdminPatients015	Patients page layout and navigation check	The "Patients" link should be active and the page title should be visible	"Patients" link was active and the page heading displayed properly	Pass	Page layout displayed correctly
TC-AdminPatients016	Patients table and filter visibility test	The table and all filter options should be visible for user interaction	Table and all filter options were visible as expected	Pass	Filters and table rendered correctly
TC-AdminPatients017	Search functionality validation	Page URL should update with the search query and display filtered results	URL updated with search=test and filtered results appeared correctly	Pass	Search executed successfully
TC-AdminPatients018	Patient data table content check	Table should display column headers and at least one patient record	Table displayed correctly with proper headers and visible rows	Pass	Patient records loaded properly
TC-AdminPatients019	Patient row "View" button visibility test	Each patient row should contain a working "View" button	"View" button was visible and functional in the first table row	Pass	Button displayed with no rendering delay
TC-AdminPatients020	Pagination visibility test	Pagination controls should be visible for navigating multiple pages	Pagination controls and page links were visible as expected	Pass	Pagination displayed correctly



**BLACK-BOX TESTING OF THE ALPHA TESTING OF CITY HEALTH WORKER
REPORTS CASE**

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/ Fail	Comments
TC-AdminReports021	Reports page layout and navigation check	Set Reports as the active nav link, and show the title "Animal Bite Reports" with a brief description underneath.	Navigation link was active; title and description displayed correctly	Pass	Page displayed as expected
TC-AdminReports022	Reports table and filter visibility test	All filter fields and options should be visible to the CHW user	All filters, search fields, and date inputs were visible as expected	Pass	Filters and fields rendered correctly
TC-AdminReports023	Search functionality validation	Page URL should include the search parameter and filter results accordingly	URL updated to include search=test, confirming successful search execution	Pass	Search function worked correctly
TC-AdminReports024	Reports table structure and data check	Table should display correct column headers and visible report records	Table headers and data rows were visible and formatted correctly	Pass	Records displayed properly
TC-AdminPatients025	Patient row "View" button visibility test	Each patient row should contain a working "View" button	"View" button was visible and functional in the first table row	Pass	Button displayed with no rendering delay
TC-AdminReports026	Report action buttons visibility test	Each report entry should have working View and Edit buttons	"View" and "Edit" buttons were visible and loaded correctly	Pass	Buttons functional and responsive
TC-AdminReports027	Pagination controls visibility test	Pagination controls should be visible for navigating multiple reports	Pagination and page links were displayed properly	Pass	Pagination rendered with no issues



BLACK-BOX TESTING OF THE ALPHA TESTING OF STAFF AUTHENTICATION CASE

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-StaffAuth028	Staff login form visibility test	Email and password inputs, "Login" button, and "Staff Login" heading should be visible	All form elements and title displayed correctly	Pass	Form components loaded properly
TC-StaffAuth029	Staff login branding display check	The page should display "BHW Portal" logo, "Welcome Back!" heading, and login description	All branding text and the logo were visible as expected	Pass	Branding displayed correctly
TC-StaffAuth030	Staff login form submission validation	After clicking "Login," the system should redirect to staff_login.php for authentication	The login form redirected correctly to staff_login.php and URL updated as expected	Pass	Login processing worked properly

BLACK-BOX TESTING OF THE ALPHA TESTING OF STAFF DASHBOARD CASE

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail	Comments
TC-StaffDash031	Welcome section display test	"Welcome" heading, overview message, and "New Bite Report" button should be visible	Welcome text and "New Bite Report" button displayed correctly	Pass	Welcome elements visible
TC-StaffDash032	Encoding a new Animal Bite Report	The user fills out all required patient information fields and clicks Save.	The system accepted all input, saved the new bite report, and redirected to the confirmation/records page	Pass	All input fields accepted valid entries, record saved successfully, and no validation or system errors occurred



“Appendix E”

Sample Black Box Testing – Beta Testing

Test runner interface showing test results for the 'Admin Login' page. The test suite is 'Admin Login Tests' and the test is 'should display admin login form elements'. The test results show that the test passed, with the following assertions:

- get('input[name="email"]') - expected 'input#email.form-control' to be visible
- get('input[name="password"]') - expected 'input#password.form-control' to be visible
- get('button[type="submit"]') - expected 'button.btn.btn-primary.btn-login' to be visible
- contains('a', 'Admin Login') - expected 'a[href="/admin/login"]' to be visible

Test runner interface showing test results for the 'Admin Dashboard' page. The test suite is 'Admin Dashboard Tests' and the test is 'should display welcome section'. The test results show that the test passed, with the following assertions:

- url - expected 'http://localhost/Animal-Bite/src/admin/admin_dashboard.php' to include '/admin/admin_dashboard.php'
- visit('/src/admin/admin_dashboard.php') - expected 'h2.admin-welcome' to be visible
- contains('p', 'Here's an overview of animal bite cases and statistics') - expected 'p.text-muted.mb-3' to be visible
- contains('a.btn', 'View All Reports') - expected 'a.btn.btn-primary' to be visible

Test runner interface showing test results for the 'Geomapping Analysis' page. The test suite is 'Admin Geomapping Tests' and the test is 'should display geomapping page layout'. The test results show that the test passed, with the following assertions:

- contains('a.nav-link', 'Geomapping') - expected 'a.nav-link.active' to have class 'active'
- contains('h2', 'Geomapping Analysis') - expected 'h2.mb-3' to be visible
- contains('p', 'Visualize animal bite cases by location for better decision making') - expected 'p.text-muted.mb-3' to be visible



Admin Geomapping Tests
should display map container and controls

```
1 -click  
(Page sub) --submitting form--  
(page load) --page loader--  
(new url) http://localhost/Animal-Bite/src/admin/geomapping.php  
2  
3 [passed] expected http://localhost/Animal-Bite/src/admin/admin_dashboard.php to include /admin/admin_dashboard.php  
4  
5 visit /src/admin/geomapping.php  
6  
7 TEST BODY  
8  
9 get #map  
10 [passed] expected <div>map, leaflet-container, leaflet-touch, leaflet-fade-anim, leaflet-grab, leaflet-touch-drag, leaflet-touch-zoom to be visible  
11  
12 ~contains button, Heatmap
```

Admin Geomapping Tests
should display map filters and statistics

```
1 [passed] expected <select>category, form-select to be visible  
2  
3 get <select[name='status']>  
4 [passed] expected <select>status, form-select to be visible  
5  
6 get <input[name='data_from']>  
7 [passed] expected <input>data_from, form-control to be visible  
8  
9 get <input[name='data_to']>  
10 [passed] expected <input>data_to, form-control to be visible  
11  
12 ~contains button, Apply Filters  
13  
14 [passed] expected <button.btn.btn-primary> to be visible  
15  
16 ~contains a, Reset Filters  
17  
18 [passed] expected <a.btn.btn-outline-secondary.no-2> to be visible
```

Admin Geomapping Tests
should handle map interactions

```
1  
2 get #map  
3 [passed] expected <div>map, leaflet-container, leaflet-touch, leaflet-fade-anim, leaflet-grab, leaflet-touch-drag, leaflet-touch-zoom to be visible  
4  
5 ~contains button, Heatmap  
6  
7 -click  
8  
9 get #map  
10 [passed] expected <div>map, leaflet-container, leaflet-touch, leaflet-fade-anim, leaflet-grab, leaflet-touch-drag, leaflet-touch-zoom to be visible  
11  
12 ~contains button, Heatmap  
13  
14 -click  
15  
16 get #map  
17  
18 [passed] expected <div>map, leaflet-container, leaflet-touch, leaflet-fade-anim, leaflet-grab, leaflet-touch-drag, leaflet-touch-zoom to be visible
```

[illegible]

The screenshot displays a Cypress test runner on the left and a web application on the right. The test runner shows a sequence of commands: `get('table th')`, `expect()` assertions, and `get('table th')` commands. The web application, titled 'Animal Bite Admin Portal', features a navigation bar with links like 'Dashboard', 'Mapping', 'Reports', and 'Admin'. Below the navigation bar is a search section with filters for 'Name', 'Date From', and 'Date To'. The main content area contains a table with columns: 'ID', 'Owner Name', 'Contact', 'Breed', 'Animal', 'Age', 'Category', 'Status', 'Report Date', and 'Action'. The table lists several animals, including 'Anthony Angkor', 'Andrea Ramon', 'Higuel Sanchez', and 'Anthony Oak One'. Each row has a corresponding 'Action' button with a plus icon.



Admin Staff Tests
should display staff page layout

http://localhost/Animal-Bites/src/admin/view_staff.php

1280x720

Develop

Admin Staff Tests
should display staff page layout

to include /admin/admin_dashboard.php

visit /src/admin/view_staff.php

TEST BODY

- contains Link, Staff
- expected Link.active to have class active
- contains h3, Staff Management
- expected <h3.h3> to be visible
- contains a, View and manage herange health workers
- expected <p.test-outside-ab-6> to be visible
- contains a.btn, Add New Staff
- expected Add New Staff to be visible

Animal Bite Admin Portal

Staff Management

View and manage herange health workers

Search Staff

Search by Name, email, or contact number

Search

Clear

Name	Contact Number	Email	Birth Date	Actions
Anthony Kingdon	020207890	anthony.kingdon@gmail.com	Mar 21, 2028	View Edit Delete
Jake Ben Spindler	020497890	jakeben.spindler@gmail.com	Jun 10, 2020	View Edit Delete

Showing 1 to 2 of 2 staff members

© 2022 Herange Health System Management System

Filtered Highlights

View All Reports

Authentication Tests > Staff Login Form
should display staff login form elements

http://localhost/Animal-Bites/src/login/staff_login.html

1280x720

BEFORE EACH (2)

- visit /src/login/staff_login.html

TEST BODY

- get input[name="email"]
- expected <input type="email"> to be visible
- get input[name="password"]
- expected <input type="password"> to be visible
- get button[type="submit"]
- expected <button type="submit"> to be visible
- contains h3, Staff Login

BHW Portal

Welcome Back!

Log in to access the Herange Health System Portal.

- Submit health reports
- Track patient records
- Access health resources

Staff Login

OR Social Network

Remember

Log In

or

Continue with Google

Need Help?

If you're having trouble logging in, please contact your administrator for assistance.

© Herange Health System Management System

Staff Dashboard Tests
should display statistics cards

http://localhost/Animal-Bites/src/staff/dashboard.php

1280x720

Staff Dashboard Tests
should display statistics cards

- contains .state-card h3, Total Cases
- expected <h3.h3> to be visible
- contains .state-card h3, Today's Cases
- expected <h3.h3> to be visible
- contains .state-card h3, Pending
- expected <h3.h3> to be visible
- contains .state-card h3, Category III
- expected <h3.h3> to be visible
- get .state-number
- expected { <div.state-number>, 3 more... } to be visible

BHW Animal Bite Portal

Dashboard Reports All Patterns Search

Welcome, Anthony!

Here's an overview of animal bite cases and activities

New File Report

Total Cases 101 All time	Today's Cases 0 New submissions	Pending 21 Awaiting action	Category III 5 Severe cases
--------------------------------	---------------------------------------	----------------------------------	-----------------------------------

Quick Actions

- New File Report
Review a new animal bite
- Add Patient
- Search Records



“Appendix F”

WHITE BOX TESTING - ALPHA TESTING

White-Box Testing of the Alpha Testing of Admin Authentication Case

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result
TC-AdminAuth001	Function: adminLogin()	Test login with correct username and password	Username: "admin", Password="correct123"	Should authenticate user and redirect to dashboard	Matches expected output	Pass
TC-AdminAuth002	Function: adminLogin()	Test login with incorrect password	username="admin", password="wrongpass"	Should return "Invalid credentials"	Matches expected output	Pass
TC-AdminAuth003	Function: adminLogin()	Test empty username handling	username="", password="test123"	Should output "Username required"	Matches expected output	Pass
TC-AdminAuth004	Function: adminLogin()	Test empty password handling	username="admin", password=""	Should output "Password required"	Matches expected output	Pass
TC-AdminAuth005	Function: adminLogin()	Test SQL injection prevention	username="OR 1=1 --", password="none"	Inputs should be sanitized; login should fail	Matches expected output	Pass
TC-AdminAuth006	Function: adminLogin()	Test login attempt for deactivated admin	username="admin2", password="valid123"	System shows "Account disabled"	Matches expected output	Pass
TC-AdminAuth007	Function: adminLogin()	Test failed login rate-limiting (failed attempts)	incorrect login attempts	Account should be temporarily locked	Matches expected output	Pass



TC-AdminAuth008	Function: adminLogin()	Test case sensitivity of username	username="ADMIN", password="correct123"	Login should fail if case-sensitive	Matches expected output	Pass
TC-AdminAuth009	Function: adminLogin()	Test trimming of leading/trailing spaces	username=" admin ", password="correct123 "	Should trim spaces and authenticate	Matches expected output	Pass
TC-AdminAuth010	Function: adminLogin()	Handling of special characters	username="admin!@#", password="test!@#"	Should accept special characters and validate normally	Matches expected output	Pass

White-Box Testing of the Alpha Testing of Staff Authentication Case

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result
TC-StaffAuth001	Function: staffLogin()	Test valid staff login	username="staff1", password="correct123"	Authenticate user and redirect to staff dashboard	Matches expected output	Pass
TC-StaffAuth002	Function: staffLogin()	Handle incorrect password	username="staff1", Password="wrong pass"	Display "Invalid credentials"	Matches expected output	Pass
TC-StaffAuth003	Function: staffLogin()	Test login for deactivated staff account	username="staffDeactivated", password="valid123"	System should display "Account disabled"	Matches expected output	Pass
TC-StaffAuth004	Function: staffLogin()	Test SQL injection handling	username=" OR 1=1 --", password="none"	Input sanitized; login must be rejected	Matches expected output	Pass



White-Box Testing of the Alpha Testing of Authentication Error Handling Case

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result
TC-AuthError001	Function: adminLogin() / staffLogin()	Test handling of missing input fields	Username="", password=""	System must return "All fields required"	Matches expected output	Pass
TCAuthError002	Function: adminLogin() / staffLogin()	Test invalid characters in username	username="adm!n<>", password="test123"	Reject and show "Invalid input format"	Matches expected output	Pass
TC-AuthError003	Function: adminLogin() / staffLogin()	Test large input size	username="a"*300, password="b"*300	System must reject overly long puts and prevent overflow	Matches expected output	Pass

White-Box Testing of the Alpha Testing of Authentication Error Handling Case

Test Case ID	Tested Code Segment	Test Description	Input Values	Expected Behavior	Actual Behavior	Result
TC-DataConn001	Function: Database::connect()	Test successful database connection	Valid DB credentials	Connection should be established without errors	Matches expected output	Pass
TCDDataConn002	Function: Database::connect()	Test network timeout handling	Incorrect DB password	System should throw Connection error	Matches expected output	Pass
TCDDataConn003	Function: Database::connect()	Test persistent connection reuse	host=""	Should fail and return "Unable connect to host"	Matches expected output	Pass
TC-DataConn004	Function: Database::connect()	Test network timeout handling	Simulated network delay	Should trigger connection timeout error	Matches expected output	Pass



“Appendix G”

Sample White Box Testing – Beta Testing

White-box testing using the PEST PHP Framework Results:

```
229     test('validates error handling in staff queries', function () {
230         $exception = new PDOException('Table staff does not exist');
231
232         expect($exception)->toBeInstanceOf(PDOException::class);
233
234         // Fallback behavior
235         $fallback = ['firstName' => 'User', 'lastName' => ''];
236         expect($fallback)->toBeArray();
237         expect($fallback['firstName'])->toBe('User');
238     });
239 });
```

```
133     test('handles new patient creation logic', function () {
134         $firstName = 'Jane';
135         $lastName = 'Smith';
136         $middleName = 'Marie';
137         $dateOfBirth = '1990-01-01';
138         $gender = 'Female';
139         $contactNumber = '09876543210';
140         $email = 'jane@example.com';
141         $address = '456 Oak St';
142         $barangay = 'Tabunok';
143         $city = 'Talisay City';
144         $province = 'Cebu';
145         $previousRabiesVaccine = 'No';
146
147         // Simulate new patient data
148         $patientData = [
149             $firstName,
150             $lastName,
151             $middleName,
152             $dateOfBirth,
153             $gender,
154             $contactNumber,
155             $email,
156             $address,
157             $barangay,
158             $city,
```

```
23     test('constructs PDO connection string correctly', function () {
24         $host = 'localhost';
25         $dbname = 'animal_bite';
26         $dsn = "mysql:host={$host};dbname={$dbname}";
27
28         expect($dsn)->toBe('mysql:host=localhost;dbname=animal_bite');
29         expect($dsn)->toContain('mysql:host=');
30         expect($dsn)->toContain('dbname=');
31     });
```



```
199 test('logs failed login attempts', function () {
200     $email = 'admin@example.com';
201     $ipAddress = '192.168.1.1';
202
203     // Simulate error Log message
204     $logMessage = "Failed admin login attempt for email: {$email} from IP: {$ipAddress}";
205
206     expect($logMessage)->toContain($email);
207     expect($logMessage)->toContain($ipAddress);
208 });
```

```
185 test('constructs staff name correctly from first and last name', function () {
186     $staff = [
187         'firstName' => 'Jane',
188         'lastName' => 'Smith'
189     ];
190
191     $fullName = $staff['firstName'] . ' ' . $staff['lastName'];
192
193     expect($fullName)->toBe('Jane Smith');
194 });
```

```
159 test('sets staff session variables correctly', function () {
160     $staff = [
161         'staffId' => 1,
162         'firstName' => 'John',
163         'lastName' => 'Doe',
164         'email' => 'staff@example.com'
165     ];
166
167     $_SESSION['staffId'] = $staff['staffId'];
168     $_SESSION['staffName'] = $staff['firstName'] . ' ' . $staff['lastName'];
169     $_SESSION['staff_email'] = $staff['email'];
170     $_SESSION['login_time'] = time();
171
172     expect($_SESSION['staffId'])->toBe(1);
173     expect($_SESSION['staffName'])->toBe('John Doe');
174     expect($_SESSION['staff_email'])->toBe('staff@example.com');
175 });
```

```
121 test('validates email with various formats', function () {
122     $validEmails = [
123         'admin@example.com',
124         'user.name@example.com',
125         'user+tag@example.co.uk',
126         'admin123@test-domain.com'
127     ];
128
129     foreach ($validEmails as $email) {
130         expect(filter_var($email, FILTER_VALIDATE_EMAIL))->not->toBeFalse();
131     }
132 });
```



```
145     test('validates commit after successful operations', function () {
146         $operations = ['insert_patient', 'insert_report', 'insert_images'];
147         $allSuccessful = true;
148
149         foreach ($operations as $op) {
150             if (!$op) {
151                 $allSuccessful = false;
152             }
153         }
154
155         if ($allSuccessful) {
156             $shouldCommit = true;
157             expect($shouldCommit)->toBeTrue();
158         }
159     });
```

```
33     test('validates all required fields are filled', function () {
34         $_POST = [
35             'firstName' => 'John',
36             'lastName' => 'Doe',
37             'contactNumber' => '09123456789',
38             'address' => '123 Main St',
39             'barangay' => 'Poblation',
40             'city' => 'Talisay City',
41             'animalType' => 'Dog',
42             'biteDate' => '2024-01-15',
43             'biteLocation' => 'Hand'
44         ];
45
46         $requiredFields = ['firstName', 'lastName', 'contactNumber', 'address', 'barangay', 'city', 'animalType', 'biteDate',
47             $missingFields = [];
48
49         foreach ($requiredFields as $field) {
50             if (empty($_POST[$field])) {
51                 $missingFields[] = $field;
52             }
53         }
54
55         expect($missingFields)->toBeEmpty();
56     });
```

```
345     test('validates commit after successful operations', function () {
346         $operations = ['insert_patient', 'insert_report', 'insert_images'];
347         $allSuccessful = true;
348
349         foreach ($operations as $op) {
350             if (!$op) {
351                 $allSuccessful = false;
352             }
353         }
354
355         if ($allSuccessful) {
356             $shouldCommit = true;
357             expect($shouldCommit)->toBeTrue();
358         }
359     });
```



```
133     test('handles new patient creation logic', function () {
134         $firstName = 'Jane';
135         $lastName = 'Smith';
136         $middleName = 'Marie';
137         $dateOfBirth = '1990-01-01';
138         $gender = 'Female';
139         $contactNumber = '09876543210';
140         $email = 'jane@example.com';
141         $address = '456 Oak St';
142         $barangay = 'Tabunok';
143         $city = 'Talisay City';
144         $province = 'Cebu';
145         $previousRabiesVaccine = 'No';
146
147         // Simulate new patient data
148         $patientData = [
149             $firstName,
150             $lastName,
151             $middleName,
152             $dateOfBirth,
153             $gender,
154             $contactNumber,
155             $email,
156             $address,
157             $barangay,
158             $city.
```

```
229     test('validates error handling in staff queries', function () {
230         $exception = new PDOException('Table staff does not exist');
231
232         expect($exception)->toBeInstanceOf(PDOException::class);
233
234         // Fallback behavior
235         $fallback = ['firstName' => 'User', 'lastName' => ''];
236         expect($fallback)->toBeArray();
237         expect($fallback['firstName'])->toBe('User');
238     });
239 });
```



```
149  test('handles file upload error codes', function () {
150      $errorCodes = [
151          UPLOAD_ERR_OK => 0,
152          UPLOAD_ERR_INI_SIZE => 1,
153          UPLOAD_ERR_FORM_SIZE => 2,
154          UPLOAD_ERR_PARTIAL => 3,
155          UPLOAD_ERR_NO_FILE => 4,
156          UPLOAD_ERR_NO_TMP_DIR => 6,
157          UPLOAD_ERR_CANT_WRITE => 7,
158          UPLOAD_ERR_EXTENSION => 8
159      ];
160
161      foreach ($errorCodes as $constant => $value) {
162          expect($value)->toBeInt();
163          expect($value)->toBeGreaterThanOrEqual(0);
164      }
165  });
```



“Appendix H”

GRAMMARIAN CERTIFICATE



Carlos Hilado Memorial State University
Alijis Campus • Binalbagan Campus • Fortune Towne Campus • Talisay (Main) Campus
A leading GREEN institution of higher learning in the global community by 2030
(Good governance, Research-oriented, Extension-driven, Education for Sustainable Development, and Nation-building)

Library and Information Services

CERTIFICATION

This is to certify that the generated capstone output of **Anthony A. Angeles, Meg Ryan D. Rojo, Julie Ann D. Panolino** with the research title **“Animal Bite Incident Mapping and Classification with Decision Support System”** has gone through Grammarly and Plagiarism Check with **4%** rating.

Attaching herewith is the original copy of the manuscript with the following comments and findings issued by the Grammarly software.

This certification is issued for whatever legal purpose it may serve them best.

Issued this 22nd Day of January 2026.


MA. SALUD P. DIONE, RL, MLIS
Readers Services Librarian



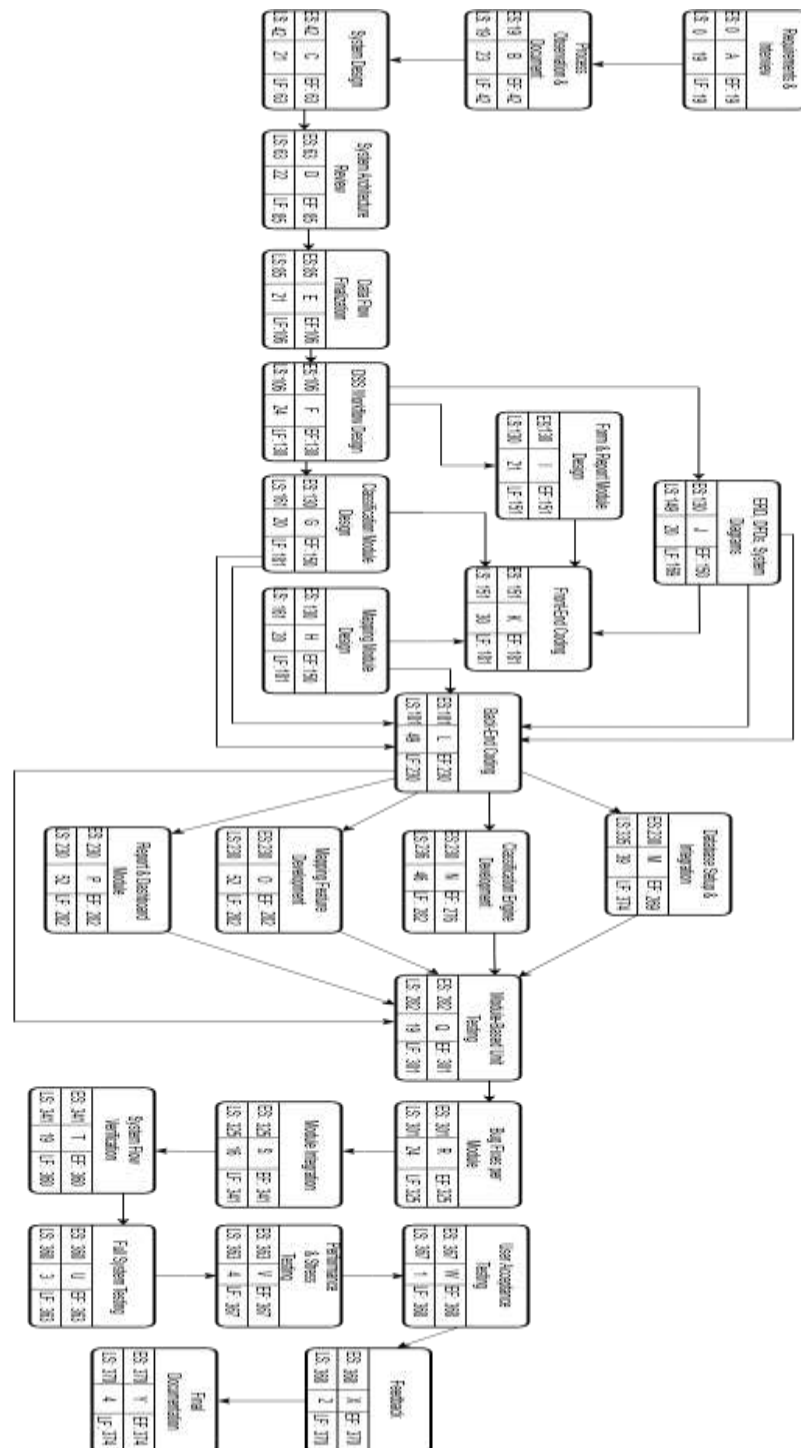
library.talisaychmsu.edu.ph
712 0006 local TUP
chmsc.edu.ph

GREEN CHMSU ExCELSIOR!

Excellence • Compassion • Environmentalism • Love of Country • Social Responsibility • Integrity • Openness • Resilience

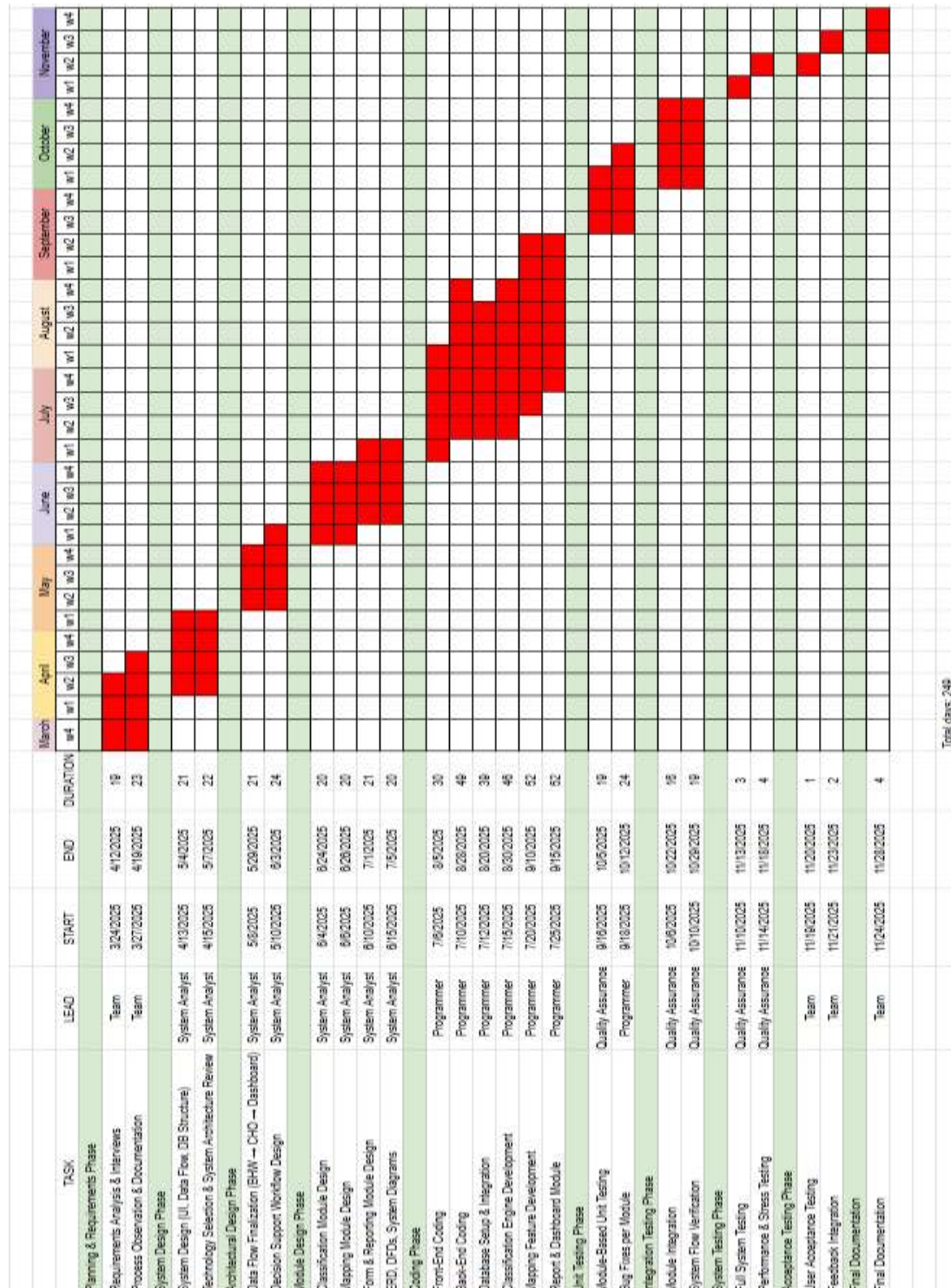
“Appendix I”

PERT – CPM



“Appendix J”

GANTT CHART





“Appendix K”

USER’S GUIDE

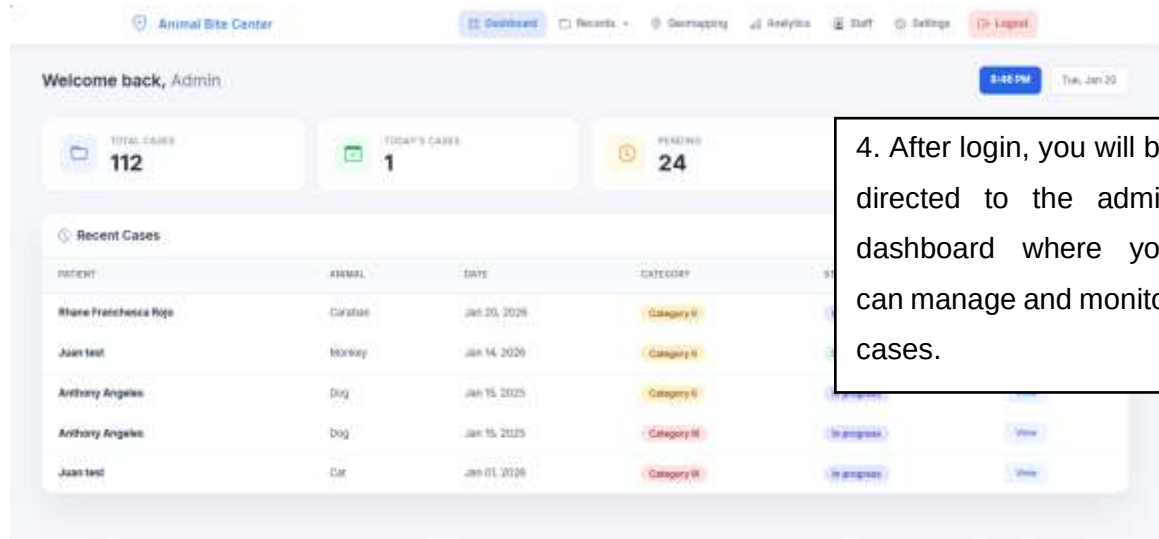
Animal Bite Incident Mapping and Classification with DSS – How to Login as Admin

1. Provide your email and password, then click 'Sign In' to access the portal

2. Signing in with a Google account is also supported.

3. If your credentials are valid, a success message will appear after logging in.

The screenshots show the 'Animal Bite Portal' interface for Talisay City. The login form includes fields for 'Email Address' (admin@example.com) and 'Password' (Enter your password), a 'Sign In' button, and a 'Continue with Google' button. The success message is 'Login successful Redirecting to dashboard...'.



STEPS:

1. Provide your email and password, then click 'Sign In' to access the portal.
2. Signing in with a Google account was also supported.
3. If your credentials were valid, a success message will appear after logging in.
4. After login, you will be directed to the admin dashboard where you can manage and monitor cases.



Animal Bite Admin – Creating a New Report Form

Animal Bite Center | Dashboard | **Records** | Geomapping | Analytics | Staff | Settings | Logout

Welcome back, Admin | 9:03 PM | Tue, Jan 20

TOTAL CASES: 112 | TODAY'S CASES: 1 | PENDING: 24 | CATEGORY III: 11

1. Navigate to the 'Records' section using the top menu bar.

PATIENT	ANIMAL	DATE	CATEGORY	STATUS	
Rhane F...		Jan 20, 2026	Category II	In progress	View
Juan tes...		Jan 14, 2026	Category II	Completed	View
Anthony Angeles	Dog	Jan 15, 2025	Category II	In progress	View
Anthony Angeles	Pink	Jan 15, 2025	Category II	In progress	View

Animal Bite Center | Dashboard | Records | Geomapping | Analytics | Staff | Settings | Logout

Welcome back, Admin | 9:21 PM | Tue, Jan 20

TOTAL CASES: 112 | TODAY'S CASES: 1 | PENDING: 24 | CATEGORY III: 11

2. From the dropdown menu, choose the 'New Report' option.

Records dropdown menu:

- Patients
- Reports
- New Report**

PATIENT	ANIMAL	DATE	CATEGORY	STATUS	
Rhane F...		Jan 20, 2026	Category II	In progress	View
Juan tes...		Jan 14, 2026	Category II	Completed	View
Anthony Angeles	Dog	Jan 15, 2025	Category II	In progress	View
Anthony Angeles	Pink	Jan 15, 2025	Category II	In progress	View



Animal Bite Center

Dashboard Records Geomapping Analytics Staff Settings Logout

New Animal Bite Report

Patient Information

Patient Selection

☒ Existing Patient ☐ New Patient

Search Patient *

Type to search for a patient

Animal Bite Details

Date of Bite *

01/20/2026

3. The system will redirect you to the new report form for case entry details.

4. Select whether the patient is already registered or is a new entry.

Animal Bite Center

Dashboard Records Geomapping Analytics Staff Settings Logout

New Animal Bite Report

Patient Information

Patient Selection

☒ Existing Patient ☐ New Patient

Search Patient *

Type to search for a patient

Animal Bite Details

Date of Bite *

01/20/2026

Animal Type *

-- Select Animal --

5. For existing patients, the search function will automatically populate their basic information in the form.



Animal Bite Center

Dashboard Records Geomapping Analytics Staff Settings Logout

Patient Information

Patient Selection
Existing Patient + New Patient

First Name *

Last Name *

Gender
-- Select Gender --

Date of Birth
mm/dd/yyyy

Contact Number
+63 9123456789
Enter 10 digits only; +63 is fixed

Barangay
Select Barangay

Address

6. For new patients, a simplified form will appear. Enter the patient's basic personal information in the fields provided.

Animal Bite Center

Dashboard Records Geomapping Analytics Staff Settings Logout

Animal Bite Details

Date of Bite * 01/21/2026

Animal Type * -- Select Animal --

Animal Ownership * -- Select Ownership --

Animal Status * Alive

Animal Vaccinated * Unknown

Was the bite provoked? * Unknown

☐ Multiple Bites

Bite Location & Classification

Bite Location (Body Part) *
e.g., right hand, left leg, face

Bite Category *
-- Select Category --

☐ Override AI classification (use with c

7. Complete the Animal Bite Details by entering the bite date, choosing the animal type, ownership, vaccination status, and provocation, then check the box if multiple bites occurred.



Animal Bite Center

Dashboard Records Geomapping Analytics Staff Settings Logout

Bite Location & Classification

Bite Location (Body Part) *

e.g., right hand, left leg, face

Bite Category *

-- Select Category --

Override AI classification (use with caution)

Bite Categories:

Category I: Touching or feeding of animals, licks on intact skin

Category II: Nibbling of uncovered skin, minor scratches or abrasions without bleeding

Category III: Single or multiple transdermal bites or scratches, contamination of mucous membrane or broken skin with saliva from animal licks, exposures due to direct contact with bats

8. Record the Bite Location & Classification by entering the affected body part, selecting the bite category, and applying the override only when

Animal Bite Center

Dashboard Records Geomapping Analytics Staff Settings Logout

Treatment Information

☐ Wound washed with soap and water ☐ Rabies vaccine administered

☐ Anti-tetanus administered

☐ Antibiotics prescribed

☐ Referred to hospital

Follow-up Date

Case Status *

☐ Additional Notes

Notes

Any additional information or observations about

9. Fill out the Treatment and Case Details section by choosing the treatment, setting the follow-up date, selecting the case status, and adding notes.

Cancel Save Report



10. Once all required sections are completed, click the 'Save Report' button to finalize and submit the entry.

STEPS

1. Navigate to the 'Records' section using the top menu bar.
2. From the dropdown menu, choose the 'New Report' option.
3. The system will redirect you to the new report form for case entry details.
4. Select whether the patient was already registered or was a new entry.
5. For existing patients, the search function will automatically populate their basic information in the form.
6. For new patients, a simplified form will appear. Enter the patient's basic personal information in the fields provided.
7. Complete the Animal Bite Details by entering the bite date, choosing the animal type, ownership, vaccination status, and provocation, then check the box if multiple bites occurred

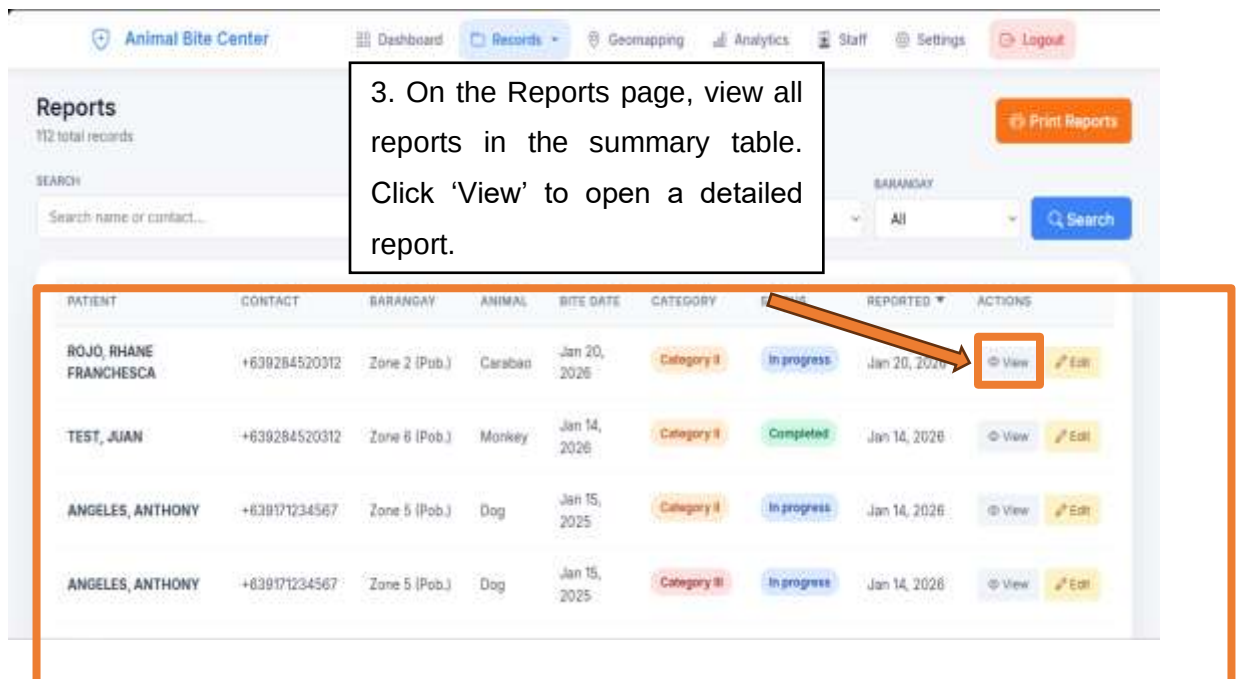
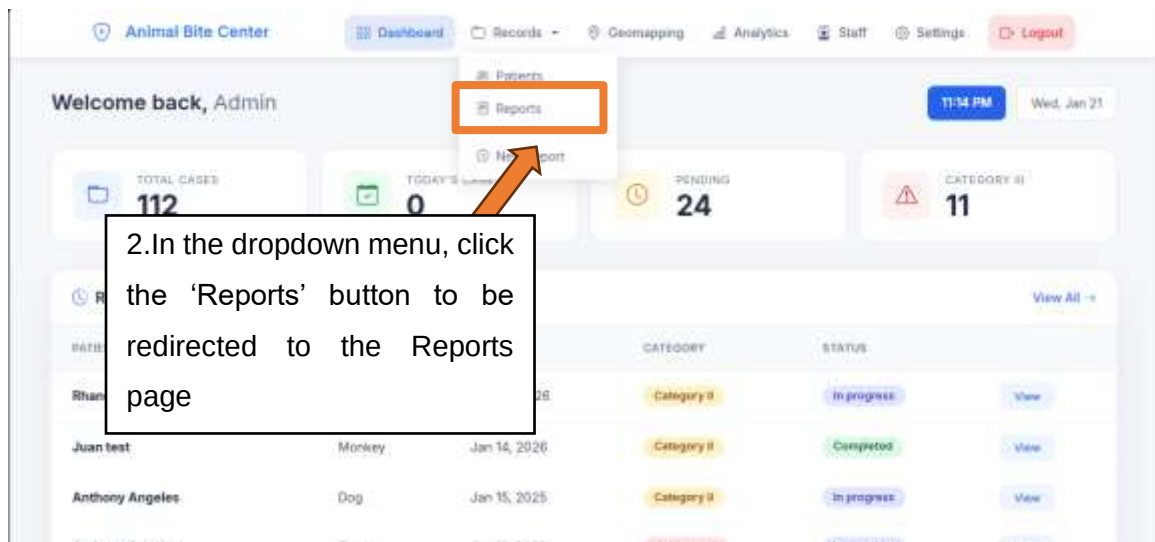


8. Record the Bite Location & Classification by entering the affected body part, selecting the bite category, and applying the override only when necessary.
9. Fill out the Treatment and Case Details section by choosing the treatment, setting the follow-up date, selecting the case status, and adding notes.
10. Once all required sections were completed, click the 'Save Report' button to finalize and submit the entry.

Animal Bite Admin – How to view reports.

1. Navigate to the 'Records' section using the top menu bar

NAME	SPECIES	DATE	CATEGORY	STATUS	
Rhane Franciska Rops	Carabao	Jan 20, 2026	Category II	In progress	View
Juan test	Monkey	Jan 14, 2026	Category II	Completed	View
Anthony Angeles	Dog	Jan 15, 2025	Category II	In progress	View
Anthony Angeles	Pig	Jan 15, 2025	Category II	In progress	View



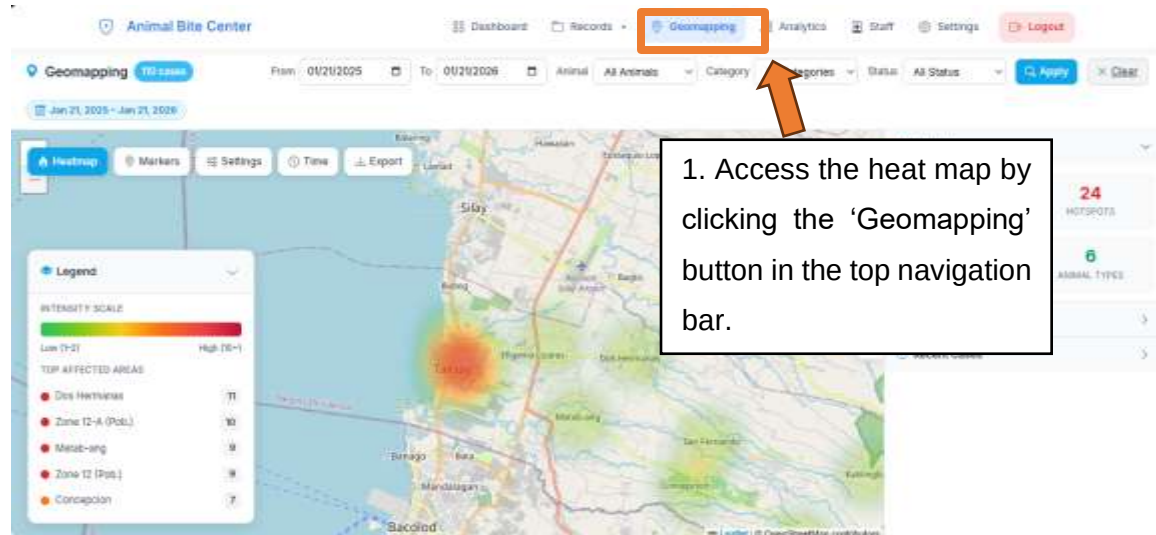
STEPS

1. Navigate to the 'Records' section using the top menu bar.



2. In the dropdown menu, click the 'Reports' button to be redirected to the Reports page.
3. On the Reports page, view all reports in the summary table. Click 'View' to open a detailed report.

Animal Bite Admin – How to navigate the heat mapping function





2. The filter section lets you select which data to display on the map. This keeps it short, clear, and consistent with the rest of your guide.

3. Click the 'Apply' button to activate the selected filter on the map.

4. The legend indicates the intensity scale, ranging from low (green) to high (red), and highlights the top affected areas.

5. Two modes are available: heatmap (color intensity) and markers (precise barangay locations).

6. Three collapsible sections present summarized information that can be easily viewed and interpreted.

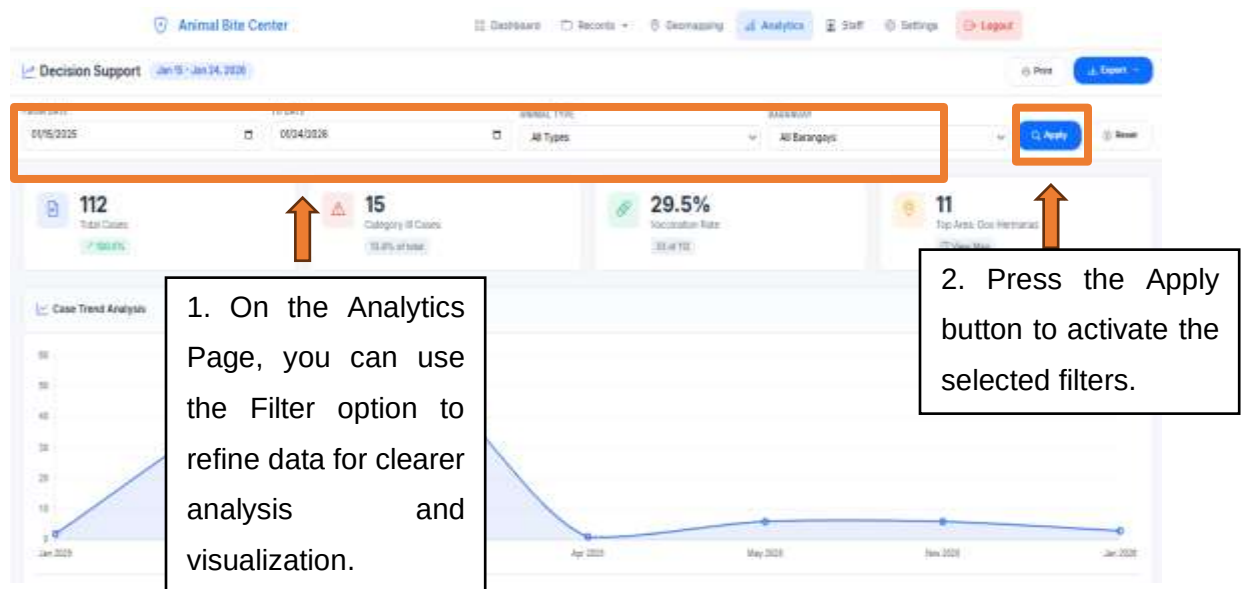
BARANGAY	CASES	RISK
Dos Hermanas	11	CRITICAL
Zone 12-A (Pob.)	10	CRITICAL
Metabo-ang	9	CRITICAL
Zone 12 (Pob.)	9	CRITICAL
Concepcion	7	HIGH
Zone 6 (Pob.)	7	HIGH

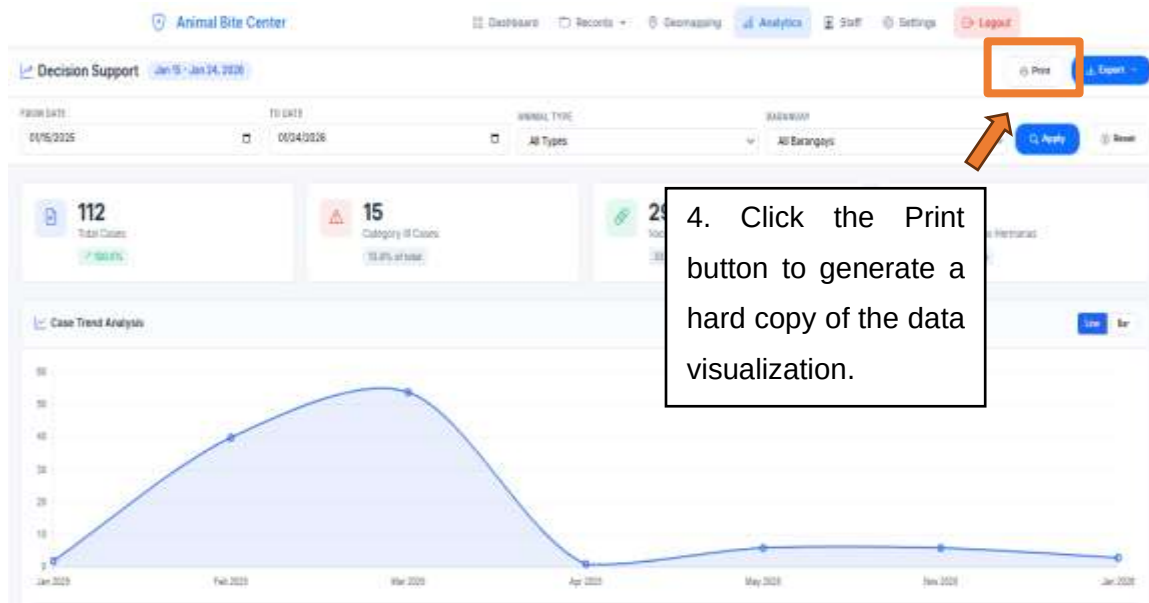
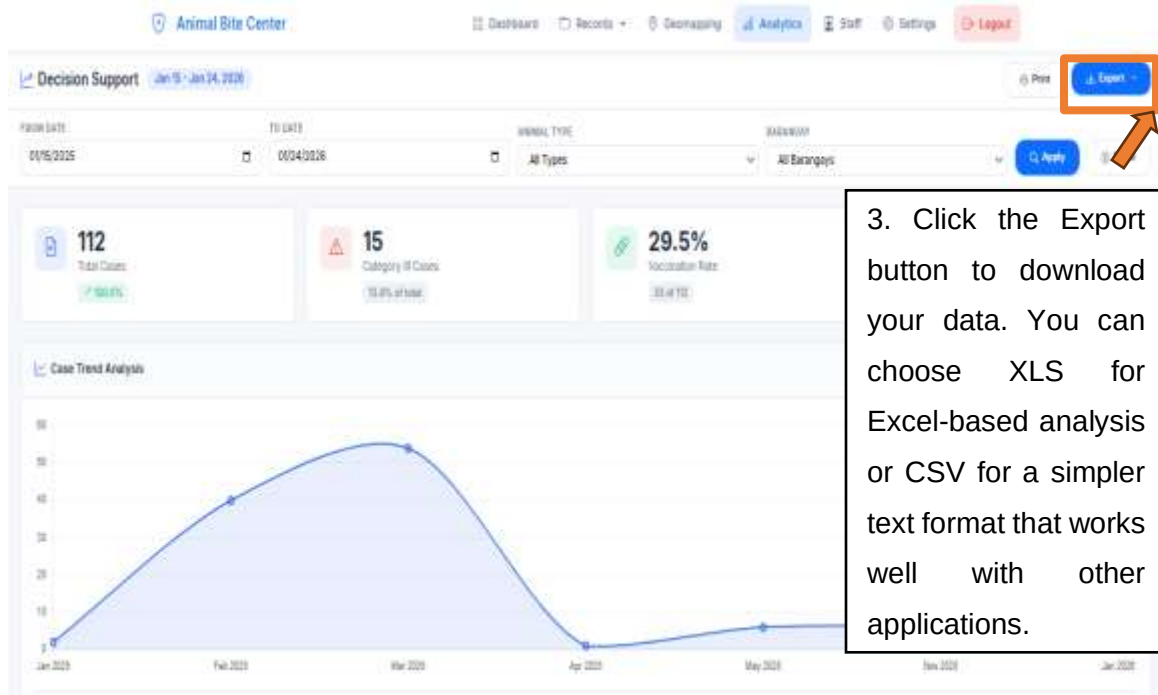


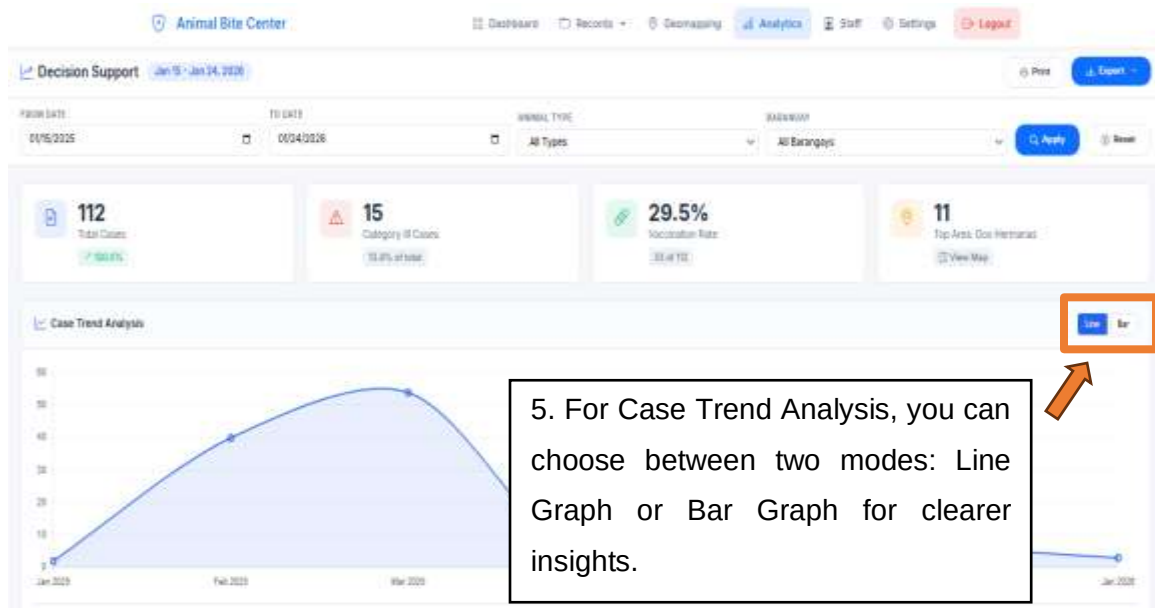
STEPS

1. Access the heat map by clicking the 'Geomapping' button in the top navigation bar.
2. The filter section lets you select which data to display on the map.
3. Click the 'Apply' button to activate the selected filter on the map.
4. The legend indicates the intensity scale, ranging from low (green) to high (red), and highlights the top affected areas.
5. Two modes were available: heatmap (color intensity) and markers (precise barangay locations).
6. Three collapsible sections present summarized information that can be easily viewed and interpreted.

Animal Bite Admin – Decision Support Page







STEPS

1. On the Analytics Page, you can use the Filter option to refine data for clearer analysis and visualization.
2. Press the Apply button to activate the selected filters.



3. Click the Export button to download your data. You can choose XLS for Excel-based analysis or CSV for a simpler text format that works well with other applications.
4. Click the Print button to generate a hard copy of the data visualization.
5. For Case Trend Analysis, you can choose between two modes: Line Graph or Bar Graph for clearer insights.
6. At the bottom of the page, a Recommendation Section displays real-time updates and provides tailored suggestions.

Animal Bite Admin – Adding New Staff

Animal Bite Center | Dashboard | Records | Geomapping | Analytics | **Staff** | Settings | Logout

Staff Management

View and manage barangay health workers

2 TOTAL STAFF | **2** ACTIVE MEMBERS | **1** TOTAL PAGES

SEARCH STAFF
Search by name, email, or contact number | Search

STAFF MEMBER	CONTACT NUMBER	EMAIL	
Anthony Angeles ID: 2	+631234567890	aaangeles.chmsu@gmail.com	Mar 31, 2025 Edit Delete
Julie Ann Panolino ID: 1	+631234567890	jspanolino.chmsu@gmail.com	Jun 30, 2003 Edit Delete

Showing 1 to 2 of 2 staff members

1. Click the 'Add New Staff' button at the top-right of the Staff Management page to open the form.



Animal Bite Center

Dashboard Records Geomapping Analytics Staff Settings Logout

Add New Staff Member

Register a new barangay health worker and assign them to a specific barangay

Back to Staff

Staff Information

PERSONAL INFORMATION

First Name Last Name

Birth Date mm/dd/yyyy

Complete Address

CONTACT INFORMATION

2. On the Add New Staff page, complete the form by entering all required personal information.

Animal Bite Center

Dashboard Records Geomapping Analytics Staff Settings Logout

CONTACT INFORMATION

Contact Number

+63 9123456789

10 digits only; prefix is fixed to +63

Email Address

SARANGAY ASSIGNMENT

Assigned Barangay

Select a barangay

The staff member will be assigned to this barangay

3. Below the Personal Information section, enter the Contact Information—including contact number and a valid email address.



Animal Bite Center

Dashboard Records Geomapping

Email Address

BARANGAY ASSIGNMENT

Assigned Barangay

Select a barangay

Please select a barangay.

☐ The staff member will be assigned to this barangay.

ACCOUNT CREDENTIALS

Password

Confirm Password

4. From the dropdown menu, select the assigned location or barangay where the staff or health worker will be designated.

Animal Bite Center

Dashboard Records Geomapping Analytics Staff Settings Logout

BARANGAY ASSIGNMENT

Assigned Barangay

Select a barangay

Please select a barangay.

☐ The staff member will be assigned to this barangay.

ACCOUNT CREDENTIALS

Password

Confirm Password

Cancel Add Staff Member

5. Lastly, create a secure password for the staff account to ensure safe access



Animal Bite Center

Dashboard Records Geomapping Analytics Staff Settings Logout

Assigned Barangay

Select a barangay

Please select a barangay.

☐ The staff member will be assigned to this barangay.

ACCOUNT CREDENTIALS

Password

6. After completing the form, click the Add Staff Member button to submit and process the entry.

Add Staff Member

STEPS

1. Click the 'Add New Staff' button at the top-right of the Staff Management page to open the form.
2. On the Add New Staff page, complete the form by entering all required personal information.
3. Below the Personal Information section, enter the Contact Information—including contact number and a valid email address.
4. From the dropdown menu, select the assigned location or barangay where the staff or health worker will be designated.
5. Lastly, create a secure password for the staff account to ensure safe access
6. After completing the form, click the Add Staff Member button to submit and process the entry.



Animal Bite Admin – Edit Staff Information

Animal Bite Center

Dashboard Records Geomapping Analytics Staff Settings Logout

Staff Management

View and manage barangay health workers

+ Add New Staff

2 TOTAL STAFF

2 ACTIVE MEMBERS

1 TOTAL PAGES

SEARCH STAFF

Search by name, email, or contact number

Search Clear

STAFF MEMBER	CONTACT	ACTIONS
Anthony Angeles ID: 1	+631	Edit Delete
Julie Ann Panolino ID: 1	+631	Edit Delete

Showing 1 to 2 of 2 staff members

1. Click the **Edit** button in the Staff Page table to modify the selected staff member's details.

Animal Bite Center

Dashboard Records Geomapping Analytics Staff Settings Logout

Edit Staff Member

Update Staff Information

Edit Staff Information

Personal Information

First Name: Anthony

Last Name: Angeles

Birth Date: 03/31/2025

Address: Pani

Assigned Barangay:

2. The page will be redirected to the edit staff member form.



3. Update any personal information fields that require modification, such as name, birthdate, or address.

4. Click the 'Save Changes' button to save the newly updated staff information into the system.

STEPS

1. Click the **Edit** button in the Staff Page table to modify the selected staff member's details.
2. The page will be redirected to the edit staff member form.
3. Update any personal information fields that require modification, such as name, birthdate, or address.



4. Click the 'Save Changes' button to save the newly updated staff information into the system.

Animal Bite Admin – Adding New Barangay Selection

1. Navigate to the Settings page by clicking Settings in the top bar menu.

BARANGAY	LATITUDE	LONGITUDE	ACTIONS
Bubog	10.736188	123.873488	Edit Delete
Cabatangan	10.678296	123.184234	Edit Delete
Concepcion	10.693767	123.899871	Edit Delete
Dos Hermanas	10.742179	123.833440	Edit Delete
Efigenio Lizares	10.743488	123.885425	Edit Delete

2. Click the 'Add Barangay' Button



Animal Site Center

Dashboard Records Geomapping Analytics Staff Settings Logout

Barangay Coordinates
Manage geographic coordinates for heatmap visualization

+ Add Barangay Refresh

3. The form will appear directly below the button once clicked.

BARANGAY NAME LATITUDE LONGITUDE

Enter barangay name e.g., 10.735000 e.g., 122.865000

Tip: To get coordinates, right click on Google Maps at the desired location and select "What's here?" - the coordinates will appear at the bottom.

Cancel Save Barangay

BARANGAY	LATITUDE	LONGITUDE	ACTIONS
Subog	10.710188	122.873488	Edit Delete

Animal Site Center

Dashboard Records Geomapping Analytics Staff Settings Logout

Barangay Coordinates
Manage geographic coordinates for heatmap visualization

+ Add Barangay Refresh

4. Start by filling up the name of the barangay.

BARANGAY NAME LATITUDE LONGITUDE

Enter barangay name e.g., 10.735000 e.g., 122.865000

Tip: To get coordinates, right click on Google Maps at the desired location and select "What's here?" - the coordinates will appear at the bottom.

Cancel Save Barangay

BARANGAY	LATITUDE	LONGITUDE	ACTIONS
Subog	10.710188	122.873488	Edit Delete



Animal Site Center | Dashboard | Records | Geomapping | Analytics | Staff | Settings | Logout

Barangay Coordinates

Manage geographic coordinates for heatmap visualization

5. Copy the latitude and longitude coordinates of the barangay from Google Maps, then paste them into the form fields

LATITUDE: e.g., 10.735000 | LONGITUDE: e.g., 122.865000

Cancel | Save Barangay

BARANGAY	LATITUDE	LONGITUDE	ACTIONS
Subog	10.735000	122.865000	Edit Delete

Animal Site Center | Dashboard | Records | Geomapping | Analytics | Staff | Settings | Logout

Barangay Coordinates

Manage geographic coordinates for heatmap visualization

+ Add Barangay | Refresh

BARANGAY NAME:

LATITUDE: e.g., 10.735000 | LONGITUDE: e.g., 122.865000

Tip: To get coordinates, right click on Google Maps at the location and select "What's here?" - the coordinates will appear at the bottom.

Cancel | Save Barangay

6. Click the Save Barangay button to store the newly created barangay in the system.

BARANGAY	LATITUDE	LONGITUDE	ACTIONS
Subog	10.735000	122.865000	Edit Delete

STEPS

1. Navigate to the Settings page by clicking Settings in the top bar menu.
2. Click the 'Add Barangay' Button.
3. The form will appear directly below the button once clicked.
4. Start by filling up the name of the barangay.



5. Copy the latitude and longitude coordinates of the barangay from Google Maps, then paste them into the form fields
6. Click the Save Barangay button to store the newly created barangay in the system.

Animal Bite Admin – Add new animal type

1. Navigate in settings page and scroll down.

2. Click the 'Add Animal Type' button.

ANIMAL TYPE	Report(s)	Action
Dog	52 report(s)	Delete
Cat	33 report(s)	Delete
Rat	3 report(s)	Delete
Monkey	3 report(s)	Delete
Bat	0 report(s)	Delete



Animal Bite Center

Dashboard Records Geomapping Analytics Staff Settings Logout

Animal Types

Manage animal types for bite reports

[Add Animal Type](#) [Refresh](#)

ANIMAL NAME

Note: Once added, this animal type will appear in the dropdown when creating or editing bite reports.

[Cancel](#) [Save Animal Type](#)

ANIMAL TYPE	REPORTS COUNT	ACTIONS
Dog	11 Report(s)	Delete

3. The form will appear after clicking the button.

Animal Bite Center

Dashboard Records Geomapping Analytics Staff Settings Logout

Animal Types

Manage animal types for bite reports

[Add Animal Type](#) [Refresh](#)

ANIMAL NAME

Note: Once added, this animal type will appear in the dropdown when creating or editing bite reports.

[Cancel](#) [Save Animal Type](#)

ANIMAL TYPE	REPORTS COUNT	ACTIONS
Dog	11 Report(s)	Delete

4. Type a valid animal name in the designated field (e.g., Dog, Cat, Carabao).



Animal Bite Center

Dashboard Records Geomapping Analytics Staff Settings Logout

Animal Types
Manage animal types for bite reports

+ Add Animal Type Refresh

ANIMAL NAME
Enter animal name (e.g., Monkey, Bat)

Note: Once added, this animal type will appear in the dropdown

Cancel Save Animal Type

ANIMAL TYPE

Dog

Report(s) Delete

5. Click the 'Save Animal Type' button to store the newly created animal type into the system.

STEPS

1. Navigate in settings page and scroll down.
2. Click the 'Add Animal Type' button.
3. The form will appear after clicking the button.
4. Type a valid animal name in the designated field (e.g., Dog, Cat, Carabao).
5. Click the 'Save Animal Type' button to store the newly created animal type into the system.



BHW / Staff – Sign in

Animal Bite Portal

Talisay City Health

1. Fill up the form with a valid email address and password created for you.

Welcome back
Sign in to access your dashboard

Email Address
admin@example.com

Password
Enter your password

Sign In

Animal Bite Portal

Talisay City Health Animal Bite Center

2. Click the 'Sign In' button to proceed.

Welcome back
Sign in to access your dashboard

Email Address
admin@example.com

Password
Enter your password

Sign In

Continue with Google

Secure access for authorized personnel only

STEPS

1. Fill up the form with a valid email address and password created for you.
2. Click the 'Sign In' button to proceed.



BHW / Staff – Create New Report

The image shows two screenshots of the Animal Bite Center web application. The top screenshot is the dashboard for a user named Anthony. It features a 'Welcome back, Anthony' message and a 'Create New Report' section. A green button labeled '+ Start New Report' is highlighted with an orange box. An orange arrow points from this button to a text box containing the instruction: '1. Click the Start New Report button to be redirected to the New Report Form.' Below this, a 'How to Report' section lists two steps: '1 Find the Patient' and '2 Record Incident Details'. The bottom screenshot shows the 'New Animal Bite Report' form. The 'Patient Information' section is highlighted with an orange box. It contains fields for 'First Name', 'Last Name', 'Gender' (a dropdown menu), 'Date of Birth' (a date picker), 'Contact Number' (with a country code dropdown and a text input), 'Barangay' (a dropdown menu), and 'Address'. An orange arrow points from the text box '2. Enter the patient's basic information: name, age, gender, and address.' to the 'Patient Information' section.

Animal Bite Center

New Report

Welcome back, Anthony
Report and document animal bite incidents in your barangay

Create New Report
Document a new animal bite incident

+ Start New Report

1. Click the Start New Report button to be redirected to the New Report Form.

How to Report

1 Find the Patient
Use the patient search to locate existing records or create a new patient entry.

2 Record Incident Details
Document the bite location, animal type, circumstances, and patient information.

Animal Bite Center

Back to Dashboard

New Animal Bite Report

Patient Information

First Name *
Last Name *

Gender
-- Select Gender --

Date of Birth
mm/dd/yyyy

Contact Number
+63 9123456789
Enter 10 digits only; +63 is fixed.

Barangay
Zone 1 (Pop.)

Address

2. Enter the patient's basic information: name, age, gender, and address.



Animal Bite Center [Back to Dashboard](#)

Animal Bite Details

Date of Bite * 01/24/2026

Animal Type * -- Select Animal --

Animal Ownership * -- Select Ownership --

Animal Status * Alive

Animal Vaccinated * Unknown

Was the bite provoked? * Unknown

☐ Multiple Bites

[Save to Dashboard Dashboard](#)

3. Fill in the Animal Bite Details form with the animal type, incident date and animal status.

Animal Bite Center [Back to Dashboard](#)

Bite Location & Classification

Bite Location (Body Part) *
e.g., right hand, left leg, face

Bite Category *
-- Select Category --

☐ Override AI classification (use with caution)

Bite Categories:

Category I: Touching or feeding of animals, licks on intact skin

Category II: Nibbling of uncovered skin, minor scratches or abrasions without bleeding

Category III: Single or multiple transdermal bites or scratches, contamination of mucous membrane or broken skin with saliva from animal licks, exposures due to direct contact with bats

4. Write the bite location (e.g., head, right arm, left leg) and select the bite category from I to III severity.



Animal Bite Center ← Back to Dashboard

DO NOT DISCHARGE CONTACT WITH REBIS

Treatment Information

☐ Wound washed with soap and water ☐ Rabies

☐ Anti-tetanus administered

☐ Antibiotics prescribed

☐ Referred to hospital

Follow-up Date:

Case Status:

☐ Additional Notes

5. Click the radio buttons to indicate if the wound was washed, if anti-tetanus was administered, and other treatments, then update the case status and set the follow-up date.

Animal Bite Center

Follow-up Date:

☐ Additional Notes

Notes

Any additional information or observations about the case

6. Write any additional notes you find relevant for the patient's case.



Animal Bite Center

Follow-up Date: mm/dd/yyyy

Case Status: Pending

☐ Additional Notes

Notes: Any additional information

7. Click the Save Report button to save the completed form and record the patient's incident report.

Save Report

STEPS

1. Click the Start New Report button to be redirected to the New Report Form.
2. Enter the patient's basic information: name, age, gender, and address.
3. Fill in the Animal Bite Details form with the animal type, incident date and animal status.
4. Write the bite location (e.g., head, right arm, left leg) and select the bite category from I to III severity.
5. Click the radio buttons to indicate if the wound was washed, if anti-tetanus was administered, and other treatments, then update the case status and set the follow-up date.
6. Write any additional notes you find relevant for the patient's case.
7. Click the Save Report button to save the completed form and record the patient's incident report.



Appendix L”

CURRICULUM VITAE



ANTHONY ANGELES

Programmer

Contact



09078060026



anthonyangeles223@gmail.com



Blk. 11 Lot 3 Menlo Garden
Phase 1, Brgy. Zone 12-A,
Talisay City, Negros
Occidental, Philippines

Profile

Seeking a challenging opportunity where I will be able to use my knowledge and skills inclined with my studied profession and the ability to work well with people which will allow me to grow professionally.

Personal Information

- | | |
|---------------------------|----------------------------|
| • Place and date of birth | Bacolod City, Feb 27, 2003 |
| • Citizenship | Filipino |
| • Marital Status | Single |
| • Religion | Roman Catholic |

Certificates

- Career Essentials in Data Analysis
- Microsoft Copilot for Productivity
- IP Awareness and Patent Search Training for Information System Students
- Career Essentials in System Administration
- Career Essentials in Cybersecurity

Education

- **SECONDARY SCHOOL**
 - Rafael B. Lacson Memorial High School / 2015 - 2021
- **COLLEGE**
 - Carlos Hilado Memorial State University 2022 - Present

Skills

- Data Encoding and Data Entry
- Computer Literate (MS Word, MS Excel and MS PowerPoint)
- System Thinking
- Adaptability
- Internet or Data Research
- Programming



MEG RYAN ROJO

Project Manager

Contact



09284520312



mdrojo.chmsu@gmail.com



Blk 11, lot 20, Menlo Phase
2, Brgy Zone-10 Talisay
City Negros Occidental

Profile

Seeking a challenging opportunity to apply my knowledge and skills, work effectively under pressure, collaborate well with others, and continuously improve through problem-solving and professional growth.

Personal Information

- | | |
|---------------------------|----------------------------|
| • Place and date of birth | Bacolod City, Sept 1, 2003 |
| • Citizenship | Filipino |
| • Marital Status | Single |
| • Religion | Roman Catholic |

Certificates

- Career Essentials in Data Analysis
- Microsoft Copilot for Productivity
- IP Awareness and Patent Search Training for Information System Students
- Career Essentials in System Administration
- Career Essentials in Cybersecurity

Education

- **SECONDARY SCHOOL**
 - Rafael B. Lacson Memorial High School / 2015 - 2022
- **COLLEGE**
 - Carlos Hilado Memorial State University 2022 - Present

Skills

- Data Encoding and Data Entry
- Computer Literate (MS Word, MS Excel and MS PowerPoint)
- System Thinking
- Adaptability
- Internet or Data Research
- Programming



JULIE ANN PANOLINO

System Analyst

Contact



09945469383



julieannpanolino.chmsu@gmail.com



Talisay Towne Heights
Brgy. Matabang, Talisay
City, Negros Occidental,
Philippines

Profile

Motivated and detail-oriented individual seeking an opportunity to apply my skills and knowledge while contributing to a productive team environment.

Personal Information

- Place and date of birth Bacolod City, Dec 5, 2003
- Citizenship Filipino
- Marital Status Single
- Religion Roman Catholic

Certificates

- Career Essentials in Data Analysis
- Microsoft Copilot for Productivity
- IP Awareness and Patent Search Training for Information System Students
- Career Essentials in System

Education

- **SECONDARY SCHOOL**
 - Domingo Lacson National High School 2021-2022
- **COLLEGE**
 - Carlos Hilado Memorial State University 2022 - Present

Skills

- Data Encoding and Data Entry
- Computer Literate (MS Word, MS Excel and MS PowerPoint)
- System Thinking
- Adaptability
- Internet or Data Research